Random-Sign IBLTs: Calibration-Free Self-Sizing After Decoding Failure

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Abstract

In a two-round set reconciliation protocol based on an Invertible Bloom Lookup Table (IBLT), when the fixed-capacity difference table of the first round suffers a decoding failure because the difference is overloaded, the receiver must estimate a safe upper bound on the unknown difference size d from the sum of squared counts of the difference table, so as to set the second-round table size adaptively. The standard IBLT, however, uses the set direction of each element directly as the accumulation sign, which couples the hypergraph-overlap bias selected by failure with the sign composition of the two-sided difference; a one-sided difference amplifies the estimation bias in the same direction, forcing existing schemes to rely on tedious offline simulation grid calibration.

This paper proposes the *random-sign IBLT*. By introducing a pseudorandom sign independent of the position mapping, the design decouples failure selection from the sign composition in principle, while fully preserving the number of cell fields, their widths, and the peeling recovery capability. We prove that the estimator is exactly unbiased on any fixed failed hypergraph structure; we prove a Gaussian limit for the estimation error conditioned on failure when the difference size is large, and derive a closed-form Gaussian upper bound; we further characterize a discrete limit distribution dominated by duplicate edge pairs at small difference scales, and construct a Skellam upper-bound rule that strictly controls the conditional miss rate for all $d \geq 2$ and requires no offline calibration. Five hundred million paired trials and end-to-end real protocol experiments spanning several orders of magnitude of load show that the random-sign IBLT removes direction sensitivity while strictly meeting the preset risk, and that the total two-round communication cost approaches the ideal lower bound for a known difference size.

1 Introduction

In many core scenarios of distributed systems, two ends of a network each hold a set A and B and need to find their symmetric difference $A \triangle B$, the elements that exist on only one side. This problem is called set reconciliation. The Invertible Bloom Lookup Table (IBLT)[19, 15] is the standard data structure for this task: each side builds a table of M cells, and each element is written through hashing into the count, keySum, and checksum fields of k cells; after the receiver subtracts the two tables cell by cell, elements shared by both sides write the same value to the same group of cells and cancel exactly, leaving only the contributions of the $d = |A \triangle B|$ differing elements in the difference table. An element present only in A has direction $+1$, and an element present only in B has direction -1. When the difference size d is not too large relative to the table size M (the load d/M is below the peeling threshold), the receiver can repeatedly find a pure cell containing a single differing element and strip away all differing elements one by one through the peeling process; once d exceeds the table's capacity, peeling stalls partway and a decoding failure occurs.

Using an IBLT raises a basic tension: the table size M must be fixed when the table is built, while the difference size d is precisely the quantity unknown before synchronization completes.

In real systems, d for the same pair of nodes can vary by several orders of magnitude across time: during routine synchronization the two sides are almost identical and d is small; after network partition recovery or a batch write, d can grow sharply. If a large table is always allocated for the worst case, most routine synchronizations waste bandwidth; if a small table is allocated, a large difference causes a decoding failure. The two-round protocol[38] provides a natural way out: the first round sends a fixed small table, so that a routine small difference completes directly within one round; when the first round suffers a decoding failure because the difference is too large, the difference table is not actually wasted, and its count array still retains the traces left by all d differing elements superimposed. The receiver only needs to estimate an upper bound U on d from the sum of squared counts of the first-round difference table, and the second round allocates a table of sufficient capacity according to U, completing the synchronization within two rounds. The key to this mechanism is controlling the underestimate risk under failure (the conditional miss rate): once $U < d$ (a miss occurs), the second-round table faces the danger of failure again because its capacity is insufficient; conversely, if U is over-amplified for safety, the second-round transmission bandwidth is wasted.

The work of [38] proposed an analysis method that uses the count sum and the sum of squared counts of the first-round difference table to construct an unbiased estimator $\widehat{d}$ and invert an upper bound U. However, when estimating d from a difference table that suffered failure, the real difficulty is that the decoding failure itself selects the sample, and this selection couples with the sign composition of the set difference (the ratio of the number of differing elements from A to those from B). When estimating d with the sum of squared counts of that method, each element's own contribution is fixed, and the estimation error comes entirely from the cross terms produced when different differing elements fall into the same cell. In the standard IBLT, the difference table directly uses each element's set-membership direction ($+1$ or -1) as the accumulation sign of count: when two same-direction elements fall into the same cell, their counts add in the same direction and the sum of squares increases; when two opposite-direction elements fall into the same cell, they cancel. At the same time, peeling fails precisely because the position mapping of differing elements to cells has too much overlap. These two factors combine to produce a systematic bias in the failure sample. Take the one-sided difference (all differing elements come from the same side, common in one-way data synchronization such as a replica missing only the data newly written to the primary) as an example: all cross terms have the same sign, and the high overlap selected by failure is amplified in the same direction into the sum of squared counts, making the estimate deviate markedly; in two-way synchronization where each side writes new data, the two sides' differences are comparable, called the balanced difference, in which case the positive and negative cross terms largely cancel and the post-failure estimated mean is almost unchanged. More troublesome is that the magnitude of this bias varies continuously with the two sides' difference ratio, and its sign reverses with the number of hashes k (Section 2). Because the receiver at decoding failure does not know whether the unknown difference comes entirely from one side or from both, faced with the observed sum of squared counts it cannot tell how much comes from the true d and how much from the direction bias brought by failure selection. Therefore, when constructing an upper bound on the standard IBLT, [38] has to rely on offline simulation calibration by configuration, except in the deep overload region.

The core insight of this paper is: since the bias originates from the standard IBLT treating the set-membership direction directly as the in-table accumulation sign, decoupling the element's set direction from the in-table accumulation sign removes this coupling at its source. Accordingly, this paper proposes the random-sign IBLT. The approach is lightweight: the two sides share in advance a random-sign hash independent of the position mapping, assigning each key a random $+1$ or -1; when inserting an element, the element's random sign is multiplied uniformly into the count, keySum, and checksum fields.

This modification satisfies the requirements of both decoding and estimation. On the decoding side, elements shared by the two sides are multiplied by the same random sign in both tables and still cancel exactly after subtraction. When a cell contains only one unrecovered differing element, its count is $+1$ or -1; multiplying keySum by the count cancels the random sign exactly (since $(\pm 1)^2 = 1$) and reads out the element's key as is; after reading the key, computing its random sign and comparing it with the count recovers whether it originally belonged to set A or B. Therefore, under ideal purity, the random-sign IBLT can strip exactly the same elements as the standard IBLT at every step, and the decoding threshold, the failure probability, and the number and byte width of the cell fields all remain unchanged. On the statistical estimation side, whether the true set difference is one-sided or balanced, the coding coefficient of each differing element in the difference table becomes an independent uniform random sign. More importantly, whether peeling succeeds is determined only by which cells the elements fall into (the mapping), while the cancellation of the estimation error is determined by the random signs; the two are fully independent. No matter how special a mapping the failure event selects, once that mapping is fixed, the random signs placed on it remain independent and uniform, cancelling out. Using random signs to cancel cross terms is a classical idea dating back to the AMS sketch and CountSketch[1, 8]; this paper implements it as a peelable data-structure design, giving a single table the ability of both difference recovery and difference-size estimation, and thereby enabling a rigorous statistical analysis of the post-failure estimator.

Main results and contributions. Building on the decoupling of encoding and sign, this paper first presents the encoding and peeling algorithm of the random-sign IBLT (Section 3), proving that under ideal purity its decoding behavior is identical to that of the standard IBLT, and that the joint distribution of the difference-table statistic and the mapping is the same for all sign compositions. We then prove that the estimator $\widehat{d}$ is exactly unbiased on every fixed mapping, so that its conditional mean after failure equals d, and fully characterize the conditional distribution of the estimation error over all load scales (Sections 4 and 5): at large difference scales we derive a Gaussian rule based on the conditional Gaussian limit; at small difference scales we characterize a discrete limit distribution, constructing a Skellam rule that requires no offline simulation calibration and uniformly controls the conditional miss rate for all difference sizes, together with an efficient planted sampling method for the rare failure at low load. The experiments (Section 6) comprehensively verify that the random-sign encoding has the same decoding behavior as the standard encoding, show that the Skellam rule keeps the conditional miss rate strictly within the target risk over the entire load region spanning several orders of magnitude, reliably complete synchronization when attached to the two-round protocol, and achieve a total transmission cost close to the ideal lower bound of knowing the true d.

Structure of the paper. Section 2 introduces the standard IBLT model and analyzes its post-failure estimation bias; Section 3 proposes the random-sign IBLT and the two-round protocol; Section 4 proves the exact unbiasedness given the mapping and gives the Gaussian rule at large scales; Section 5 derives the small-scale discrete limit distribution and the Skellam rule for all scales; Section 6 reports the experimental evaluation; Sections 7 and 8 discuss related work, rule selection, and computational cost. Complete proofs and supplementary experiments are in the appendix.

2 Model and post-failure estimation bias

This section introduces the standard IBLT and the ideal peeling model, defines the requirement of the two-round protocol on the conditional miss rate, characterizes the structure of failure at

low load, and analyzes the mechanism by which the post-failure bias of the standard IBLT varies with the sign composition. Notation used throughout the paper is collected in Appendix A.

2.1 The standard IBLT and the ideal peeling model

The two sides hold sets A and B, respectively, and need to find the symmetric difference $A \triangle B$, whose size $d = |A \triangle B|$ is unknown in advance. The standard IBLT consists of M cells, each storing three fields: count, keySum, and checksum. When inserting a key x, the cell hash selects k distinct cells from $[M]$, and accumulates 1, x, and $\mathrm{chk}(x)$ into the three fields of these cells, respectively. After each side builds its table and the two are subtracted cell by cell, a shared key falls into the same cells and writes the same value in both tables, so all three fields cancel; only the contributions of the d differing keys remain in the difference table, with a key present only in A carrying direction $s_i = +1$ and a key present only in B carrying $s_i = -1$. Denote the numbers of keys in the two directions by d_+ and d_-; the ordered pair (d_+, d_-) is called the *sign composition*, and $b = d_+ - d_- = |A| - |B|$ is the difference between the two set sizes. Throughout the paper we use two extreme sign compositions: when all differing keys come from the same side ($|b| = d$), it is called the *one-sided difference*, and when the two sides have equal numbers of differences ($d_+ = d_-$, $b = 0$), it is called the *balanced difference*. These two cases correspond to common synchronization scenarios. The one-sided difference arises from one-way synchronization, where only one side inserts or deletes entries and the other side stays unchanged. The balanced difference arises from two-way synchronization, where each side modifies a small number of entries and the insertions and deletions roughly cancel. Real loads usually lie in between, and the specific position depends on how much each side changed. Decoding uses the peeling process: repeatedly find a pure cell containing a single unrecovered differing key, read out this key by multiplying keySum by the count, verify it with the checksum, and remove its contribution from all k cells it belongs to; the removal may turn other cells pure, and the iteration continues until the table has no pure cell.

Under ideal purity (a cell with count ± 1 indeed contains a single unrecovered differing key), whether peeling succeeds is determined entirely by the mapping structure of differing keys to cells. Viewing the M cells as vertices and the k cells selected by each differing key as a hyperedge, the difference table corresponds to a random k-uniform hypergraph: the d differing keys independently and uniformly choose k-subsets $S_i \subseteq [M]$, and different keys may choose the same subset. Denote by $\mathcal{G}$ the unsigned mapping with key labels, and by $H_{ij} = |S_i \cap S_j|$ the overlap of two edges, the number of cells jointly selected by differing keys i and j. On $\mathcal{G}$, a pure cell is exactly an unsigned degree-one vertex, and peeling is equivalent to repeatedly deleting edges incident to degree-one vertices on the hypergraph. The deletion order does not affect the terminal state, and peeling always stops at the 2-core of the hypergraph; when the 2-core is nonempty, a decoding failure occurs, denoted as event F, whose probability $\Pr(F)$ is called the failure probability. Hence F depends only on the unsigned mapping $\mathcal{G}$. When the load d/M is below the critical value α_c (about 0.818 for $k = 3$ and about 0.772 for $k = 4$[30, 14]), peeling succeeds with high probability, and the rare failure is determined by small cores such as duplicate edge pairs, decaying polynomially in M; in the critical window of width of order $M^{-1/2}$ around α_c, the failure probability rises from near zero to near one[13]; past α_c, peeling fails almost surely.

2.2 The two-round protocol and the conditional miss rate

Because the capacity M must be fixed before d is known, the two-round protocol[38] first sends a small table of fixed size M_1: if the difference is small, peeling succeeds directly in the first round; if the first round suffers failure, the receiver reads an upper bound U on d from the count array of the first-round difference table, chooses the second-round size $m(U)$ according to U, and re-encodes and decodes with a fresh independent cell hash.

Denote by C_v the count of cell v in the difference table. Following [38], define the estimator $\widehat{d}$ of d from the count sum $\sum_v C_v$ and the sum of squares $\sum_v C_v^2$ saved before peeling begins:

$$\widehat{d} = \frac{1}{k(1-k/M)}\Big(\sum_v C_v^2 - \frac{1}{M}\Big(\sum_v C_v\Big)^2\Big), \tag{2.1}$$

and denote by $s(n)$ its unconditional standard-deviation scale:

$$s(n) = \sqrt{\frac{2n(n-1)}{M-1}}. \tag{2.2}$$

Under the standard IBLT, $\widehat{d}$ is exactly unbiased for d, has variance $s(d)^2$, and this holds for any sign composition; the expectation here is taken over all runs and is not conditioned on failure F, so it is called unconditional. If the upper bound U were evaluated identically over all runs, adding a few standard deviations to $\widehat{d}$ would control the underestimate probability. But in the two-round protocol, when the first round succeeds the protocol ends immediately and U need not be computed; only when the first round suffers a decoding failure (event F) does the receiver compute U to decide the second-round table size. Once $U < d$ occurs, the second-round table's capacity falls below the preset margin and faces the risk of failing again. We call the underestimate event $U < d$ a *miss*, and the underestimate probability given that the first round suffered failure, $\Pr(U < d \mid F)$, the *conditional miss rate*. Hence the sizing rule must control the conditional miss rate:

$$\Pr(U < d \mid F) \le u, \tag{2.3}$$

where u is the target risk, which in practical deployment can be 0.01 or 0.05, i.e. a 1 % or 5 % underestimate risk.

From the perspective of the entire two-round protocol, a synchronization only misses when the first round suffers failure and the upper bound underestimates d, whose probability is the *joint miss rate* $\Pr(F, U < d) = \Pr(F)\Pr(U < d \mid F)$. Keeping the conditional miss rate within u guarantees that whether the first round lies in the overload region where failure is almost certain or in the low-load region where failure occurs occasionally, the joint miss rate never exceeds $u\Pr(F)$; on the failure samples without a miss ($U \ge d$), the second-round table has a sufficient capacity margin and hence completes decoding with high probability. Section 5.6 gives the complete decomposition of this two-round total failure rate.

2.3 Failure at low load and the planted distribution

Eq. (2.3) is required to hold for all loads, and the low-load region needs separate treatment. When d is much smaller than M, the failure probability decays polynomially in M, and failure is a rare event: the distribution conditioned on F is hard to analyze directly, and direct simulation must generate many mappings to obtain one failure sample. The failure structure in this region, however, is simple. Call two edges with the same mapping a duplicate edge pair. A single edge cannot form a 2-core, and two edges form a 2-core if and only if they are a duplicate edge pair; when $d = o(M)$, the 2-cores formed by three or more edges are negligible relative to duplicate edge pairs. Hence failure is nearly the same event as the existence of a duplicate edge pair, and conditioning on F is close to planting a pair of duplicate edges into a random mapping. This motivates the *planted distribution*: choose a pair of keys uniformly, make them select the same uniformly random k-subset, and let the mappings of the remaining keys be independent and uniform.

Proposition 2.1 (Failure is determined by duplicate edge pairs when $d = o(M)$)**.** *Fix $k \ge 3$ and let $2 \le d = o(M)$. Then the failure probability is asymptotically equal to the expected number of duplicate edge pairs, namely $\Pr(F) \sim \binom{d}{2}/\binom{M}{k}$; and the total variation distance between the distribution of $\mathcal{G}$ conditioned on F and the planted distribution tends to zero.*

Duplicate edge pairs are the main source of IBLT failure; this mechanism was pointed out by Goodrich and Mitzenmacher[19], and the finite-length analysis of Yugawa and Wadayama[40] also shows that stopping sets of size 2 dominate. Proposition 2.1 advances it to a total-variation approximation of the whole mapping, valid for all $d = o(M)$; the quantitative error bound and proof are in Lemma D.2 of Appendix D. Proposition 2.1 is useful both theoretically and experimentally. Theoretically, it reduces the problem conditioned on failure to a problem under the planted distribution, which carries no failure condition and is easy to compute directly; the limit theorems of Sections 4 and 5 handle the failure condition through it when $d = o(M)$, and the risk decomposition of Section 5.6 also uses it to estimate the first-round failure probability. Experimentally, it allows sampling directly from the planted distribution, every sample of which produces a failure; Section 6.1 uses this planted sampling to evaluate low-load configurations that direct sampling cannot cover, and checks its finite-size accuracy on configurations where both samplings are feasible.

2.4 Post-failure bias of the standard IBLT

The difficulty in controlling Eq. (2.3) is that the failure event F selects the mapping $\mathcal{G}$. Failure samples concentrate on mappings with large overlap H_{ij}, i.e. different differing keys jointly select many cells: at low load they are mainly mappings containing duplicate edge pairs (Proposition 2.1), and in the critical window and at higher load they are large cores with dense overlap. Consequently, any statistic depending on H_{ij} deviates from its unconditional distribution once conditioned on F.

In the standard IBLT, the difference-table count is $C_v = \sum_{i:v\in S_i} s_i$, and the coding coefficient of each differing key is its fixed set direction $s_i \in \{+1, -1\}$. When summing C_v^2, the sum of the diagonal terms of individual keys is fixed at kd, and the estimation error comes entirely from the cross terms $s_i s_j H_{ij}$ produced by different keys on shared cells. The excess overlap of same-sign pairs raises $\widehat{d}$, and that of opposite-sign pairs lowers $\widehat{d}$. Hence the conditional relative bias of $\widehat{d}$ decomposes exactly into the product of two factors.

Proposition 2.2 (Bias decomposition of the standard IBLT)**.** *In the standard IBLT, for any sign composition* (d_+, d_-)*,*

$$\frac{\mathbb{E}[\widehat{d} \mid F]}{d} - 1 = c_F\Big(\frac{b^2}{d} - 1\Big), \qquad c_F = \frac{\mathbb{E}[H_{12} \mid F] - k^2/M}{k(1 - k/M)}. \tag{2.4}$$

The proof is in Appendix B. Because F is invariant under permutation of the key labels, $\mathbb{E}[H_{ij} \mid F]$ is the same for all $i \neq j$, and the formula takes the pair of differing keys 1 and 2 as a representative. c_F depends only on the selection of the mapping by failure and measures how much the inter-edge overlap in the failure sample exceeds its unconditional mean k^2/M; $b^2/d - 1$ depends only on the sign composition (d_+, d_-). Under the one-sided difference ($|b| = d$), the sign product of every pair of intersecting edges is $+1$, so $b^2/d - 1 = d - 1$, and the excess overlap adds in the same direction and is amplified fully onto $\widehat{d}$; under the balanced difference ($b \approx 0$), same-sign and opposite-sign pairs are comparable in number, the positive and negative cross terms cancel, and only a negligible $-c_F$ bias remains.

Figure 2.1 shows this decomposition at $M = 16384$ with five levels of failure probability (0.10 to 0.90) each for $k = 3$ and $k = 6$; the five sign compositions on the same configuration share the same batch of mappings, so peeling succeeds or fails identically. Since $b^2/d - 1 = d(b/d)^2 - 1$, the relative bias at each level is a linear function of $(b/d)^2$ with slope determined by c_F. After determining c_F from the one-sided point, the predictions of Eq. (2.4) for the remaining levels differ from the measured values by no more than twice the standard error of the difference. For $k = 3$ at failure probability 0.10, the relative bias of the one-sided difference is $+0.70\%$, about 0.64 times the standard deviation $s(d)$; the higher the failure probability, the closer the failure sample is to the full sample and the weaker the selection, so $|c_F|$ decreases. The sign of c_F also

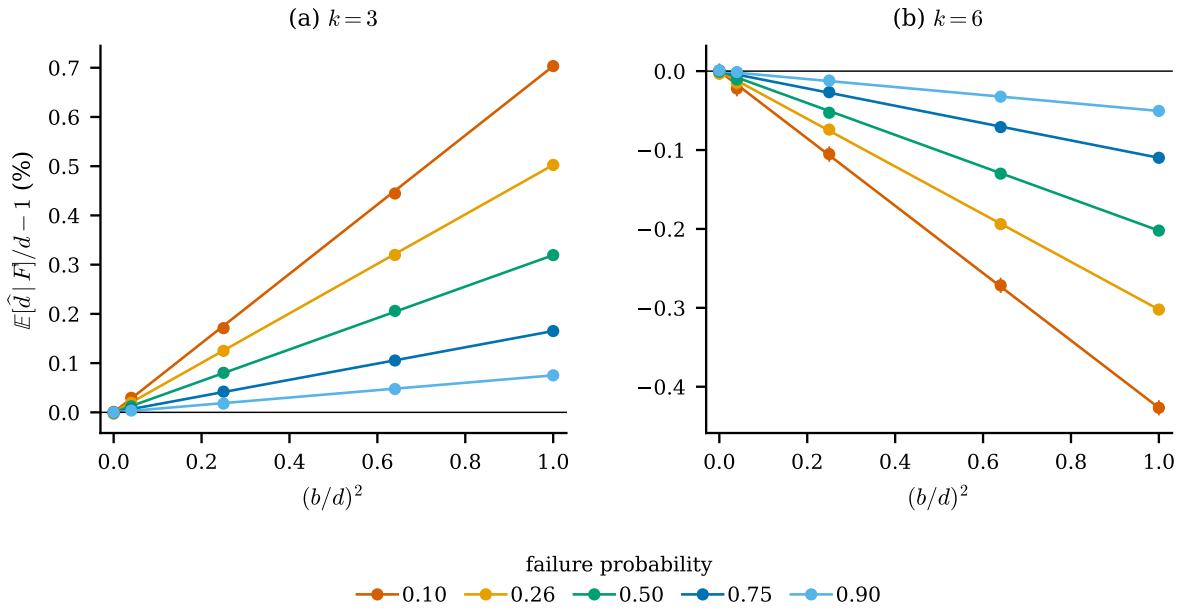


Figure 2.1: Post-failure relative bias of the standard IBLT as a function of $(b/d)^2$ ($M = 16384$). (a) $k = 3$; (b) $k = 6$. Each color corresponds to one level of failure probability; all configurations lie in the critical window, with d/M in 0.812–0.822 for $k = 3$ and 0.633–0.639 for $k = 6$. Points are measured values, and lines are Eq. (2.4).

changes with k: for $k = 6$, $c_F < 0$, and the one-sided failure sample is instead biased low, with relative bias -0.43% at failure probability 0.10 (Figure 2.1(b)). The sign of the bias changes with k, and its magnitude varies with the sign composition between zero and the one-sided value, so the receiver cannot offset it with a fixed correction (Figure 6.2).

In real synchronization, the receiver does not know (d_+, d_-) in advance, and from the count array of the difference table alone it cannot distinguish whether a larger observed $\widehat{d}$ comes from a larger true d or from the failure-induced bias under a one-sided difference; the upper-bound rule of [38] therefore needs simulation calibration by load interval outside the deep overload region. From Eq. (2.4) one sees that the bias arises because the standard IBLT directly treats the set direction s_i as the accumulation sign of count, and the sign composition determines the factor by which c_F is amplified; once the positive and negative signs are balanced, this factor approaches -1, and the offset brought by failure selection vanishes. This suggests a natural idea: if the set difference of any sign composition could carry balanced random signs in the difference table while peeling still recovers the original set direction of each differing key, then the post-failure estimator would no longer be affected by the sign composition. Section 3 proposes the random-sign IBLT accordingly.

3 The random-sign IBLT

This section proposes the random-sign IBLT. The core change of the design is to multiply each key by a random sign shared by the two sides on insertion and deletion, while the cell fields, their widths, and the peeling process all remain unchanged. We first give the encoding and decoding rules, prove that it has the same ideal decoding capability as the standard IBLT and that the coding coefficients are independent of the mapping, and then explain how the two-round protocol uses the lower bound $|b|$ to determine the second-round capacity.

3.1 Encoding rules, decoding equivalence, and sign independence

The two sides share in advance a sign hash $\varepsilon : \mathcal{U} \to \{-1, +1\}$ independent of the cell hash, generated directly from the key and a shared seed, requiring no per-key storage. Algorithm 3.1 gives the complete encoding and decoding procedure; the lines marked $\star$ are the entire difference from the standard IBLT: the change is limited to multiplying in the sign on insertion and deletion, and recovering the sign when outputting the direction, while the number of cell fields, their widths, and the control flow of peeling are unchanged. We describe the encoding and decoding parts in turn.

Algorithm 3.1 Encoding and decoding of the random-sign IBLT. Lines marked $\star$ differ from the standard IBLT; field arithmetic is modulo 2^{64}.

1: **procedure** INSERT(x)
2: **for** each cell v of x **do**
3: $\text{count}_v \leftarrow \text{count}_v + \varepsilon(x)$ ▷ $\star$ standard IBLT adds 1
4: $\text{keySum}_v \leftarrow \text{keySum}_v + \varepsilon(x)\, x$ ▷ $\star$ standard IBLT adds x
5: $\text{checksum}_v \leftarrow \text{checksum}_v + \varepsilon(x)\, \text{chk}(x)$ ▷ $\star$ standard IBLT adds $\text{chk}(x)$
6: **procedure** DELETE(x)
7: same as INSERT(x), with the three fields subtracting the same quantities
8: **procedure** PEEL(difference table)
9: add cells with count ± 1 to the candidate queue
10: **while** the candidate queue is nonempty **do**
11: remove cell v; if $C_v \neq \pm 1$, process the next candidate
12: $x \leftarrow C_v\, \text{keySum}_v$
13: **if** $\text{checksum}_v = C_v\, \text{chk}(x)$ and the cells of x include v **then**
14: output key x and direction $C_v\, \varepsilon(x)$ ▷ $\star$ standard IBLT outputs C_v
15: if the direction is $+1$, call DELETE(x), otherwise call INSERT(x)
16: add cells whose count becomes ± 1 to the candidate queue
17: **return** whether all three fields of every cell are zero

Encoding corresponds to lines 1–7 of Algorithm 3.1. When inserting a key x, accumulate $\varepsilon(x)$, $\varepsilon(x)\, x$, and $\varepsilon(x)\, \text{chk}(x)$ into its k cells, and subtract the same quantities on deletion; keySum and checksum are accumulated modulo 2^{64} (in the set model, keySum and checksum can also use the bitwise XOR implementation common in the standard IBLT, see Appendix C.1; the analysis and experiments of this paper use the additive form). After the two sides subtract cell by cell, the three-field contributions of a shared key are identical and cancel exactly; the coding coefficient of a differing key i in the difference table is $s_i \varepsilon_i$, where $\varepsilon_i = \varepsilon(x_i)$. The three fields of cell v in the difference table are thus

$$C_v = \sum_{i : v \in S_i} s_i \varepsilon_i, \qquad \text{keySum}_v = \sum_{i : v \in S_i} s_i \varepsilon_i\, x_i, \qquad \text{checksum}_v = \sum_{i : v \in S_i} s_i \varepsilon_i\, \text{chk}(x_i). \tag{3.1}$$

In the analysis we treat ε_i as iid uniform signs, independent of the mapping $\mathcal{G}$.

Decoding corresponds to lines 8–17 and follows the peeling of the standard IBLT. When cell v is a pure cell containing a single unrecovered differing key x_i, its count is $C_v = s_i \varepsilon_i \in \{-1, +1\}$, and multiplying keySum_v by C_v cancels the sign $s_i \varepsilon_i$ to read out the candidate key x_i (line 12). After passing the checksum and cell-membership checks (line 13), multiply C_v by $\varepsilon(x_i)$; using $\varepsilon_i^2 = 1$ recovers the original set direction $s_i = C_v\, \varepsilon(x_i)$ (line 14). Then call DELETE or INSERT according to the recovered direction s_i (line 15): both operations multiply $\varepsilon(x_i)$ again into all k cells of x_i, so what is actually subtracted from the difference table is exactly the coding coefficient $s_i \varepsilon_i = C_v$ times the three-field contribution, and hence after each peeling step the three fields still equal the encoded sum of the unrecovered keys. The success criterion of peeling

is also the same as in the standard IBLT, namely that all three fields of every cell become zero (line 17).

Figure 3.1 uses a small one-sided example ($s_1 = \cdots = s_4 = +1$) to compare the standard IBLT and the random-sign IBLT under the same mapping. At the peeling level, the pure cells of both encodings are the degree-1 cells 1, 4, 7, and 8 (shaded in the figure), and their counts all lie in $\{\pm 1\}$; for example, in pure cell 4, although the random sign makes $C_4 = -1$, multiplying back by $\varepsilon_2 = -1$ still recovers the true direction of x_2 as $s_2 = C_4\varepsilon_2 = +1$, and the peeling processes of the two are identical. At the sum-of-squares level, the sum of the single-key diagonal terms of both encodings is $kd = 12$, but the ways different keys superimpose on the shared cells (cells 2, 3, 5, and 6) are completely different: the same-sign accumulation of the standard IBLT raises the counts of all shared cells to 2, the cross terms are all positive, and $\sum_v C_v^2 = 20$; in the random-sign IBLT, the solid lines ($\varepsilon_i = +1$) and dashed lines ($\varepsilon_i = -1$) cancel on the shared cells, and in this example each shared cell consists of exactly one positive and one negative key, so the cross terms are all negative, the counts cancel to 0, and $\sum_v C_v^2 = 4$. The two propositions below generalize the decoding equivalence and sign independence shown in Figure 3.1.

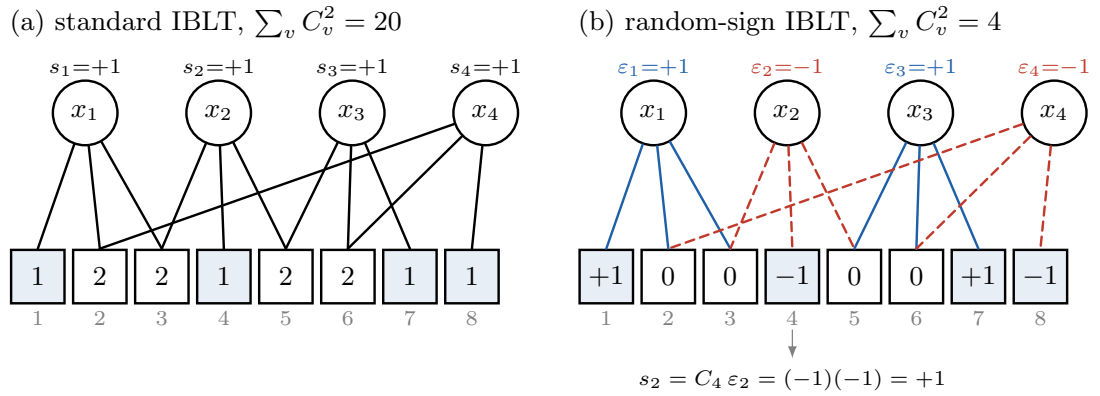


Figure 3.1: The standard IBLT and the random-sign IBLT under the same mapping ($k = 3$, $M = 8$, one-sided difference $d = 4$). The shaded cells have count ± 1; in (b), solid and dashed lines correspond to $\varepsilon_i = +1$ and $\varepsilon_i = -1$, respectively.

The random sign introduces a situation that cannot occur in the standard IBLT under a one-sided difference: a degree-three cell whose three coding coefficients are $(+1, +1, -1)$ also has count $+1$. The candidate key read out from such a cell fails the checksum and cell-membership checks, so peeling skips it until it contains a single key; hence the sign cancellation neither makes peeling delete an extra key nor makes it stop early. Proposition 3.1 formalizes this point.

Proposition 3.1 (Ideal decoding equivalence)**.** *Assume the pure-cell test has no false positives and field arithmetic does not overflow. Then at every intermediate state, the keys removable by the random-sign IBLT are exactly those removable by unsigned degree-one peeling. Hence the random-sign IBLT and the standard IBLT leave the same 2-core under the same cell hash, the failure event F depends only on the unsigned mapping $\mathcal{G}$, and the decoding threshold, the failure probability, and the recovered result are all identical to those of the standard IBLT.*

Proof. Induct on the number of peeling steps: the field invariant guarantees that a degree-one cell has count ± 1 and passes the check, cells of degree other than one are excluded by ideal purity, and the two processes can adopt the same cell-selection order; the full argument is in Appendix C.2. □

In a real implementation, the finite width of the 64-bit checksum gives a nonzero false-acceptance probability: under the ideal hash, each check of a non-pure cell passes falsely with probability at most 2^{-64}. Section 5.6 accounts for this term separately in the risk upper bound of

the two-round protocol, Section 6.2 verifies the implementation of this property in five hundred million real IBLT paired trials, and Table H.4 in the appendix summarizes the success-failure agreement and recovery checks on each configuration; the time cost of decoding is in Section 6.6.

Another point that may raise questions is conditioning. Failure selects mappings with large overlap, and conditioning on failure usually changes the distribution of random variables associated with the mapping; the bias of the standard IBLT under a one-sided difference arises exactly from this. The random sign is not affected by this selection: by Proposition 3.1, failure is determined only by the mapping, while the signs are generated by a sign hash independent of the cell hash, so after fixing the mapping the signs remain iid uniform. Proposition 3.2 gives a precise statement of this property.

Proposition 3.2 (Signs are independent of the mapping)**.** *For any fixed direction vector (s_i), the coding coefficients $(s_i\varepsilon_i)$ are iid uniform signs and are independent of the mapping $\mathcal{G}$. Hence failure F is independent of the coding coefficients: given F, or even given the entire $\mathcal{G}$, the coding coefficients remain iid uniform signs. Any statistic built from (C_v), $\mathcal{G}$, and F has a joint distribution that is the same for all sign compositions (d_+, d_-).*

Proof. Given (s_i), the map $(\varepsilon_i) \mapsto (s_i\varepsilon_i)$ is a bijection on $\{\pm 1\}^d$ that sends the uniform distribution to the uniform distribution; because (ε_i) is independent of $\mathcal{G}$, $(s_i\varepsilon_i)$ is an iid uniform sign sequence independent of $\mathcal{G}$, and the joint distribution of $(s_i\varepsilon_i, \mathcal{G})$ does not depend on (s_i). By Proposition 3.1, F is a function of $\mathcal{G}$, so the distribution of the coding coefficients is unchanged given F or given $\mathcal{G}$. By Eq. (3.1), (C_v) is determined by $(s_i\varepsilon_i)$ and $\mathcal{G}$, so the joint distribution of any statistic built from (C_v), $\mathcal{G}$, and F does not depend on (s_i) and is the same for all (d_+, d_-). □

The numerical manifestation of this property in the protocol is in Section 6.5 and Table H.10 of the appendix: after fixing the configuration and calibration factor, the random-sign encoding gives close conditional miss rates under one-sided and balanced differences. Figure 6.1 visually contrasts the post-failure distributions of the standard and random-sign encodings.

Propositions 3.1 and 3.2 together are the starting point of all the analysis in this paper: the success of decoding is determined only by $\mathcal{G}$, the randomness of the estimator comes from the signs, and the two are independent. Hence the post-failure analysis can always first fix the mapping and then study the quadratic form of iid signs on the overlap graph; the resulting statements hold automatically for all sign compositions. Since the joint distribution of the coding coefficients $s_i\varepsilon_i$ does not depend on the direction vector (s_i), the subsequent analysis need not distinguish $s_i\varepsilon_i$ from ε_i, and below we uniformly write them as ε_i.

3.2 The two-round protocol flow and the lower bound $|b|$

When the random-sign IBLT is used in a two-round protocol, the receiver runs peeling after subtracting cell by cell to obtain the difference table. In the standard IBLT, the count sum of the difference table $\frac{1}{k}\sum_v C_v = b = d_+ - d_-$ directly gives the deterministic lower bound $d \geq |b|$; in the random-sign IBLT, however, $\frac{1}{k}\sum_v C_v = \sum_i \varepsilon_i$ depends only on the random signs and no longer carries the set-size difference b. For this reason the sender attaches the set size $|A|$ (8 bytes) to the first-round table, and the receiver thereby obtains $b = |A| - |B|$ directly.

If the first-round peeling succeeds, the protocol ends immediately. If the first round suffers failure, the receiver computes an upper bound U from the raw counts (C_v) of the difference table and, using the known lower bound $|b|$, takes $U_b = \max(U, |b|)$ (for the one-sided difference, $|b| = d$). It then sets the second-round table size $m(U_b) = \lceil U_b/(0.8\alpha_c) \rceil$ and re-encodes and decodes with a fresh cell hash independent of the first round: as long as $U_b \geq d$, the second-round load is at most $0.8\alpha_c$, below the peeling threshold with a margin left, so the second round succeeds with high probability. Any constant target load below α_c works for the analysis that follows, and 0.8 is the value adopted in this paper. Sections 4 and 5 analyze the distribution of the sum of squared counts after failure and give the rule for computing the upper bound U.

4 Unbiased estimation after failure and the Gaussian upper bound

This section uses the sign and mapping independence established in Section 3 to analyze the statistical properties of $\widehat{d}$ after decoding failure and to construct an upper bound for d. Because the random signs of different keys cancel out on any fixed mapping, we first prove that $\widehat{d}$ is exactly unbiased for every fixed mapping (Proposition 4.1), and hence remains unbiased for any rare failure. To further control the conditional miss rate within the target risk u, we also need to characterize the distribution of the fluctuation of $\widehat{d}$ around d after failure: when $d \gg \sqrt{M}$, the standardized estimation error converges to the standard Gaussian distribution (Theorem 4.2), from which the Gaussian rule follows directly (Definition 4.3).

4.1 Conditional unbiasedness and conditional variance given the mapping

Expanding the sum of squares and the total sum in Eq. (2.1) by key: each differing key occupies k cells and $\varepsilon_i^2 = 1$, so the diagonal parts of the two terms combine to give exactly the true value d, and the estimation error reduces to the cross terms between different keys:

$$\widehat{d} - d = \frac{2}{k(1-k/M)} \sum_{i<j} \varepsilon_i \varepsilon_j \Big(H_{ij} - \frac{k^2}{M} \Big). \tag{4.1}$$

The right-hand side is a quadratic form in the random signs, whose coefficient $H_{ij} - k^2/M$ is the difference between the overlap and its unconditional mean, determined entirely by the mapping $\mathcal{G}$. In the same expansion of the standard IBLT, the sign products are replaced by the fixed $s_i s_j$, so the selection of the overlap by failure translates directly into bias (Proposition 2.2). Under random signs, even after fixing a mapping on which failure occurs, each $\varepsilon_i \varepsilon_j$ still takes ± 1 with equal probability (Proposition 3.2), so the cross terms have mean zero on every mapping. Intuitively, failure can select the overlap but cannot select the signs: it keeps mappings with large overlap and only amplifies the coefficients of the cross terms, while the sign of each term is still decided with equal probability, so the center of the error remains at zero and failure changes only the scale of the error. Proposition 4.1 writes this intuition as the conditional mean and conditional variance given the mapping.

Proposition 4.1 (Conditional unbiasedness and conditional moments given the mapping)**.** *For every fixed mapping $\mathcal{G}$,*

$$\mathbb{E}[\widehat{d} \mid \mathcal{G}] = d, \qquad \mathrm{Var}(\widehat{d} \mid \mathcal{G}) = \frac{4}{k^2(1-k/M)^2} \sum_{i<j} \Big(H_{ij} - \frac{k^2}{M} \Big)^2 \le 2d(d-1). \tag{4.2}$$

Hence for any mapping event $\mathcal{S}$ of positive probability, $\mathbb{E}[\widehat{d} \mid \mathcal{S}] = d$. In particular, $\mathbb{E}[\widehat{d} \mid F] = d$ holds for all finite M, all sign compositions, and any rare failure. Averaging over the random mapping $\mathcal{G}$, $\mathbb{E}\widehat{d} = d$ and $\mathrm{Var}\,\widehat{d} = s(d)^2$.

Proof sketch. Given $\mathcal{G}$, (ε_i) are iid uniform signs, and taking expectations term by term in Eq. (4.1) using $\mathbb{E}[\varepsilon_i \varepsilon_j \mid \mathcal{G}] = 0$ gives the conditional unbiasedness given the mapping; the conditional variance in Eq. (4.2) then follows because the sign products of different unordered pairs (i, j) are pairwise uncorrelated, and its upper bound $2d(d-1)$ is given by the Cauchy–Schwarz inequality. The complete proof is in Appendix C.3. □

After fixing the mapping, $\widehat{d}$ is a Rademacher quadratic form on the Gram matrix of the centered incidence vectors: the conditional mean equals the trace, and the conditional variance comes only from the off-diagonal terms, the same identity as in the Hutchinson trace estimator[21]. Proposition 4.1 realizes this identity on the overlap coefficients and thereby gives conditional unbiasedness for any mapping event.

Figure 6.2 gives a numerical illustration of this unbiasedness: the post-failure bias of the standard encoding changes direction with k, while the mean of the random-sign encoding stays near zero. Further comparisons of conditional moments on more configurations are in Section 6.2 and Appendix H.5.

4.2 The Gaussian limit after failure when $d \gg \sqrt{M}$

Proposition 4.1 fixes the center of the estimator after failure; constructing an upper bound also requires knowing the shape of the distribution of the error around this center. Failure affects $\widehat{d}$ only through the mapping, and by Eq. (4.2) the scale of the error given the mapping is determined by the sum of squared overlaps $\sum_{i<j}(H_{ij} - k^2/M)^2$; the selection of the mapping by failure changes this sum and may therefore change the distribution of the error. We standardize the error by the unconditional standard deviation $s(d)$ as

$$X = \frac{\widehat{d} - d}{s(d)}, \tag{4.3}$$

and study the conditional distribution of X given F.

Fix a mapping first. By Eq. (4.1), the error is a weighted sum of the sign products $\varepsilon_i\varepsilon_j$, whose main weight comes from key pairs sharing a cell, that is, the intersecting edge pairs with $H_{ij} \geq 1$ in $\mathcal{G}$, whose expected number is about $k^2d^2/(2M)$. The sign products of different key pairs are pairwise uncorrelated, and no few terms dominate the sum. The number of intersecting edge pairs grows with d^2/M and tends to infinity when $d \gg \sqrt{M}$; the error is then a sum of many ± 1 perturbations, and its distribution tends to Gaussian. When d is of the same order as $\sqrt{M}$, there are only finitely many intersecting edge pairs, the error consists of a few ± 1 terms, and the distribution remains discrete. The scale parameter that distinguishes these two regimes is written as

$$c = \frac{d}{\sqrt{M}}. \tag{4.4}$$

The expected number of intersecting edge pairs is about $k^2c^2/2$, so c plays the role of the collision scale in the birthday paradox[2, 14]; it and the load d/M are two independent parameters.

What failure changes is the mapping, and this structure of the signs is preserved: given any mapping on which failure occurs, (ε_i) are still iid uniform signs (Proposition 3.2). As long as the mapping on which failure occurs still contains many intersecting edge pairs, the Gaussian shape is preserved. The same holds for rare failures: under a subcritical load, a typical failure is caused by a duplicate edge pair, which raises a single H_{ij} to k and leaves the overall structure of the intersecting edge pairs almost unchanged.

Theorem 4.2 (Gaussian limit after failure)**.** *Fix $k \geq 3$ and let $d/\sqrt{M} \to \infty$, with no requirement on the limit of d/M. Then*

$$\sup_x \left|\Pr(X \leq x \mid F) - \Phi(x)\right| \longrightarrow 0. \tag{4.5}$$

This includes the critical window, rare failures under a fixed subcritical load, and arbitrary overload.

Proof sketch. The proof has two steps. In the first step we fix the mapping and take randomness only over the signs: if the number of intersecting edge pairs is of the same order as M, and each key participates in overlaps of size at most $O(\log M)$, then the higher cumulants of the standardized quadratic form Eq. (4.1) tend to zero, the conditional distribution of X tends to $\mathcal{N}(0,1)$, and the convergence is uniform over such mappings. The second step shows that after conditioning on F, the mapping still satisfies the conditions of the first step with probability tending to 1, in three cases according to the limit of d/M. When d/M has a positive finite limit,

$\Pr(F) \geq \binom{M}{k}^{-1}$, the probability of a mapping violating the conditions is superpolynomially small, and remains negligible after conditioning. When $d = o(M)$, by Proposition 2.1 the distribution of the mapping under failure is asymptotically the same as the planted distribution in TV distance, and the Gaussian limit holds for the latter. When $d > M$, failure occurs with certainty and the conditional distribution is the unconditional one. The complete proof is in Appendix E. □

Figure 6.1 compares X on failure samples and all samples at a configuration with $c \approx 104$ (for the standard encoding, X is computed in the same way from its counts $C_v = \sum_{i:v\in S_i} s_i$): the failure samples of the random-sign encoding almost coincide with all samples, while the failure samples of the standard encoding are shifted as a whole.

4.3 The Gaussian rule and its conditional miss rate

By Theorem 4.2, after failure $\widehat{d}$ is approximately Gaussian with mean d and standard deviation $s(d)$. If d were known, the probability that $\widehat{d}$ falls below $d + s(d)\Phi^{-1}(u)$ would be about u, which sets the margin accordingly. The difficulty is that the scale $s(d)$ (Eq. (2.2)) depends on the unknown d. Substituting $s(\widehat{d})$ directly would make the margin fluctuate together with the observation: when $\widehat{d}$ is low, $s(\widehat{d})$ is also small, giving a smaller margin precisely on the samples that need a larger one, and the risk analysis would have to handle this additional error.

Test inversion is a classical construction of confidence sets[31]; it reasons in the reverse direction and tests candidate values one by one. Assuming the true value is n, the scale $s(n)$ is a known constant, and the probability that $\widehat{d}$ falls below $n + s(n)\Phi^{-1}(u)$ is about u. If the observation falls below this lower bound, then n is too large as the true value, so we exclude n; the largest of the unexcluded candidates is the upper bound U. The risk of an upper bound constructed this way depends on a single event: $U < d$ means that the true value d itself is excluded, and when the true value is d the probability of this event is about u. Definition 4.3 replaces $\Phi^{-1}(u)$ with a more accurate quantile under finite M, and is otherwise the same as the process above.

Definition 4.3 (Gaussian rule)**.** *Given an observation $\widehat{d}$ and a target risk $u \in (0, 1/2)$. For each candidate integer $n \geq 2$, assume the true value is n, in which case the scale $s(n)$ is completely known. Let $\chi^2_{M-1,u}$ denote the lower u quantile of the chi-square distribution with $M-1$ degrees of freedom, and set the acceptance lower bound to*

$$t(n) = n + s(n)\,\frac{\chi^2_{M-1,u} - (M-1)}{\sqrt{2(M-1)}}. \tag{4.6}$$

Accept candidate n if $\widehat{d} \geq t(n)$, and reject it otherwise. Let U be the largest accepted candidate; if no candidate is accepted, return $+\infty$.

Under deep overload, $\widehat{d}/d$ tends to a standardized $\chi^2(M-1)$[38], so the formula replaces $\Phi^{-1}(u)$ with the standardized chi-square quantile $(\chi^2_{M-1,u} - (M-1))/\sqrt{2(M-1)}$, which describes the skewness more accurately under finite M; the two agree as $M \to \infty$. In the range $M \geq 64$, $u \geq 0.01$ used in the experiments, $t(n)$ is strictly increasing in $n \geq 2$ (Appendix E.5), so U can be found by monotone binary search; outside this range one checks candidates one by one by the definition. The whole rule depends only on the observation $\widehat{d}$ and the known parameters (M, k, u).

Corollary 4.4 (Conditional miss rate of the Gaussian rule)**.** *Under the conditions of Theorem 4.2, for any fixed $u \in (0, 1/2)$, $\limsup \Pr(U < d \mid F) \leq u$.*

Proof sketch. Take $n = d$; by Theorem 4.2 and $(t(d) - d)/s(d) \to \Phi^{-1}(u)$, the rejection probability of the true candidate is $u + o(1)$; the conclusion follows from $\{U < d\} \subseteq \{\widehat{d} < t(d)\}$. The complete proof is in Appendix E. □

Corollary 4.4 shows that in the range $d \gg \sqrt{M}$ where Theorem 4.2 holds, the conditional miss rate of the Gaussian rule is asymptotically at most u, and this range includes the critical window and the overload region. Figure 6.4 shows that as c grows the miss/u of the Gaussian rule approaches the target level. When d falls to the same order as $\sqrt{M}$ or below, the Gaussian approximation no longer holds and the actual conditional miss rate exceeds u; Section 5 analyzes the reason and gives the Skellam rule, whose conditional miss rate is asymptotically at most u uniformly over all $d \geq 2$.

5 The Skellam rule for all loads

The guarantee of the Gaussian rule on the conditional miss rate requires $d \gg \sqrt{M}$. When the IBLT table size is configured according to the expected load, d can be far below $\sqrt{M}$; although failure is then rare, once it occurs the two-round protocol still has to determine the second-round table size from the post-failure observation. In this range the conditional miss rate of the Gaussian rule can reach several times the target u (Table H.7 in the appendix).

The reason is the shape of the post-failure error under low load. Here failure has essentially a single source: two differing keys select exactly the same k cells. These cells contain only these two keys, there is no pure cell in the table, and peeling cannot start. Under random signs these two keys are equally likely to have the same or opposite signs. If they were each placed in the table alone, each key would leave a count ± 1 in its own k cells, with a total sum of squares $2k$, which divided by k gives exactly the contribution 2 that the two keys should have. When the two have opposite signs, the counts of the k shared cells all cancel to 0, this $2k$ sum of squares disappears entirely, and the estimate is 2 below the true value; when the two have the same sign, each shared cell has count ± 2, whose square is 4, the sum of squares becomes $4k$, and the estimate is 2 above the true value. This jump is equally likely upward or downward, and its size is independent of M. The remaining keys occasionally share a cell too, and when they do they almost always share exactly one, each time shifting the estimate by $2/k$ with direction again equally likely; the number of such key pairs is approximately Poisson[2, 4], and is finite when c is bounded. The post-failure error thus consists of one large jump of size 2 and several small jumps of size $2/k$.

The Gaussian cannot describe these two jumps. When d is small relative to $\sqrt{M}$, the standard deviation $s(d)$ of the Gaussian rule tends to zero, while the downward jump of size 2 stays unchanged, lying an arbitrary number of standard deviations outside the Gaussian tail, so the probability of rejecting the true value tends to one half. When c is of order one, there are only finitely many small jumps, the error is still a lattice distribution, and it shows a visible deviation from the Gaussian in the lower tail where the target risk lies (Appendix H.6). The null distribution in this section keeps both jumps, tends naturally to the Gaussian used by the Gaussian rule as c grows (Figure 5.1), and the Skellam rule built from it controls the conditional miss rate for all $d \geq 2$. The complete proof and numerical implementation are in Appendices F and G.

5.1 The lattice-preserving test statistic $\widetilde{d}$

To make the jumps above land exactly on lattice points, the test statistic itself must remain discrete. The correction term $(\sum_v C_v)^2/M$ in $\widehat{d}$ moves its values off the lattice; this section therefore uses the normalized sum of squares without subtracting the mean term,

$$\widetilde{d} = \frac{1}{k} \sum_{v=1}^{M} C_v^2. \tag{5.1}$$

Expanding the sum of squares, the diagonal terms give d, and the error contains only the overlap itself:

$$\widetilde{d} - d = \frac{2}{k} \sum_{i<j} H_{ij}\, \varepsilon_i \varepsilon_j. \tag{5.2}$$

Every term on the right-hand side is an integer multiple of $2/k$, so $\widetilde{d} - d$ takes only integer multiples of $2/k$. Both jumps can be read off from this expansion: for a duplicate edge pair $H_{ij} = k$, contributing exactly $2\varepsilon_i\varepsilon_j$, that is, ± 2 with equal probability; for an edge pair sharing a single cell $H_{ij} = 1$, contributing $\pm 2/k$.

After removing the correction term, $\widetilde{d}$ is still exactly unbiased for every mapping.

Proposition 5.1 (Conditional unbiasedness of $\widetilde{d}$)**.** *For every fixed mapping $\mathcal{G}$, $\mathbb{E}[\widetilde{d} \mid \mathcal{G}] = d$. Hence for any mapping event $\mathcal{S}$ of positive probability, $\mathbb{E}[\widetilde{d} \mid \mathcal{S}] = d$; in particular, $\mathbb{E}[\widetilde{d} \mid F] = d$ holds for all finite M, all sign compositions, and any rare failure.*

Proof. Given $\mathcal{G}$, H_{ij} is a constant; by Proposition 3.2 the signs are still iid uniform, and $\mathbb{E}[\varepsilon_i\varepsilon_j \mid \mathcal{G}] = 0$ for $i \neq j$. Taking expectations term by term in Eq. (5.2) gives the result. □

This property also comes from the random signs. In the standard IBLT the sign products of the cross terms are the fixed $s_i s_j$, and the same expansion gives the unconditional mean of the sum of squares as $d + \frac{k}{M}(b^2 - d)$, which depends on the unknown b; the correction term in $\widehat{d}$ exists exactly to remove this term. The random signs make the cross terms have mean zero on every mapping, so the sum of squares itself is already unbiased and unbiasedness no longer relies on the correction term.

$\widetilde{d}$ and $\widehat{d}$ differ only by a correction term with weight k/M, namely the square of the count sum; it depends only on the random signs and is the classical AMS second-moment estimator[1] (Appendix C.5). After standardization by $s(d)$ the difference between the two tends to zero, so Theorem 4.2 holds for $\widetilde{d}$ as well (Appendix E).

5.2 Constructing the null distribution from the two jumps

This section gives, for each candidate n, a reference distribution of $\widetilde{d}$ after failure when the true value is n, namely the null distribution; Section 5.4 uses it to test candidates one by one. This section writes down the number of occurrences of the two jumps.

The large jump occurs exactly once, its direction decided by the sign product of the duplicate edge pair, with contribution $2\varepsilon_1\varepsilon_2$.

The small jumps come from the $n-2$ keys other than the duplicate edge pair. They form $\binom{n-2}{2}$ pairs, each of which shares a cell with probability about k^2/M and produces a small jump. The small jumps are equally likely upward or downward, so the expected numbers of upward and downward jumps are each

$$\lambda_n = \frac{1}{2}\binom{n-2}{2}\frac{k^2}{M} = \frac{k^2(n-2)(n-3)}{4M}. \tag{5.3}$$

The key pairs sharing a cell are sparse, and the upward and downward counts are approximately two independent $\mathrm{Pois}(\lambda_n)$ counts (Poisson splitting[25]). The upward count minus the downward count is the net number of steps of the small jumps, which follows a Skellam distribution[35], denoted $Z_n \sim \mathrm{Skellam}(\lambda_n, \lambda_n)$; each step has size $2/k$.

Adding the two jumps to the candidate n gives the null distribution.

Definition 5.2 (null distribution)**.** *For $\lambda \geq 0$, let*

$$J_\lambda = 2\varepsilon_1\varepsilon_2 + \frac{2}{k}Z, \qquad Z \sim \mathrm{Skellam}(\lambda, \lambda), \tag{5.4}$$

where $\varepsilon_1\varepsilon_2$ is ± 1 with equal probability, independent of Z. Call $2\varepsilon_1\varepsilon_2$ the ± 2 term and $\frac{2}{k}Z$ the Skellam term. For a candidate integer $n \geq 2$, take λ_n as in Eq. (5.3); the null distribution of candidate n is the distribution of

$$n + J_{\lambda_n} = n + 2\varepsilon_1\varepsilon_2 + \frac{2}{k}Z_n \tag{5.5}$$

whose CDF is denoted G_n.

Section 5.3 proves that the distribution of $\widetilde{d}$ after failure does tend to G_d, uniformly over all $d \geq 2$ (Theorems 5.3 and 5.4).

When $n = 2, 3$ there are fewer than two remaining keys, $\lambda_n = 0$, and the null distribution is $n \pm 2$ each with probability one half. In general the mean of the null distribution is n; by $\operatorname{Var} Z_n = 2\lambda_n$, its variance is $4 + 2(n-2)(n-3)/M$.

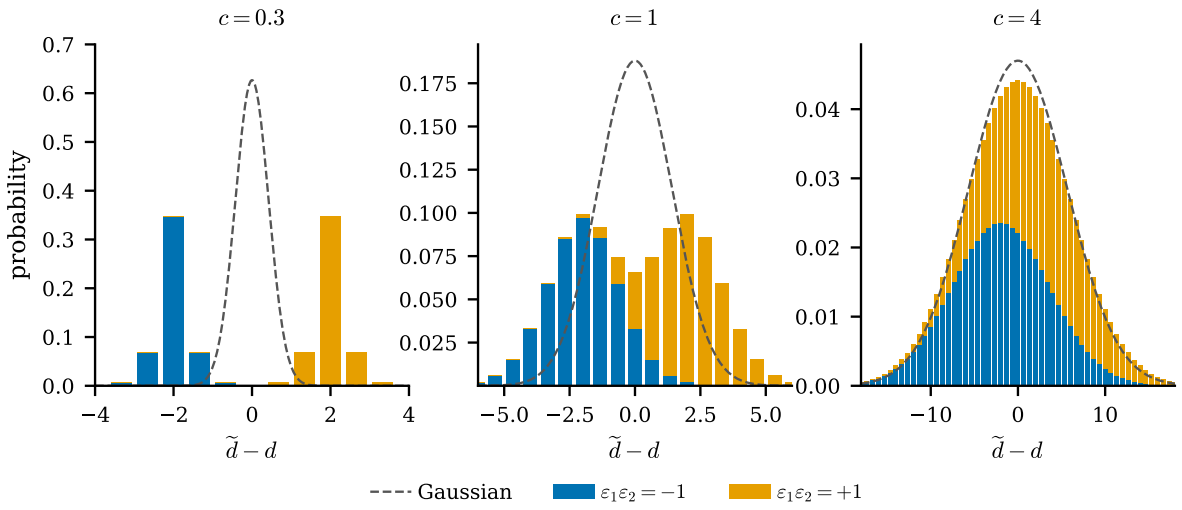


Figure 5.1: The shape of the null distribution at three values of $c = d/\sqrt{M}$ ($k = 3$, with λ_d taking its limit value $k^2c^2/4$). Blue and orange are the two cases of opposite and equal signs of the duplicate edge pair, and the dashed line is the Gaussian used by the Gaussian rule.

First consider, from the shape, why this family of distributions can cover all loads. Figure 5.1 plots the null distribution as c varies. When c is small the Skellam term is almost zero, the distribution takes only the two points ± 2, and the Gaussian used by the Gaussian rule concentrates in the middle, with both points outside its tails. When $c \approx 1$ the two cases of opposite and equal signs each expand into a lattice distribution with spacing $2/k$ and overlap each other, and the shape still deviates clearly from the Gaussian. As c grows further, the width of the Skellam term far exceeds 2, the ± 2 term occupies only a small part, and the null distribution tends to the same Gaussian used by the Gaussian rule. The same family of distributions thus connects the small-c region and the region where the Gaussian rule applies. The rigorous approximation guarantee is in Section 5.3.

5.3 The null distribution approximates the post-failure distribution at all loads

The null distribution of Definition 5.2 comes from a counting intuition; before using it as the reference to test candidates, we must prove that it indeed describes the distribution of $\widetilde{d}$ after failure. Moreover, this description must hold for all d: the receiver does not know which load

region d falls in and cannot switch rules by load, so the same family G_d must apply for both small and large d. This section explains this in two cases and then combines them.

When d is of the same order as $\sqrt{M}$ or smaller, the overlap is sparse. G_d regards the post-failure mapping as a duplicate edge pair plus several scattered single-cell overlaps among the remaining keys. The real mapping can be more complex: the duplicate edge pair shares a cell with a remaining key, two keys share more than one cell, or one key shares cells with two other keys at the same time. These three cases are all much rarer than a single-cell overlap, and their probabilities tend to zero as M grows. Removing them, the real mapping is exactly what G_d envisions, so the real distribution is close to G_d at every lattice point (Theorem 5.3), corresponding to the first two shapes in Figure 5.1.

When $d \gg \sqrt{M}$, the intersecting key pairs are many and interwoven, the scattered structure above no longer holds, yet G_d still applies because the two sides tend to the same Gaussian. The real $\widetilde{d}$ is a sum of many small terms with equal probabilities of each sign, tending to Gaussian by Theorem 4.2; G_d also tends to Gaussian, as shown by the third shape in Figure 5.1, with width matching $s(d)$. The two sides differ in detail and coincide at the Gaussian limit. Theorem 5.4 combines the two cases into an approximation uniform over all $d \geq 2$.

Theorem 5.3 (Discrete limit distribution when c is bounded)**.** *Fix $k \geq 3$ and a finite constant C.*

(i) On $2 \leq d \leq C\sqrt{M}$, the total variation distance between the distribution of $\widetilde{d}$ under F and the null distribution G_d tends to zero uniformly, and is at most

$$O_k\Big(\frac{d}{M} + \frac{d^3}{M^2}\Big). \tag{5.6}$$

(ii) If $d/\sqrt{M} \to c \in (0, \infty)$, then the distribution of $\widetilde{d} - d$ under F converges in total variation to the distribution of $J_{k^2c^2/4}$, whose variance is $4 + 2c^2$.

(iii) The limits in (ii) satisfy at the two ends

$$J_{k^2c^2/4} \Rightarrow 2\varepsilon_1\varepsilon_2 \quad (c \to 0), \qquad \frac{J_{k^2c^2/4}}{\sqrt{2}\,c} \Rightarrow \mathcal{N}(0,1) \quad (c \to \infty).$$

Figure 6.3 shows the lower tail of this discrete limit distribution using planted samples with c from 0.25 to 8, contrasted with the Gaussian approximation; Section 6.3 also reports the distance between the empirical and limit distributions at fixed c as M grows.

Proof sketch. (i) proceeds in two steps. The first step replaces the real failure with a duplicate edge pair: by Proposition 2.1, the mapping under failure can be replaced by the planted distribution, with error $O_k(d/M^2 + M^{1-k})$ in the range of the theorem (Lemma D.2). The second step excludes the three cases above on the planted mapping, with probabilities $O_k(d/M)$, $O_k(d^2/M^2)$, and $O_k(d^3/M^2)$. After the exclusion, the remaining structure is a duplicate edge pair plus several mutually disjoint single-cell overlaps: the duplicate edge pair gives the ± 2 term; the Chen–Stein local dependence bound[2, Theorem 1] shows that the number of overlaps is approximately $\mathrm{Pois}(2\lambda_d)$, and the independent signs then split it into two independent $\mathrm{Pois}(\lambda_d)$ counts, whose difference gives the Skellam term, so $\widetilde{d} - d$ is close to J_{λ_d}. The Poisson approximation has error $O_k((d^2 + d^3)/M^2)$. Since $2 \leq d \leq M$ and $k \geq 3$, the quantities d/M^2, d^2/M^2, and M^{1-k} do not exceed d/M, and $(d^2 + d^3)/M^2$ does not exceed $2d^3/M^2$; combining all terms gives Eq. (5.6). (ii) By (i), it suffices to show that J_{λ_d} converges in total variation to $J_{k^2c^2/4}$; when $d/\sqrt{M} \to c$ we have $\lambda_d \to k^2c^2/4$, which follows from the continuity of the Poisson parameter. Since the two terms are independent, $\mathrm{Var}\, J_\lambda = 4 + 8\lambda/k^2$, and substituting $\lambda = k^2c^2/4$ gives $4 + 2c^2$. (iii) As $c \to 0$, $\lambda \to 0$, the Skellam term tends to zero in probability, and only the ± 2 term of J_λ remains; as $c \to \infty$, $\mathrm{Var}\, J_{k^2c^2/4} \sim 2c^2$, the Skellam term satisfies the CLT, and the ± 2 term divided by $\sqrt{2}\,c$ tends to zero. The complete argument is in Appendix F. □

Theorem 5.4 (Uniform CDF approximation)**.** *Fix $k \geq 3$. The Kolmogorov distance between the distribution of $\widetilde{d}$ under F and the null distribution G_d satisfies*

$$\sup_{d\geq 2} \sup_{x\in\mathbb{R}} \left|\Pr(\widetilde{d} \leq x \mid F) - G_d(x)\right| \longrightarrow 0 \quad (M \to \infty). \tag{5.7}$$

Proof sketch. When $d \gg \sqrt{M}$ there is no need to extend the matching structure to a dense intersection graph. By Theorem 4.2, the conditional distribution of $\widetilde{d}$ standardized by $s(d)$ tends to the standard Gaussian; the null distribution is that of $d + J_{\lambda_d}$, and when $\lambda_d \to \infty$ the Skellam term of J_{λ_d} satisfies the CLT, the ratio of $\operatorname{Var} J_{\lambda_d} = 4+2(d-2)(d-3)/M$ to $s(d)^2$ tends to one, and the ± 2 term divided by $s(d)$ tends to zero. The two tend to the same continuous Gaussian CDF, so the Kolmogorov distance tends to zero. This step applies to $d = o(M)$, a positive load ratio, and unbounded overload. Uniformity is obtained by a subsequence argument: take a subsequence along which $d/\sqrt{M}$ converges; finite limits are handled by Theorem 5.3, and infinite limits by the Gaussian argument above (Appendix F). □

Thus the same family of null distributions G_d describes the distribution of $\widetilde{d}$ after failure for all $d \geq 2$: for small d it is close pointwise on the lattice, and for large d it tends to the same Gaussian as the real distribution, so using it does not require knowing which load region we are in. Section 5.4 builds the Skellam rule from it.

5.4 The Skellam rule controls the conditional miss rate at all loads

The Skellam rule follows the test inversion of the Gaussian rule (Section 4.3), replacing only the Gaussian lower tail with the lattice lower tail of G_n: when the observation falls in the lower u tail of G_n, n is too large as the true value, and candidate n is rejected; U is the largest candidate not rejected.

Definition 5.5 (Skellam rule)**.** *Given an observation $\widetilde{d}$ and a target risk $u \in (0, 1/2)$. For each candidate integer $n \geq 2$, accept candidate n if $G_n(\widetilde{d}) > u$, and reject it otherwise. Take the upper bound to be the supremum of all accepted candidates,*

$$U = \sup\{n \geq 2 :\ G_n(\widetilde{d}) > u\}, \tag{5.8}$$

with $U = +\infty$ when the accepted candidates are unbounded.

The null distribution of candidate 2 takes only 0 and 4, and $\widetilde{d} \geq 0$, so candidate 2 is always accepted and $U \geq 2$. On realistic table sizes U is also finite (for $u = 0.01$ it suffices that $M > 18$), and it can be found exactly by a finite search, with a cost per computation roughly proportional to λ_d (Appendix G).

The behavior at the two ends of the load explains why the Skellam rule works for all loads. When c is small the Skellam term is almost zero, G_n puts half its mass on $n-2$ and half on $n+2$, candidate n is accepted if and only if the smaller point $n-2$ does not exceed the observation, so $U \approx \widetilde{d}+2$: the rule adds back to the observation a possibly lost large jump. When the duplicate edge pair has opposite signs the observation is $d-2$ and U equals d exactly; when they have the same sign the observation is $d+2$ and U is about $d+4$; neither case misses. In this region the scale $s(n)$ of the Gaussian rule tends to zero and U is close to the observation itself, so when the duplicate edge pair has opposite signs it underestimates d, and the conditional miss rate tends to one half. When c is large the Skellam term dominates, G_n tends to Gaussian, and the Skellam rule returns to the Gaussian rule.

Theorem 5.6 is the main result of this section: when the table size is large enough, no matter which load region d falls in, the probability that the Skellam rule underestimates d after failure does not exceed the target risk u. This guarantee holds for any sign composition, and the rule uses only $\sum_v C_v^2$ of the same table and the known parameters (M, k, u). The Gaussian rule has this guarantee only when $d \gg \sqrt{M}$ (Corollary 4.4). Section 5.6 substitutes it into the risk decomposition of the two-round protocol.

Theorem 5.6 (Uniform asymptotic upper bound on the conditional miss rate)**.** *Fix $k \geq 3$ and $u \in (0, 1/2)$. For any sign composition, the Skellam rule satisfies*

$$\limsup_{M \to \infty} \sup_{d \geq 2} \Pr(U < d \mid F) \leq u. \tag{5.9}$$

Proof sketch. If the observation followed G_n exactly, the event of rejecting n is that the observation falls in a lower tail of total probability at most u, whose probability is at most u by the definition of the CDF, and the same holds for discrete distributions. The rejection region is a lower set of the observation, so the difference between the rejection probabilities under the real distribution and under G_n is at most their Kolmogorov distance. A miss means the true value is rejected, that is, $\{U < d\} \subseteq \{G_d(\widetilde{d}) \leq u\}$; taking $n = d$ and applying Theorem 5.4 gives Eq. (5.9). By Proposition 3.2, the joint distribution of the coding coefficients does not depend on the sign composition, so the conclusion holds for any sign composition. The complete proof is in Appendix F. □

Theorem 5.6 is a statement as the table size tends to infinity; on a finite table there are two differences. First, G_n is a lattice distribution, the rejection region can contain only whole lumps of probability, and the actual conditional miss rate can be strictly below u. Second, the null distribution omits the structure in which the duplicate edge pair shares cells with the remaining keys, whose occurrence rate is about $(d-2)k^2/M$, which can push the conditional miss rate slightly above u when the table is small. Section 6.4 checks the behavior on finite tables by experiments (Figures 6.4 and 6.5): at $M = 64$ the largest miss/u is 1.08, exceeding the sampling error; all configurations with $M \geq 256$ meet the target within sampling error. The effect of the omitted structure decreases as $1/M$, and the extra capacity also decreases as M grows. When the first round can only use a smaller table, the margin $1/(0.8\alpha_c)$ of the second-round table size already absorbs a slight underestimate (Section 5.6); when more margin is needed, one can lower the target load 0.8 to trade a larger second-round table for a lower two-round failure rate.

5.5 Combining the two rules by the first-round observation

The two rules are weak in different ranges. The Gaussian rule underestimates d only when c is bounded, and on this range the Skellam rule is cheap to compute; the cost of the Skellam rule grows roughly with λ_d, that is, with c^2, while the Gaussian rule already meets the target when c is large. A real system can therefore combine the two rules by the first-round observation: use the Gaussian rule when $\widetilde{d}$ is not below a switch point, and the Skellam rule otherwise. $\widetilde{d}$ is computed directly from the first-round counts, so the switching itself adds no cost.

As long as the switch point grows faster than $\sqrt{M_1}$ in M_1, the combined rule keeps the uniform guarantee of Theorem 5.6 (Appendix F.9). When $d/\sqrt{M_1}$ is bounded, $\widetilde{d}$ falls below the switch point with probability tending to one, and the combined rule is the Skellam rule; when $d/\sqrt{M_1} \to \infty$, both rules compare an asymptotically Gaussian statistic with the same lower u quantile and give asymptotically the same conclusion. For example, taking the switch point $1.5\alpha_c M_1$, that is, a first-round load of 1.5, sizing above load 1.5 needs only one monotone binary search, and the protocol time in the overload region is the same as the Gaussian rule (Section 6.6).

5.6 From the conditional miss rate to the failure probability of the two-round protocol

Finally, connect the post-failure conditional miss rate to the overall success rate of two-round reconciliation. The protocol runs according to the flow of Section 3.2: denote the first-round failure by F_1 and the second-round failure by F_2; denote by E the checksum misjudgment event, that is, a non-pure cell passes the check, a key passing the check does not belong to the set difference, or the core is nonempty while all fields are zero at termination.

Theorem 5.7 (Two-round risk decomposition)**.** *Under the ideal hash model, denote by* $\Pr_m(F)$ *the probability of a peeling failure of* d *differing keys on a fresh table of* m *cells. If the reserve size is at least* $m(d)$*, then*

$$\Pr\left((F_1 \cap F_2) \cup E\right) \le \Pr(F_1,\ U_b < d) + \Pr(F_1) \max_{m \ge m(d)} \Pr_m(F) + \Pr(E). \tag{5.10}$$

Proof. By the union bound it suffices to control $F_1 \cap F_2$ and E separately. U_b is a function of the first-round counts and the set sizes, and the second-round mapping is independent of it; under ideal purity, F_2 depends only on the second-round mapping and d. Hence $\Pr(F_1 \cap F_2) = \mathbb{E}[\mathbf{1}_{F_1} \Pr_{m(U_b)}(F)]$. On $\{U_b < d\}$ we upper-bound the conditional failure probability by one, and on $\{U_b \ge d\}$ we have $m(U_b) \ge m(d)$; taking the supremum gives Eq. (5.10). □

The three terms correspond to three ways of going wrong. The first term is a sizing error: after the first-round failure the upper bound underestimates d, and the second-round table is allocated according to U_b, with insufficient capacity. This is the joint miss rate $\Pr(F_1)\Pr(U_b < d \mid F_1)$. Under the Skellam rule, by Theorem 5.6, it is asymptotically at most $u\Pr(F_1)$ uniformly over all $d \ge 2$; under a one-sided difference $U_b \ge |b| = d$ and this term is zero. The conditional miss rate of the Gaussian rule can be much larger than u when c is bounded, but then failure itself is rare: by Proposition 2.1, when $d = o(M_1)$, $\Pr(F_1)$ is asymptotically equal to the expected number of duplicate edge pairs $\binom{d}{2}/\binom{M_1}{k}$, which is about 2.8×10^{-6} for $k = 3$, $M_1 = 1024$, $d = \sqrt{M_1} = 32$, so the joint miss rate is still small.

The second term is the randomness of peeling itself: the second-round table has sufficient capacity, yet the random hypergraph can still leave a nonempty 2-core with small probability. The second-round load is at most $0.8\alpha_c$, and by Lemma D.3 it tends to zero polynomially in d. The third term is a verification error: the checksum misjudges a non-pure cell as pure, or misjudges termination on a nonempty core, producing a wrong recovery result. Each check passes falsely with probability at most 2^{-64}, and the number of checks does not exceed the number of cells plus k times the number of deleted keys; the keySum and count of a nonempty core can cancel to zero simultaneously, and the probability that the checksums are then also all zero is likewise at most 2^{-64}, occurring at most once per round at termination. Hence $\Pr(E)$ is of order $2(M_1 + m + kd) \cdot 2^{-64}$, negligible at realistic scales. When the checksum is shortened to 32 bits this term is of order $2(M_1+m+kd)\cdot 2^{-32}$, which is at most 10^{-4} for $M_1 = 16384$ and load at most 3; $\widetilde{d}$ comes only from the count field, so the first term is unaffected by the checksum width, and at deployment one allocates the target failure probability between the sizing and checksum terms.

Eq. (5.10) adds the three terms directly, with the first term computed as if every miss leads to a two-round failure, and is therefore conservative. The second-round table leaves a margin, so under a slight underestimate the second round still succeeds with high probability, and the actual impact of such misses is on the same level as the second term. The protocol experiments of Section 6.5 also show that two-round failures come almost entirely from the second term, and the two-round failure rates of the two rules are comparable to that of the oracle upper bound $(U = d)$.

6 Experimental evaluation

The experiments answer three questions. First, whether the random-sign encoding removes the dependence of the post-failure estimate on the sign composition without changing decoding (Section 6.2). Second, what shape the lower tail of the post-failure statistic has (Section 6.3), and how this shape determines the conditional miss rate and the extra capacity of the two rules at finite size (Section 6.4). Third, what the reliability and communication cost are when the upper bound is placed into the two-round protocol of a real IBLT (Section 6.5), and what the

computational cost is (Section 6.6). Finally, Section 6.7 summarizes the guarantees and costs of the various rule choices and gives deployment recommendations.

The experiments are divided into two kinds, statistical experiments (Exp-STAT) and protocol experiments (Exp-PROTO). The statistical experiments study only the distribution of the post-failure statistic, with configurations in Section 6.1; the protocol experiments run the two-round protocol on a complete IBLT, with configurations in Section 6.5. A few figures additionally take a single configuration to display the shape of the distribution, with its parameters given in the figure caption.

6.1 Configurations and sampling of the statistical experiments

The statistical experiments do not need a complete IBLT: the sampler independently and uniformly generates d random hyperedges and runs peeling, and on the same mapping simultaneously computes $X = (\widehat{d} - d)/s(d)$ of the standard encoding under one-sided and balanced differences, together with the exact integers $\sum_v C_v$ and $\sum_v C_v^2$ of the random-sign encoding, and then computes the upper bound of the two rules from the same observation.

Each configuration is determined by (k, M, d). Two axes describe the position of a configuration: $c = d/\sqrt{M}$ (Eq. (4.4)) and the first-round load $d/(\alpha_c M)$, controlling different effects. The parameter c determines the statistical scale of the estimation error: when c is small the post-failure $\widehat{d} - d$ is dominated by the ± 2 offset of a few duplicate edge pairs, and when c is large it tends to a continuous distribution; the load determines the position of the peeling phase transition, subcritical below 0.95, in the critical window from 0.95 to 1.05, and overloaded above 1.05. For fixed k, M the two axes satisfy $d/(\alpha_c M) = c/(\alpha_c \sqrt{M})$, so only changing M can move the load while keeping c fixed, thereby separating the two effects.

The statistical experiments take $k = 3, 4, 5$ and $M \in \{256, 1024, 4096, 16384\}$. For each pair (k, M), d takes the union of three kinds of values: the 8 values making $c \in \{0.5, 1, 2, 4, 8, 16, 32, 64\}$, the 5 values making the load $\in \{0.95, 1, 1.05, 1.5, 3\}$, and the small differences $d = 2, 3, 6$. After removing the points with load exceeding 3 there are 189 configurations, each collecting about 20,000 failures. $M = 64$ gives 42 configurations by the same rule, used separately to test the boundary of applicability of the rules. The positions of these configurations on the two axes are shown in Figure H.1 of the appendix.

Planted sampling. Failure samples have two sources. When the load is near or above the critical value, failure is common, and it suffices to randomly generate a mapping and keep the failure samples; this direct sampling gives the true failure-conditional distribution. When the load is low the failure probability is small, and direct sampling is too expensive. By Proposition 2.1, when $d = o(M)$ failure is almost always caused by a pair of keys with completely identical mappings, so such samples can be constructed directly: randomly choose a pair of differing keys to fall on the same group of k cells, and draw the mappings and signs of the remaining keys independently. In the statistical experiments the 84 configurations with high failure rate use direct sampling, and the remaining 105 use this planted sampling, marked separately in the figures (Appendix H.2).

The planted-sampling control (Exp-PLANT) compares the two kinds of samples on 12 planted configurations where direct sampling is still affordable: among direct failures the proportion caused by a duplicate edge pair is 0.988 to 1.000, the KS distance of X is 0.0048 to 0.0079, comparable to the sampling error at this sample size, and the miss/u of the Skellam rule also agrees. The planted approximation deviates only near the critical load when M is small, and the planted configurations of the statistical experiments are not in this region (Appendix H.3).

6.2 The random-sign encoding removes the dependence on the sign composition

This subsection verifies two properties of the random-sign encoding: the post-failure estimate no longer depends on the sign composition, and the result of decoding is identical to that of the standard encoding.

Conditional bias. Under the random-sign encoding, the post-failure $\widehat{d}$ is unbiased for every sign composition; under the standard encoding, the selection of the mapping by failure introduces into $\widehat{d}$ a bias that depends on the sign composition. Figure 6.1 compares three conditions on the same batch of 4 million mappings, with the success or failure of peeling identical in each case: the failure samples of the standard encoding under a one-sided difference are shifted right as a whole by about $0.6s(d)$ and are almost unbiased under a balanced difference; the failure samples of the random-sign encoding almost coincide with all samples (numerical values in Appendix H.5).

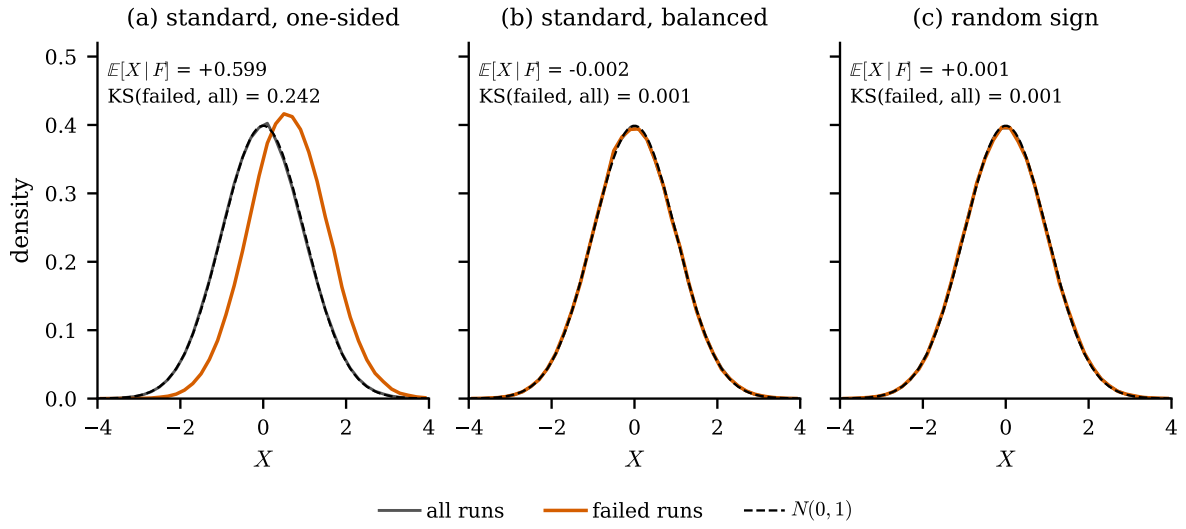


Figure 6.1: $X = (\widehat{d} - d)/s(d)$ under three conditions on the same batch of mappings ($k = 3$, $M = 16384$, $d = 13314$, failure rate 0.125).

The bias of the standard encoding depends on both the sign composition and k, and the receiver cannot correct it in advance. Figure 6.2(a) compares the two encodings at five sign compositions: the solid lines of the standard encoding are near zero at the balanced point and deviate more as the difference tilts to one side, with the direction of the deviation determined by k; for a one-sided difference it is $+0.638$ at $k = 3$ and -0.387 at $k = 6$ (mechanism in Appendix B). The receiver knows k and therefore the direction of the bias, but it does not know (d_+, d_-) in advance and cannot determine the size of the correction. The dashed lines of the random-sign encoding stay near zero, and after magnifying the vertical axis by about 40 times (Figure 6.2(b)) there is still no discernible difference among the five sign compositions. This agrees with Proposition 3.2: the coding coefficients $s_i\varepsilon_i$ are iid random signs, and their distribution is independent of (d_+, d_-). The 84 direct-sampling configurations of the statistical experiments give the same conclusion (Appendix H.5).

Decoding equivalence. The random-sign encoding does not change the set of keys recovered by decoding (Proposition 3.1), so it inherits the threshold and the failure probability of the standard IBLT. The decoding-equivalence test (Exp-DEC) verifies this with paired trials on a

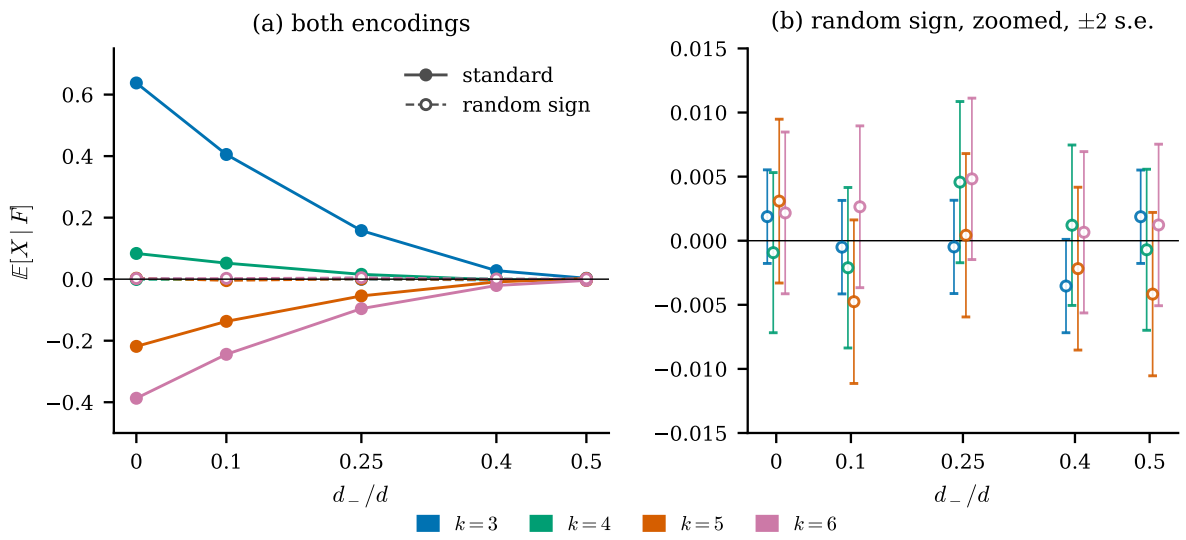


Figure 6.2: The mean of the post-failure X as the sign composition varies ($M = 16384$, failure probability about 0.1, with the two encodings sharing the same batch of mappings on the same configuration). Color indicates k. (a) Solid lines with filled markers are the standard encoding, dashed lines with open markers are the random-sign encoding; (b) the random-sign encoding data in (a) magnified about 40 times on the vertical axis, with error bars of two standard errors.

complete IBLT: the two encodings share the same batch of keys and cell hash, and each trial checks whether the success or failure of the first round, each accepted key, and the set recovered at termination agree with an independent degree-one peeling reference. In more than 500 million paired trials the number of disagreements is zero (Appendix H.4). Table 6.1 gives the failure probability near the threshold: after normalizing the load by the respective α_c, the five values of k almost coincide, and the transition band narrows as M grows, consistent with the critical window of width of order $M^{-1/2}$ in Section 2.

Table 6.1: Failure probability of first-round decoding. Columns are the load $d/(\alpha_c M)$; the two encodings agree in every trial, so the values apply to both.

		$M = 1024$			$M = 16384$			$M = 65536$		
k	α_c	0.98	1	1.02	0.98	1	1.02	0.98	1	1.02
3	0.818	0.298	0.715	0.958	0.000	0.648	1.000	0.000	0.619	1.000
4	0.772	0.266	0.743	0.977	0.000	0.654	1.000	0.000	0.620	1.000
5	0.702	0.268	0.756	0.975	0.000	0.661	1.000	0.000	0.629	1.000
6	0.637	0.284	0.740	0.971	0.000	0.665	1.000	0.000	0.627	1.000
10	0.461	0.383	0.759	0.966	0.000	0.663	1.000	0.000	0.630	1.000

The additional requirement of the random-sign encoding on the decoder is the same as when the standard IBLT handles a balanced difference. The random signs make a cell of degree greater than one possibly show a count of ± 1 (Section 3); all such candidates are rejected at the checksum step, with not a single false acceptance. The frequency with which the random-sign encoding encounters such candidates is independent of the sign composition and equals the frequency of the standard encoding under a balanced difference (Figure H.3 of the appendix). The cell fields and byte widths of the two encodings are the same; the random-sign encoding additionally sends the set size $|A|$ (8 bytes) with the table, used to recover the lower bound $|b|$.

6.3 The discrete limit distribution accurately describes the post-failure lower tail at finite size

The limit distribution of Section 5 almost coincides with the empirical post-failure lower tail without any fitting. Figure 6.3 takes six configurations with c from 0.25 to 8 at $M = 16384$ and compares the empirical CDF of the post-failure $(\widetilde{d} - d)/s(d)$, the limit distribution of Theorem 5.3, and $N(0, 1)$: on all six configurations the empirical CDF almost coincides with the limit distribution, while $N(0, 1)$ deviates clearly when c is small. The empirical distribution is computed exactly from integer sums of squares, and the limit distribution uses the G_d of the main text, with no fitted parameters. The finite-size error decreases as M grows; at $c = 1, 2, 4$ the KS distance drops from within 0.034 at $M = 256$ to within 0.012 at $M = 16384$ (Table H.6 of the appendix).

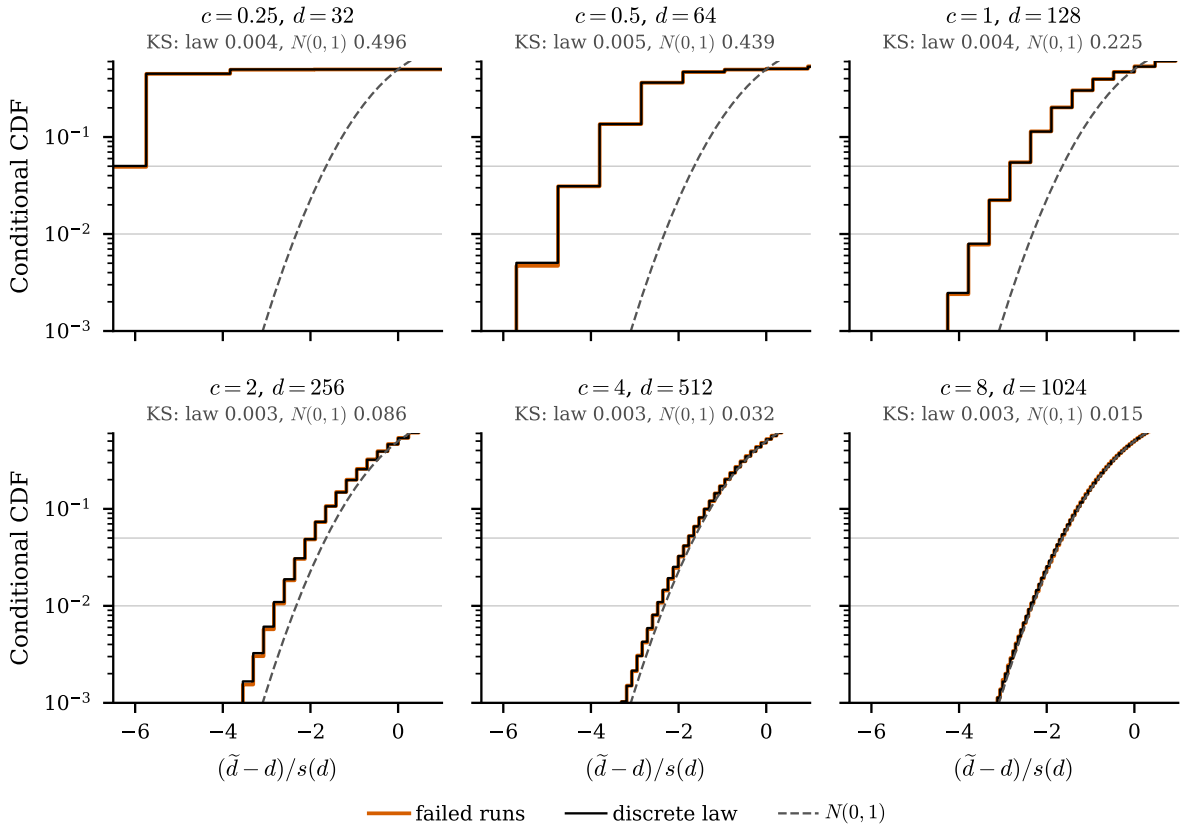


Figure 6.3: The post-failure lower tail of $(\widetilde{d} - d)/s(d)$ at different c ($k = 3$, $M = 16384$; 400,000 planted samples per configuration for $c = 1, 2, 4$, and 100,000 for the rest). The limit distribution is computed directly from Theorem 5.3, with zero fitting. Below the title are the KS distances from the empirical CDF to the limit distribution and to $N(0, 1)$, with gray lines at 1% and 5%.

The steps of the lower tail come from the duplicate edge pairs and are filled in as c grows. When c is small, failure is almost always caused by one duplicate edge pair; when the two edges have opposite signs the observation is 2 below d. This -2 offset is large relative to $s(d)$; at $c = 0.25$ it lies at a standardized error of about -5.7, and about four tenths of the failure samples fall on this step, the large step at $c = 0.25$ in Figure 6.3. As c grows, the overlap among the remaining edges brings many $\pm 2/k$ offsets that fill in the steps, and the lower tail gradually approaches the Gaussian (Appendix H.6).

This discrete structure determines the behavior of the two rules at finite size. The Gaussian rule treats the discrete left tail as a continuous tail: when c is small $s(d)$ is small, the -2 offset caused by opposite signs of the duplicate edge pair lies far in the Gaussian tail, and the probability of rejecting the true d exceeds u. The G_n of the Skellam rule counts the probability of the observation value itself and accepts the candidate when the observation falls on a lattice point, so it is conservative in the small-c and low-load regions where the values are sparse. The conditional miss rate results of the next subsection agree with these two points.

6.4 Conditional miss rate and extra capacity at finite size

This subsection tests the risk of the two rules at finite table size and measures the extra capacity paid by the upper bound. Theorem 5.6 guarantees that the conditional miss rate of the Skellam rule is asymptotically at most u for all $d \geq 2$, and the Gaussian rule has the same guarantee only when $c \to \infty$ (Corollary 4.4). The conditional miss rate is the proportion of misses ($U < d$) among failure samples, reported as miss/u; a value not exceeding 1 means the risk is within the target. The extra capacity is $(U - d)/d$.

Conditional miss rate. The Skellam rule keeps the risk near the target over the whole region. The circles in Figure 6.4 all fall near miss/u $= 1$ on the 189 configurations, and the maximum 1.04 is within the sampling error of 20,000 failures per configuration. Further, the 12 configurations closest to the risk upper bound are retested with independent samples (ExpWORST), covering bounded c, the critical window, and the overload region: in Figure 6.5 the estimated miss/u values do not exceed 0.99, and the largest confidence upper bound is $1.024u$, meeting the $1.1u$ criterion set before sampling (Appendix H.7). The low-load configurations with $c < 8$ are conservative, most clearly at $u = 0.01$, for the reason of the discrete structure of Section 6.3: the lower tail concentrates on a few lattice points, and the rule cannot use up the risk budget exactly between lattice points.

The Gaussian rule meets the target only when c is large. The triangles in Figure 6.4 are near 1 when $c \geq 8$ and rise quickly as c decreases, already exceeding the target by more than 8 times at $u = 0.01$, $c = 1$ (Table H.7 of the appendix). This excess does not disappear as the table size grows, consistent with the finite limit of Theorem 5.3. The reason is again the discrete structure: when c is very small, the post-failure observation falls near $d + 2$ and $d - 2$ with probability one half each (Proposition F.1), the Gaussian tail gives the latter an extremely small probability, the rule therefore accepts the candidate $d - 2$, and the miss rate tends to 1/2.

Extra capacity. The extra capacity is determined by the table size and decreases as $M^{-1/2}$, with the two rules almost identical. At $u = 0.01$ and $c \geq 8$, the median extra capacity of the Skellam rule drops from 25% at $M = 256$ to 2.6% at $M = 16384$, because $U - d$ is about 2.3–2.8 times $s(d)$ and $s(d)/d \approx \sqrt{2/M}$. In the two-round protocol this extra capacity appears as the increase of the second-round table size over the oracle; in Figure 6.6(a) it falls along the dotted line of slope $-1/2$ (Section 6.5). When c is bounded the difference itself is small and few extra elements appear; at $c = 4$ the median of $U - d$ is 15 (Appendix H.7).

For engineering use we recommend $M \geq 1024$. The null distribution of Section 5 omits the overlap between the duplicate edge pair and the remaining edges, whose occurrence rate $(d - 2)k^2/M$ grows as the table size decreases. At $M = 64$ this omission is no longer negligible, and the largest miss/u of the Skellam rule is 1.08, exceeding the sampling error. At $M = 256$ no miss/u exceeding the sampling error is observed, but the occurrence rate of the omitted structure is 4 times that at $M = 1024$, and the median extra capacity also reaches 25%.

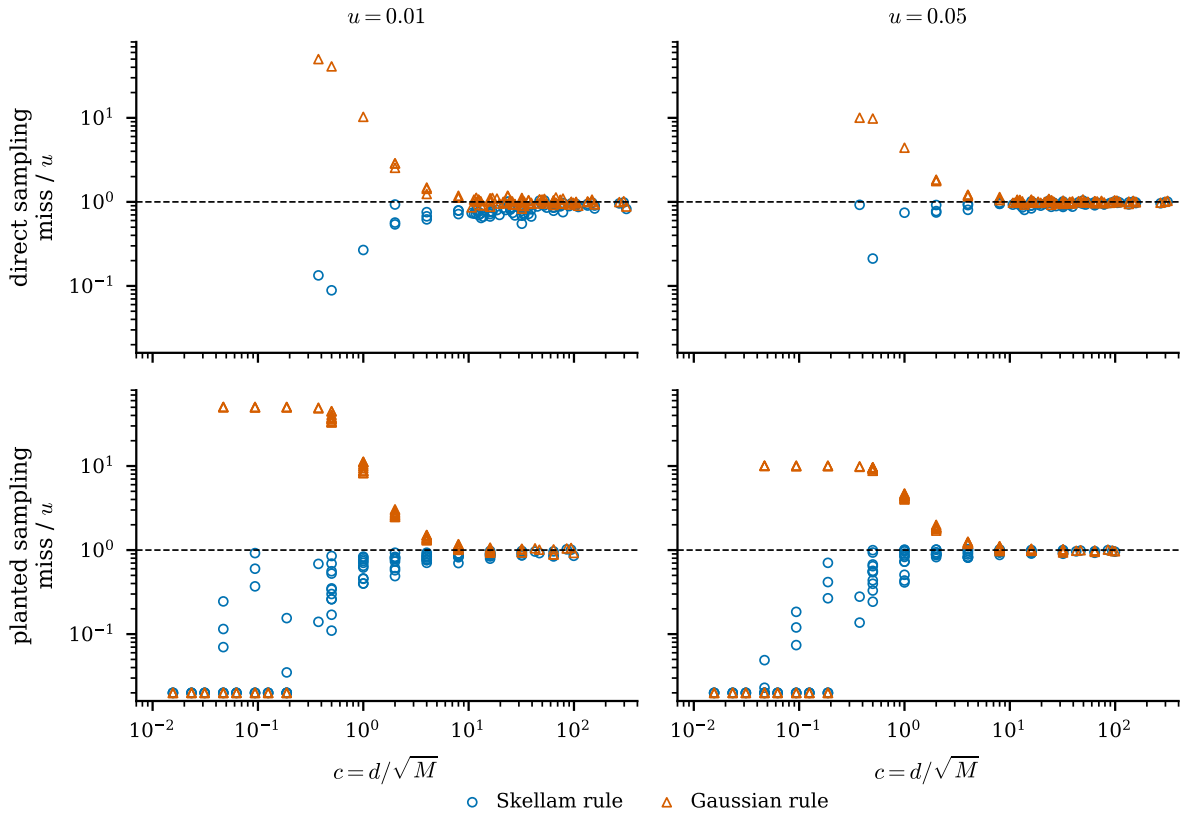


Figure 6.4: miss/u versus $c = d/\sqrt{M}$ in the statistical experiments ($k = 3, 4, 5$, $M = 256$–16384, about 20,000 failures per configuration; configurations with zero miss are drawn on the lower boundary). The top row is direct sampling, the bottom row is planted sampling; circles are the Skellam rule, triangles are the Gaussian rule.

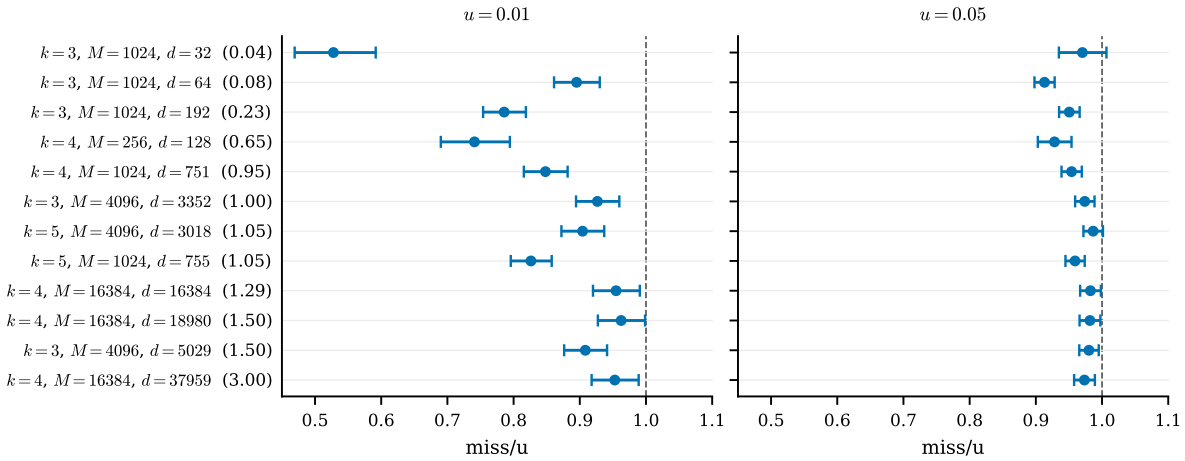


Figure 6.5: Estimated miss/u of the Skellam rule and two-sided 95% intervals (Clopper–Pearson) on the Exp-WORST configurations. Configurations are ordered by load, with the load in parentheses; the dashed line is miss/u = 1.

6.5 The two-round protocol on a real IBLT

Theorem 5.7 decomposes the two-round failure probability into three terms: the joint miss rate, the peeling failure when the capacity is sufficient, and the checksum misjudgment. This subsection runs the two-round protocol on a complete IBLT, tests the relative sizes of the first two terms in the real protocol, measures the communication cost of the random-sign IBLT, and compares it with the existing calibration rule.

The protocol experiments (Exp-PROTO) take $k = 3, 4, 5$, first-round table size $M_1 \in \{256, 1024, 4096, 16384\}$, and first-round loads 0.95, 1, 1.05, 1.5, 3, with the difference taking two sign compositions, one-sided (all on the A side) and balanced, for 120 configurations in total; the two sides additionally hold 1000 common elements. Each configuration runs until the first round accumulates 20,000 failures. After each first-round failure, three upper bounds determine the second-round table size separately and the second round runs: the Skellam rule, the Gaussian rule, and the oracle that takes the true d directly. The two rules take $U_b = \max(U, |b|)$ as in Section 3.2, the second-round table size is $\max\{\lceil U_b/(0.8\alpha_c)\rceil, 64\}$, and a fresh cell hash is used. The three upper bounds share all keys and hashes within the same trial, so the difference comes only from the upper bound itself and the three can be compared pairwise trial by trial (implementation details in Appendix H.8).

The reliability of the two-round protocol is comparable to that of the oracle. In Figure 6.6(b), the final failure rates of the two rules fall in the same range as the oracle, none exceeding 0.13% when $M_1 \geq 1024$, and at $u = 0.01$ the maximum is not higher than the oracle (Table H.9 of the appendix); the two rules are statistically indistinguishable. The final failure comes mainly from the peeling failure of the second round itself, consistent with the expectation that the second term dominates in Theorem 5.7: at $u = 0.01$, no more than 6% of the final failures follow a miss. The reason is that the second-round table size leaves a margin $1/(0.8\alpha_c)$, which absorbs most slight underestimates. The same reason also explains why the rules are no worse than the oracle: the upper bound given by the rules is usually larger than d, the second-round table is slightly larger than the oracle, and the peeling failure is therefore smaller.

The extra communication cost is determined only by M_1 and decreases as $M_1^{-1/2}$. In Figure 6.6(a), the configurations with different k, load, and sign composition at the same M_1 almost coincide and fall along the dotted line of slope $-1/2$. At $u = 0.01$, the increase of the second-round table size of the Skellam rule over the oracle drops from 25.6% at $M_1 = 256$ to 2.6% at $M_1 = 16384$, consistent with the extra capacity of Section 6.4; converted to total bytes, the overhead drops from 14.7% to 1.5%.

Figure 6.7 gives the same trend at fixed load (first-round target load 1.5, balanced, $u = 0.01$): as the table size grows, the miss/u of the Skellam rule gradually approaches 1 from below, while the Gaussian rule stays near 1 (in the top row each point merges all first-round failures of the same configuration, Appendix H.7); the total byte overhead relative to the oracle drops from 16.8% at $M_1 = 256$ to 1.7% at $M_1 = 16384$, tending to become negligible as the table size grows.

Comparison with the existing calibration rule. The existing calibration rule depends on the sign composition under the standard encoding, which is the consequence in the protocol of the bias shown in Section 6.2. We plug the calibration factor $q_{0.01}$ of [38] into the same protocol, taking $U = \lceil \widehat{d}/q_{0.01} \rceil$, with the rest of the settings the same as the two rules of this paper (Table H.10 of the appendix). Under the standard encoding, at load 0.95–1.0 the same factor gives a conditional miss rate of 0.05–0.40u for a one-sided difference and 0.86–0.94u for a balanced one, differing by an order of magnitude; the random-sign encoding brings both to 0.80–1.01u. The calibration factor must be calibrated by simulation separately for each set of parameters, and the literature gives values only for $k = 3$, $M_1 \leq 4096$, and $u = 0.01$; on these parameters it is close to the Skellam rule in final failure rate and total bytes. The Skellam rule is computed directly from the theoretical distribution of Section 5, requiring no calibration for any k, M, load, or u.

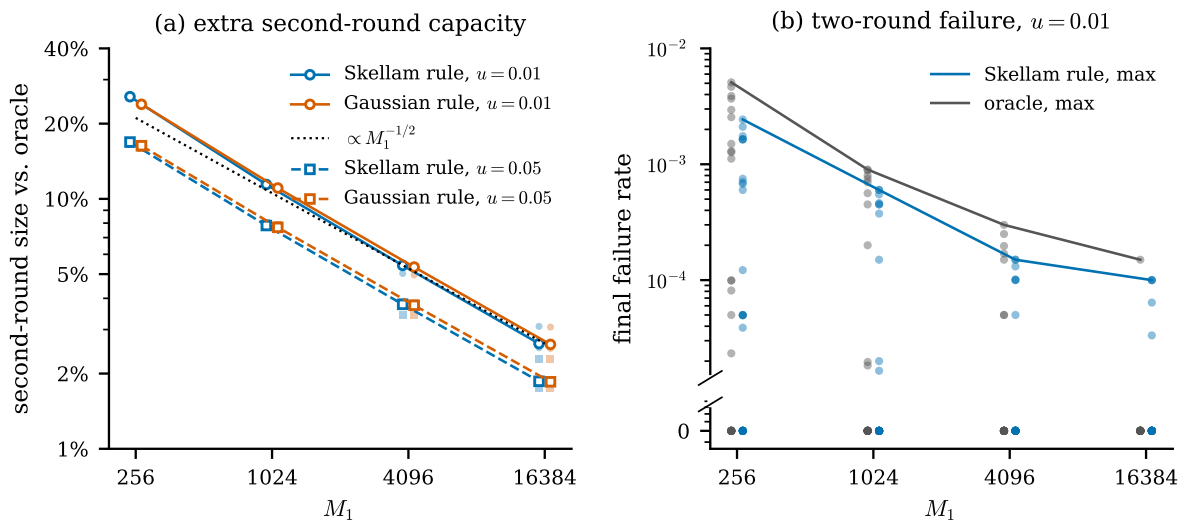


Figure 6.6: The two-round protocol as the first-round table size M_1 varies (120 configurations, 13.94 million trials per u in total). (a) The increase of the second-round table size over the oracle after a first-round failure, one point per configuration, with the dotted line of slope $-1/2$; (b) the final two-round failure rate of each configuration at $u = 0.01$, with zero configurations drawn at 0 below the axis break.

6.6 Computational cost

Compared with the standard encoding, the extra computational time comes almost entirely from the sizing of the Skellam rule, and the overhead of the random-sign encoding itself is negligible. With the Gaussian rule as the baseline, the Skellam rule increases the total protocol time by less than half when the load does not exceed 1.5; at load 3, sizing becomes the main cost and the total time is 2.8–3.8 times that of the Gaussian rule; combining the two rules by the first-round observation avoids this part of the cost (Section 5.5).

The random-sign encoding makes any difference look like a balanced difference to the decoder; under a one-sided difference the candidates checked per recovered key increase from 1.00 to 1.77 ($k = 3$), and the resulting increase in decoding time is less than 1% of the total protocol time.

The sizing cost of the Skellam rule grows with the load. Its exact CDF requires computing and looking up Poisson probability tables, which depend only on (M, k) and the candidate n, and the same receiver can cache and reuse them across synchronizations. Table 6.2 gives the time of the complete protocol at $M_1 = 16384$, including first-round encoding and decoding, sizing, and the second round: at load 1.0–1.05 the Skellam rule is 14–23% more than the Gaussian rule, and at load 1.5 it is 39–43% more; the sizing cost grows roughly as c^2, and at load 3 the total time of the Skellam rule is 57–88 ms. The conditions of the computational-cost measurement (Exp-COST), the per-sizing time under different cache conditions, and the memory use are in Appendix H.9.

6.7 Rule choice and deployment

At deployment we recommend combining the two rules by the first-round observation (Section 5.5). Table 6.3 summarizes the guarantees and costs of using each rule alone and in combination. The Gaussian rule needs only a microsecond-scale scalar search, and its guarantee range is $d \gg \sqrt{M}$, suited to applications whose workload can guarantee this condition and that pursue low latency; the table size alone is not enough to rule out bounded c and a fixed small d. The Skellam rule holds for all $d \geq 2$ and does not need to know which load region d falls in, at the

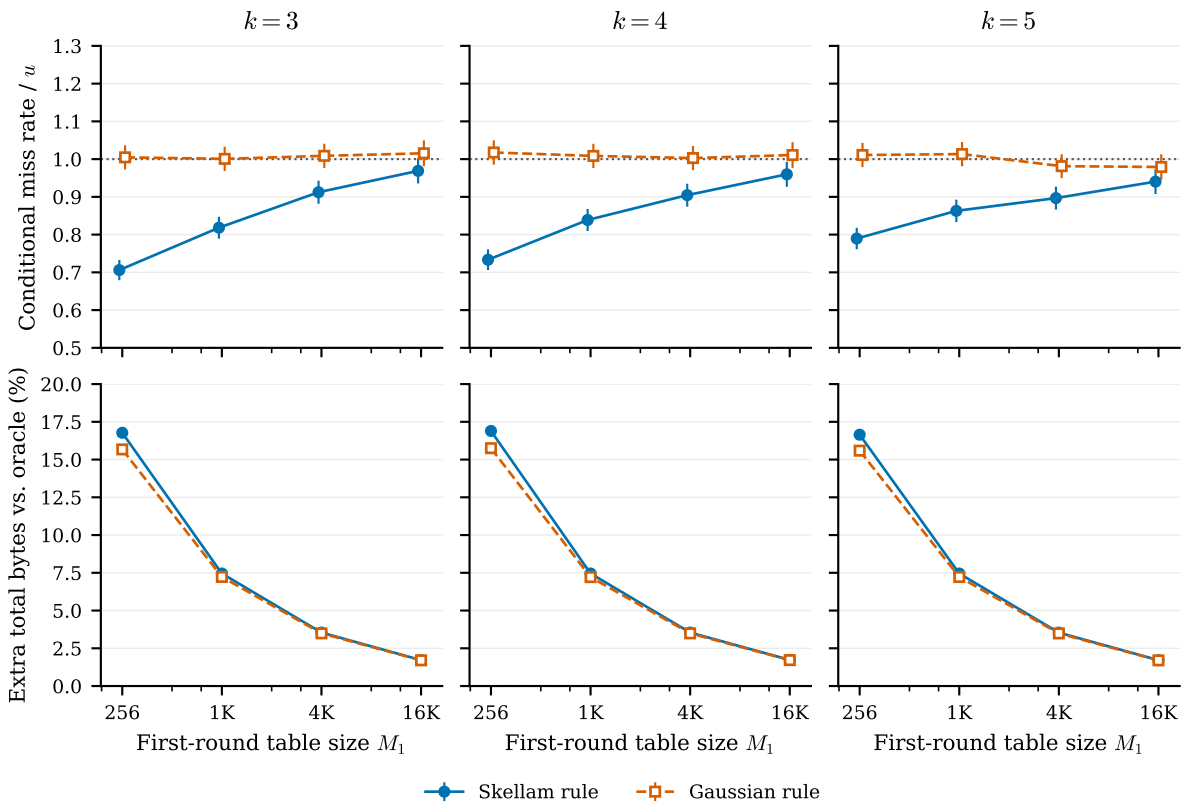


Figure 6.7: The two-round protocol at fixed load as the table size varies (balanced, $u = 0.01$, target load 1.5). One column per k; the top row merges all first-round failures of the same configuration (about 330,000 per point); the bottom row is the communication overhead relative to the oracle of the same configuration.

Table 6.2: Total protocol time per trial (ms, median of 10 repetitions; $M_1 = 16384$, one-sided difference; the Skellam rule caches the Poisson probability tables between successive trials).

	Skellam rule			Gaussian rule		
Load	$k = 3$	$k = 4$	$k = 5$	$k = 3$	$k = 4$	$k = 5$
1.0	12.7	16.6	17.6	10.9	14.0	15.4
1.05	15.9	19.2	21.5	13.1	15.6	18.4
1.5	19.7	24.0	27.4	14.0	16.8	19.7
3	57.1	84.1	88.1	18.8	22.4	25.8

cost of millisecond-scale computation that looks up Poisson tables, which becomes the main cost of the protocol when the first round is heavily overloaded (Section 6.6). The combined use takes the Gaussian rule at high load and the Skellam rule otherwise, keeping the uniform guarantee unchanged and the time in the overload region the same as the Gaussian rule.

7 Related work

IBLT and set reconciliation. IBLT[19] and the invertible Bloom filter[15] list the set difference by peeling, and their success or failure is determined by the 2-core of a random hypergraph[30, 14, 13]; at low load, failure is dominated by duplicate edge pairs[19, 40], and

Table 6.3: Choice of the sizing rule.

Rule	Range of conditional miss rate asymptotically at most u	Suitable scenario	Computational cost
Gaussian rule	$d \gg \sqrt{M}$	workload guarantees the range, low latency valued	microsecond-scale, scalar search
Skellam rule	all $d \geq 2$, uniformly	d may range from far below $\sqrt{M}$ to overload	millisecond-scale, cached Poisson tables
Combined use	all $d \geq 2$, uniformly	needs the uniform guarantee while controlling overload time	microsecond-scale at high load, otherwise as the Skellam rule

the probability of a linear core at subcritical load is exponentially small[24, 11]. Subsequent work improves the hypergraph structure[27], gives rateless codes[39, 28], or predicts and breaks the 2-core after a decoding failure[17]. Belazzougui, Kucherov, and Walzer[5] use markers over a finite field to distinguish the direction of the difference and reduce the failure probability to exponentially small. These works focus on how to reduce failure or continue decoding from failure; this paper focuses on how to read the same table after failure to decide the next-round capacity.

Estimating the difference size. The common way to estimate d is to send a separate estimator: the strata estimator[15], min-wise hashing[7], the odd sketch[29], and GXBits[23]. PBS[18] uses 128 independent ToW sketches to first estimate d and then sets the capacity to a fixed multiple of the estimate, with the estimator and the reconciliation structure transmitted and accounted separately. The estimate in this paper is read on the same set of counts used for decoding, without sending a separate estimator; the cost is that the estimate is needed only when decoding fails, and failure selects the mapping, which is exactly the object of the analysis in this paper.

Random-sign sketches and second-moment estimation. AMS[1] estimates F_2 by a sum of random signs; CountSketch[8] uses random-sign counters to support point queries; Thorup and Zhang[36] analyze its second-moment extension and give a centered second-moment estimate for a single counter array, and Fast-AGMS[10] uses the same structure for distributed queries. $\widehat{d}$ has the same algebraic form as these estimators, and its conditional unbiasedness given the mapping also has the structure of the Hutchinson trace estimator[21]. These sketches carry no decodable keySum and do not involve the conditional distribution of a decoding failure. Fischer and Nakos[16] combine hypergraph peeling with a random-sign CountSketch in a single measurement, but the tail norm is provided by an independent short sketch and treated as known in the analysis. This paper writes the shared random signs into the count, keySum, and checksum of the IBLT, so that the same table can peel and, after failure, give an estimate independent of the direction of the difference, together with a post-failure upper bound valid for all loads.

Reading the same table after failure. The work [38] reads the count energy after a first-round failure to set the second-round capacity, treating iid signs as a statistical model of the sign

composition in the analysis and obtaining a conditional mean equal to d; its upper-bound rule uses a chi-square approximation under deep overload and relies on per-configuration simulation calibration under other loads. This paper writes the random signs into the three fields of the IBLT as the encoding, making these properties hold exactly for every fixed mapping, and gives a post-failure distribution and upper bound valid for all loads.

Conditional limits and discrete approximations. The limit distribution when c (Eq. (4.4)) is bounded uses the Chen–Stein method[2] to handle the Poisson approximation of sparse overlap, and the Skellam distribution[35] describes the difference of two kinds of counts. At low load, the failure-conditional distribution is replaced by the planted distribution, which is the size-biased distribution of the duplicate edge pair count[3]. When $d \gg \sqrt{M}$, the random-sign quadratic form given the mapping is handled by the cumulant method on the dependency graph[22], needing only control of the concentration[37, 9] and the degree of the overlap graph; the Gaussian limit for deterministic coefficients is proved by an edge-exposure martingale CLT[20], and a general quadratic form CLT is in[12]. The upper-bound rule uses the classical test inversion[31].

8 Discussion and conclusion

The random-sign IBLT lets one table simultaneously perform decoding and difference-size estimation, with the cells and the decoding capability both unchanged. Whether decoding succeeds is determined only by the mapping, and the randomness of the estimate comes from the signs; this separation makes the estimate exactly unbiased for every fixed mapping and identically distributed for all sign compositions, and makes the post-failure distribution fully characterizable: Gaussian when $d \gg \sqrt{M}$, and a discrete distribution of a ± 2 term plus a Skellam term when $c = d/\sqrt{M}$ is bounded. The same family of distributions describes all scales, and the Skellam rule that takes it as the null distribution makes the conditional miss rate asymptotically at most the target risk uniformly over all $d \geq 2$.

At deployment we recommend combining the two rules by the first-round observation, which keeps the uniform guarantee for all $d \geq 2$ with the time in the overload region the same as the Gaussian rule; the guarantees and costs of the various choices are in Section 6.7. On the deployment parameters, the Skellam rule meets the target on all experimental configurations with $M \geq 256$, and for engineering use we recommend $M \geq 1024$ (Section 6.4). The checksum can be shortened to 32 bits in exchange for narrower cells, leaving the risk of sizing unaffected, and the probability of a verification error is still at most 10^{-4} at $M_1 = 16384$ and load at most 3 (Section 5.6). Storing the counts of several independent signs per cell reduces the absolute error but does not change the post-failure conditional scale, and one still performs inversion by the post-failure distribution (Appendix F.10). The random-sign encoding does not increase the space overhead of the cells, and the increase in decoding time is less than 1% of the total protocol time (Section 6.6).

Three questions remain for future work. First, before building the tables, reject more candidates with a rigorous finite-parameter tail upper bound, reducing the computational cost of the Skellam rule. Second, turn the errors of Theorem 5.3 and Theorem 4.2 into computable constants, giving a finite risk bound for a specified M. Third, extend to multiset differences whose weights can exceed one: there the random-sign count gives an unbiased estimate of the F_2 of the difference vector, and the decoding and post-failure analysis of this extension remain to be completed.

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A Notation

This appendix collects the notation used in the main text and the principal proofs, ordered by section for easy reference. Except where quantities are compared across different scales, the dependence of each quantity on M is suppressed; we fix $k \geq 3$ and let $M \to \infty$, and a subscript k denotes a constant depending only on k. The symbol $\Rightarrow$ denotes convergence in distribution. Section 3 uniformly writes the coding coefficient $s_i\varepsilon_i$ as ε_i, and this convention is followed below.

B Post-failure bias of the standard IBLT

This appendix proves Proposition 2.2 and gives a numerical verification.

In the standard IBLT the count is $C_v = \sum_{i:v\in S_i} s_i$, and the coefficient is the set direction. Expanding $\widehat{d}$,

$$\widehat{d} - d = \frac{2}{k(1-k/M)} \sum_{i<j} s_i s_j \Big(H_{ij} - \frac{k^2}{M} \Big). \tag{B.1}$$

Failure changes the distribution of the overlap: the overlap of same-sign pairs raises $\widehat{d}$, and the overlap of opposite-sign pairs lowers $\widehat{d}$, which yields Proposition 2.2.

Proof of Proposition 2.2. By Eq. (B.1),

$$\mathbb{E}[\widehat{d} \mid F] - d = \frac{2}{k(1-k/M)} \sum_{i<j} s_i s_j \, \mathbb{E}\Big[H_{ij} - \frac{k^2}{M} \;\Big|\; F \Big].$$

F depends only on the unsigned mapping and is invariant under permutation of the key labels; the random mapping $\mathcal{G}$ is exchangeable over the labels, so $\mathbb{E}[H_{ij} \mid F]$ is the same for all $i < j$ and equals $k^2/M + k(1-k/M)c_F$. The number of same-sign pairs minus the number of opposite-sign pairs is $\binom{d_+}{2} + \binom{d_-}{2} - d_+ d_- = (b^2 - d)/2$. Substituting gives $\mathbb{E}[\widehat{d} \mid F] - d = c_F(b^2 - d)$, and dividing by d yields Eq. (2.4). The proof uses only these two properties of F, so it holds for any event that depends only on the unsigned mapping, is invariant under label permutation, and has positive probability. □

Table A.1: Principal notation.

Notation	Meaning
Model and peeling	
A, B	sets held by the two sides; $\lvert A\rvert$ and $\lvert B\rvert$ are the set sizes
M, M_1	number of cells; number of cells in the first-round table
k	number of cells selected by each key
x, x_i	key; the i-th differing key
$\mathrm{chk}(x)$	checksum of key x
d	difference size $\lvert A \triangle B\rvert$
d_+, d_-	number of keys in A only and in B only
b	difference of the two set sizes, $d_+ - d_- = \lvert A\rvert - \lvert B\rvert$
S_i	the k-subset selected by key i
$\mathcal{G}$	unsigned mapping with key labels
H_{ij}	overlap $\lvert S_i \cap S_j\rvert$ of two edges
F	peeling failure on a single table; $\Pr(F)$ is the failure probability
F_1, F_2	peeling failure in the first and second rounds
E	checksum misjudgment event
α_c	peeling threshold (about 0.818 for $k = 3$ and 0.772 for $k = 4$)
Encoding and measurement	
s_i	set direction of key i, taking values ± 1
$\mathcal{U}$, $\varepsilon(\cdot)$	universe of keys, and the sign hash shared by the two sides
ε_i	random sign $\varepsilon(x_i)$ of key i; in the analysis the coding coefficient $s_i\varepsilon_i$ is written as ε_i
C_v	count of cell v, $\sum_{i:v\in S_i} \varepsilon_i$ (in the standard encoding, $\sum_{i:v\in S_i} s_i$)
$\widehat{d}$	estimator of d, see Eq. (2.1); used by the Gaussian rule
$s(n)$	standardization scale $\sqrt{2n(n-1)/(M-1)}$, see Eq. (2.2)
X	standardized estimation error $(\widehat{d} - d)/s(d)$, see Eq. (4.3)
$\widetilde{d}$	estimator of d, $\frac{1}{k}\sum_v C_v^2$, see Eq. (5.1); used by the Skellam rule
c_F	normalized amount by which the overlap exceeds its unconditional mean in the failure sample, see Eq. (2.4)
Φ, $\mathcal{N}(0,1)$	CDF and distribution of the standard Gaussian
Inversion and protocol	
u	target risk, taken as 0.01 and 0.05 in the experiments
n	candidate difference size, an integer $n \geq 2$
U, U_b	upper bound on d given by the selected rule; $\max(U, \lvert b\rvert)$ after enforcing the lower bound $\lvert b\rvert$
$t(n)$	acceptance lower bound of the Gaussian rule, see Eq. (4.6)
$\chi^2_{M-1,u}$	lower u quantile of the chi-square distribution with $M-1$ degrees of freedom
λ_n	parameter of the Skellam term in the null distribution, see Eq. (5.3)
Z_n	$\mathrm{Skellam}(\lambda_n, \lambda_n)$, the difference of two independent $\mathrm{Pois}(\lambda_n)$
J_λ	$2\varepsilon_1\varepsilon_2 + \frac{2}{k}\,\mathrm{Skellam}(\lambda, \lambda)$, see Eq. (5.4)
G_n	CDF of the null distribution $n + J_{\lambda_n}$ of candidate n (Definition 5.2)
c	$d/\sqrt{M}$ (Eq. (4.4)); its limit in Theorem 5.3
$m(\cdot)$	second-round table-size rule $m(U_b) = \lceil U_b/(0.8\alpha_c)\rceil$
$\Pr_m(F)$	failure probability on a new table with m cells
Proofs	
Ω_2	sum of squared overlaps $\sum_{i<j} H_{ij}^2$
$\mathcal{G}_M$	set of good graphs in Appendix D, whose complement has probability an arbitrary fixed negative power of M

The simulations used in Figure 2.1 of Section 2 and Figure 6.2 of Section 6.2 directly verify Proposition 2.2. The key point is that the five sign compositions on the same configuration share the same batch of mappings: F depends only on the unsigned mapping, so the failure samples of the five sign compositions are identical sample by sample and hence share the same c_F. The other factor on the right, $b^2/d - 1$, depends only on the sign composition and equals $d - 1$ under a one-sided difference. Hence the conditional bias of any sign composition equals the one-sided bias multiplied by $(b^2 - d)/(d^2 - d)$: once c_F is determined from a single one-sided point, the bias of every other level can be predicted with no fitting. Figure 2.1 plots exactly this prediction. The horizontal axis is $(b/d)^2$; the straight line of each color is determined by the one-sided point of that configuration at $(b/d)^2 = 1$, and the remaining four points fall on the line, confirming Eq. (2.4).

The full verification covers 20 configurations with $k = 3, 4, 5, 6$ and failure probability from 0.10 to 0.90 on the $M = 16384$ grid, each configuration having five levels of sign composition $d_-/d \in \{0, 0.1, 0.25, 0.4, 0.5\}$. Figure 2.1 shows ten of these configurations with $k = 3$ and $k = 6$; the standard-encoding curves in Figure 6.2(a) are four of the configurations with failure probability about 0.1, including $k = 4$ and $k = 5$. The 20 configurations have four non-one-sided sign composition levels each, 80 predictions in total. In units of $s(d)$, the difference between prediction and measurement is at most 0.0064, all within two standard errors (the standard error accounts for the sampling error of both the one-sided mean and the measured value).

C Encoding properties and proofs of exact moments

This appendix first describes the XOR implementation of keySum and checksum, then gives the proofs of the finite identities in Sections 3 and 4, summarizes the exact verification of these identities on small parameter sets, and finally discusses the AMS observation given by the count sum.

C.1 XOR implementation

In the set model, each key is inserted at most once on each side, so keySum and checksum can be accumulated by bitwise XOR as in the common implementation of the standard IBLT, with only the count multiplied by $\varepsilon(x)$. Under XOR, $-x = x$, so the keySum and checksum of a difference-table cell are the XOR of the differing keys that select the cell and of their chk values, independent of the coding coefficients. In a pure cell, keySum and checksum equal x_i and $\mathrm{chk}(x_i)$ directly, and the check becomes verifying that the checksum equals $\mathrm{chk}(x)$; when deleting x_i, the count subtracts C_v, and the other two fields are XORed once more with x_i and $\mathrm{chk}(x_i)$. The induction argument of the next subsection for the additive form still holds, the direction is still recovered by $C_v\,\varepsilon(x)$, and the process and result of peeling are the same as in the additive form. In a multiset, the same key may be inserted repeatedly, and XOR cancels repeated contributions while the count does not, so the additive form is needed there. If peeling modifies the difference table in place, the receiver can accumulate the two scalars $\sum_v C_v^2$ and $\sum_v C_v$ while subtracting cell by cell, with no need to keep the original difference table.

C.2 Proof of Proposition 3.1 (ideal decoding equivalence)

Proof. Induct on the number of peeling steps. The invariant is that after each step, the three fields of every cell equal the encoded sum of the keys not yet deleted: the count is $\sum s_i\varepsilon_i$, keySum is $\sum s_i\varepsilon_i x_i$, and checksum is $\sum s_i\varepsilon_i\,\mathrm{chk}(x_i)$, the sums ranging over the keys not yet deleted that select the cell. Initially this is the definition in Eq. (3.1).

Suppose that at some step the degree of cell v in the unsigned hypergraph is one, with the unique key x_i. Then the count is $s_i\varepsilon_i \in \{\pm 1\}$, the count times keySum equals x_i, the checksum equals the count times $\mathrm{chk}(x_i)$, and the cell hash of x_i contains v, so v passes the pure-cell

test. Conversely, a cell of degree other than one passing the test is a false positive, excluded by the ideal-purity assumption. Hence the set of cells removable by the random-sign encoding is the same as the set of unsigned degree-one cells. When deleting x_i, subtract the coding coefficient $s_i\varepsilon_i$ times the three-field contribution from its k cells; the invariant is preserved, and the corresponding edge is removed from the unsigned hypergraph. The two processes adopt the same cell-selection order and reach the same endpoint. The endpoint of peeling is independent of the selection order and is always the 2-core, so F depends only on $\mathcal{G}$.

At termination all fields must be checked: all-zero counts alone do not certify success, because under random signs the counts of a nonempty core can also all be zero. keySum can also cancel simultaneously, and success is misjudged only when the checksums are then also all zero; this event is accounted for in E of Section 5.6. □

C.3 Proof of Proposition 4.1 (conditional unbiasedness and conditional moments)

Proof. By $H_{ii} = k$ and $\varepsilon_i^2 = 1$, the sum of squared counts and the square of the sum are

$$\sum_v C_v^2 = \sum_{i,j} \varepsilon_i\varepsilon_j H_{ij} = kd + 2\sum_{i<j} \varepsilon_i\varepsilon_j H_{ij},$$
$$\frac{1}{M}\Big(\sum_v C_v\Big)^2 = \frac{k^2}{M}\Big(\sum_i \varepsilon_i\Big)^2 = \frac{k^2}{M}d + 2\sum_{i<j} \varepsilon_i\varepsilon_j \frac{k^2}{M}.$$

Dividing the first identity by k gives Eq. (5.2); subtracting the two gives $kd(1 - k/M) + 2\sum_{i<j} \varepsilon_i\varepsilon_j(H_{ij} - k^2/M)$, and dividing by $k(1 - k/M)$ gives Eq. (4.1). For a fixed mapping, $\mathbb{E}[\varepsilon_i\varepsilon_j] = 0$ $(i \neq j)$, so the conditional expectations of $\widehat{d} - d$ and $\widetilde{d} - d$ are both zero.

Below we write the cross-term coefficients as inner products. Let $w_i \in \mathbb{R}^M$ be the incidence vector of key i minus its mean k/M; then $\langle w_i, w_j\rangle = H_{ij} - k^2/M$, $\|w_i\|^2 = k(1 - k/M)$, and Eq. (4.1) becomes

$$\widehat{d} - d = \frac{2}{k(1 - k/M)} \sum_{i<j} \langle w_i, w_j\rangle\, \varepsilon_i\varepsilon_j.$$

Given $\mathcal{G}$ the inner products are constants, and the randomness lies only in the sign products $\varepsilon_i\varepsilon_j$. The expectation of a product of sign products is nonzero if and only if each subscript occurs an even number of times, in which case the expectation is one.

For the variance, expand the square and consider two unordered pairs $\{i, j\}$ and $\{i', j'\}$. When the two are the same, the product is $\varepsilon_i^2\varepsilon_j^2 = 1$; when they differ, they share at most one subscript, so some subscript occurs exactly once and the expectation is zero. Hence only the diagonal part remains,

$$\mathrm{Var}(\widehat{d} \mid \mathcal{G}) = \frac{4}{k^2(1 - k/M)^2} \sum_{i<j} \langle w_i, w_j\rangle^2,$$

which is the equality part of Eq. (4.2).

For the third central moment, expand the cube and consider ordered triples of unordered pairs. When all three are the same, the product is $\varepsilon_i\varepsilon_j$; when two are the same, the product is the sign product of the third pair; both have expectation zero. When the three are pairwise distinct, each subscript occurring an even number of times requires the three pairs to form a triangle $\{i, j\}, \{i, h\}, \{j, h\}$; each triangle corresponds to $3! = 6$ ordered triples. Hence

$$\mathbb{E}[(\widehat{d} - d)^3 \mid \mathcal{G}] = \frac{8 \cdot 6}{k^3(1 - k/M)^3} \sum_{i<j<h} \Big(H_{ij} - \frac{k^2}{M}\Big)\Big(H_{ih} - \frac{k^2}{M}\Big)\Big(H_{jh} - \frac{k^2}{M}\Big).$$

For the two upper bounds, by Cauchy–Schwarz, $\langle w_i, w_j\rangle^2 \le \|w_i\|^2\|w_j\|^2 = [k(1 - k/M)]^2$. Substituting into the variance formula, each of the $\binom{d}{2}$ unordered pairs contributes at most 4,

giving $\operatorname{Var}(\widehat{d} \mid \mathcal{G}) \le 2d(d-1)$. On the other hand, the component of the vector $\sum_i \varepsilon_i w_i$ on cell v is $C_v - \frac{1}{M}\sum_u C_u$, so

$$\widehat{d} = \frac{\|\sum_i \varepsilon_i w_i\|^2}{k(1-k/M)} \ge 0,$$

and by the triangle inequality $\|\sum_i \varepsilon_i w_i\| \le \sum_i \|w_i\| = d\sqrt{k(1-k/M)}$, giving $\widehat{d} \le d^2$.

For the random mapping $\mathcal{G}$, the conditional mean is identically d, and by the law of total variance $\operatorname{Var}\widehat{d} = \mathbb{E}[\operatorname{Var}(\widehat{d} \mid \mathcal{G})]$. The intersection size H_{ij} of two independent uniform k-subsets has a hypergeometric distribution with mean k^2/M and variance $k^2(1-k/M)^2/(M-1)$. Substituting into the variance formula, each unordered pair contributes $4/(M-1)$, giving $\operatorname{Var}\widehat{d} = 2d(d-1)/(M-1) = s(d)^2$. □

C.4 Exact verification on small parameters

The finite identities of Sections 3 and 4 are verified by exhaustive enumeration on small parameters. For the six triples (M, k, d) in Table C.1, all mappings and all signs are enumerated, 155,640 states in total; all arithmetic uses integers and rationals, and the resulting values are exact. Over all states, common elements cancel in the difference table; the keys and directions recovered by signed peeling are correct, and the unrecovered keys are exactly the unsigned 2-core; on every mapping, $\mathbb{E}[\widehat{d} \mid \mathcal{G}] = d$, the conditional variance and third central moment equal the formulas above, and $\operatorname{Var}(\widehat{d} \mid \mathcal{G}) \le 2d(d-1)$; under all direction vectors the distribution of $\widehat{d}$ over the random signs is the same. Table C.1 summarizes the unconditional and post-failure statistics.

Table C.1: Exact results obtained by exhaustive enumeration on small parameters.

(M, k, d)	states	p_F	$\operatorname{Var}\widehat{d}$	$\mathbb{E}[\widehat{d} \mid F]$	$\operatorname{Var}(\widehat{d} \mid F)$	$\Pr(\widehat{d} = 0 \mid F)$
$(4, 3, 2)$	40	1/4	4/3	2	4	1/2
$(4, 3, 3)$	160	1	4	3	4	0
$(4, 3, 4)$	560	1	8	4	8	27/512
$(5, 3, 3)$	1,760	2/5	3	3	4	0
$(5, 3, 4)$	11,440	1	6	4	6	93/8000
$(6, 3, 4)$	141,680	2029/4000	24/5	4	11256/2029	75/4058

In Table C.1, $\operatorname{Var}\widehat{d}$ equals $s(d)^2 = 2d(d-1)/(M-1)$ row by row, and $\mathbb{E}[\widehat{d} \mid F]$ equals d row by row, consistent with the conditional unbiasedness given the mapping in Eq. (4.2). Failure preserves the mean and amplifies the scale: on the three parameter sets with $p_F < 1$, $\operatorname{Var}(\widehat{d} \mid F)$ is 3 times, 4/3 times, and about 1.16 times the unconditional variance, respectively; when $p_F = 1$, failure is a certain event and the two coincide. After failure, $\widehat{d}$ takes the value zero with positive probability, which for $(4, 3, 2)$ is 1/2.

The values in the $(4, 3, 2)$ row come from a duplicate edge pair. Each key independently and uniformly chooses one of the $\binom{4}{3} = 4$ 3-subsets. When the two keys choose the same subset, the degree of these three cells is two, there is no pure cell, and peeling stops at the first step, so $p_F = 1/4$. The counts of the three cells then all equal $\varepsilon_1 + \varepsilon_2$, and the remaining cells are zero. When the two signs are equal, $\sum_v C_v^2 = 12$ and $\sum_v C_v = \pm 6$, and Eq. (2.1) gives $\widehat{d} = 4$; when they differ, the counts are all zero and $\widehat{d} = 0$. The two cases have probability 1/2 each, so $\mathbb{E}[\widehat{d} \mid F] = 2 = d$, $\operatorname{Var}(\widehat{d} \mid F) = 4$, which is 3 times the unconditional variance 4/3, and $\Pr(\widehat{d} = 0 \mid F) = 1/2$. This is the smallest-scale instance of the duplicate-edge-pair structure of Section 5.

C.5 An AMS observation independent of failure

Each key enters k cells and uses the same sign, so under random signs the count sum equals the sum of the signs sample by sample,

$$\frac{1}{k}\sum_{v} C_v = \sum_{i=1}^{d} \varepsilon_i. \tag{C.1}$$

The right-hand side depends only on the signs, and its square is the standard AMS second-moment estimator under unit weights[1]. Rearranging Eq. (2.1) and Eq. (5.1) relates $\widetilde{d}$ and $\widehat{d}$ through this AMS term:

$$\widetilde{d} = \Big(1 - \frac{k}{M}\Big)\widehat{d} + \frac{k}{M}\Big(\frac{1}{k}\sum_{v} C_v\Big)^2. \tag{C.2}$$

Proposition C.1 (AMS on the same table)**.** *The full distribution of $(\frac{1}{k}\sum_v C_v)^2$ is exactly independent of the mapping, and hence exactly independent of ideal failure: $\frac{1}{k}\sum_v C_v$ takes the value $d-2j$ with probability $2^{-d}\binom{d}{j}$, $j = 0, \dots, d$. Its mean is d and its variance is $2d(d-1)$, which is $M-1$ times the unconditional variance $s(d)^2$ of $\widehat{d}$. By Eq. (C.2), $\widetilde{d}$ is a deterministic weighted sum of it and $\widehat{d}$. Given the mapping, its conditional covariance with $\widehat{d}$ is $\frac{4}{k(1-k/M)}\sum_{i<j}(H_{ij} - k^2/M)$.*

Proof. The right-hand side of Eq. (C.1) depends only on the signs and is independent of $\mathcal{G}$, hence independent of any event that depends only on the mapping. $\sum_i \varepsilon_i = d-2j$ if and only if exactly j signs are negative, with probability $2^{-d}\binom{d}{j}$. $\mathbb{E}(\sum_i \varepsilon_i)^2 = d$, and the fourth moment is $3d^2 - 2d$, so the variance is $2d(d-1)$. Eq. (C.2) follows by directly rearranging Eq. (2.1) and Eq. (5.1). The covariance formula follows by expanding both as quadratic forms in the signs and using that products of different unordered pairs are uncorrelated: $(\sum_i \varepsilon_i)^2 - d = 2\sum_{i<j}\varepsilon_i\varepsilon_j$, multiplied pair by pair with the cross expectation of Eq. (4.1). Averaged over failure it is $2d(d-1)c_F$, generally nonzero. □

This observation explains the connection of the design to classical AMS: the count array is a random-sign linear sketch whose sum is exactly the AMS sketch. It is exactly independent of failure, and no matter how rare the failure, it does not change its distribution; the cost is precision, as its variance is $M-1$ times that of $\widehat{d}$. $\widehat{d}$ replaces the square of a single sum by the sum of squares over the M cells, reducing the variance to $2d(d-1)/(M-1)$, but it is correlated with failure and requires the post-failure analysis. By the covariance of Proposition C.1, the two are generally correlated after failure, and the risk must be allocated separately when they are combined.

Proposition C.1 is verified by the same exhaustive enumeration as Appendix C.4 on the four parameter sets $(4,3,2)$, $(4,3,4)$, $(5,3,3)$, and $(6,3,4)$, 144,040 states in total: on every mapping the distribution of $(\frac{1}{k}\sum_v C_v)^2$ is the same, with mean d and variance $2d(d-1)$.

D Hypergraph lemmas

This appendix proves the hypergraph results used by both Theorem 4.2 and the discrete limit distribution: the small-core counting, the fact that failure is determined by a duplicate edge pair when $d = o(M)$, the low-load lemma at fixed subcritical load, and the concentration of the overlap graph. These results depend only on the unsigned mapping $\mathcal{G}$, where $\mathcal{G}$ consists of d labeled independent uniform k-subsets, with duplicate edges allowed.

Fix $k \geq 3$. Let $\lambda = \binom{d}{2}/\binom{M}{k}$ be the expected number of duplicate edge pairs, let Q be the planted distribution that first picks a pair of edge labels uniformly and then conditions on this pair mapping identically, and let $b_k = \lceil 3(k-2)/2 \rceil$. Write $K = \binom{M}{k}$, let $N = \sum_{i<j} \mathbf{1}_{\{S_i = S_j\}}$ be the number of duplicate edge pairs, and let $D = \{N \geq 1\}$.

D.1 Small-core counting and duplicate edge pairs

Lemma D.1 (Small-core counting)**.** *A 2-core subgraph with e edges uses at most $v_e = \lfloor ke/2 \rfloor$ vertices. For $d \le A_0 M$ and $e \le \varepsilon M$ (with $\varepsilon < 1/k$), the first-moment upper bound on its occurrence probability satisfies*

$$B_e = \binom{d}{e}\binom{M}{v_e}\Big(\binom{v_e}{k}\Big/\binom{M}{k}\Big)^e \le \big[C(e/M)^{(k-2)/2}\big]^e,$$

where C depends only on k and A_0. When the factor in d is kept, $B_e \le (C_k d/M)^e (e/M)^{\lceil (k-2)e/2 \rceil}$.

The proof of Theorem 1 in [19] uses the same first-moment counting to obtain a failure probability $O(t^{-k+2})$; the analogous counting of stopping sets appears in [40, 32]. Here we keep the factor in d, for use by Lemma D.2 when $d = o(M)$.

Proof. For a fixed set I of e edge labels and a set V of v_e vertices, the event $A_{I,V} = \{S_i \subseteq V,\ i \in I\}$ contains every core lying in V. Summing over I and V gives the first moment B_e in the lemma. Using $\binom{d}{e} \le (3d/e)^e$, $\binom{M}{v_e} \le (3M/v_e)^{v_e}$, $\binom{v_e}{k}/\binom{M}{k} \le (v_e/M)^k$, together with $ke - v_e = \lceil ke/2 \rceil$ and $v_e \le ke/2$,

$$B_e \le 3^{e+v_e}\Big(\frac{d}{e}\Big)^e\Big(\frac{v_e}{M}\Big)^{\lceil ke/2 \rceil} \le \Big(\frac{C_k d}{M}\Big)^e\Big(\frac{e}{M}\Big)^{\lceil (k-2)e/2 \rceil},$$

which is the bound that keeps the factor in d. Substituting $d \le A_0 M$ and dropping the ceiling since $e/M < 1$ gives the first bound. Let $a = (k-2)/2$; keeping the ceiling at $e = 3$ gives $B_3 = O(M^{-b_k})$. For $e \ge 4$, the ratio of consecutive terms is at most $Ce^a((e+1)/M)^a$. For fixed $r \ge 0$, choose ε small enough that the ratio of consecutive terms of $e^r B_e$ is at most $1/2$ on $4 \le e < \varepsilon M$, so that

$$\sum_{e=3}^{\lfloor \varepsilon M \rfloor} e^r B_e = O(M^{-b_k}).$$

This sum also covers the intermediate range with $e \to \infty$ and $e = o(M)$. □

The following lemma is the quantitative form of Proposition 2.1: when $d = o(M)$, failure is almost always caused by a pair of duplicate edges, so the failure-conditioned mapping is close to the planted distribution Q in TV. Duplicate edge pairs are the bottleneck of IBLT failure, a mechanism pointed out in [19]; Q has density N/λ with respect to the original distribution, the size-biased distribution of N [3].

Lemma D.2 (failure is determined by a duplicate edge pair when $d = o(M)$)**.** *Let $2 \le d = o(M)$; $d \to \infty$ is not required. Then*

$$\Pr(F \setminus \{\textit{there is a duplicate edge pair}\}) \le C_k\Big(\frac{d^3}{M^{b_k+3}} + \frac{d^4}{M^{2k}}\Big) = o(\lambda),$$

$$\big\|\mathcal{L}(\mathcal{G} \mid F) - Q\big\|_{TV} \le C_k\Big\{\lambda + dM^{k-b_k-3} + \frac{d^2}{M^k}\Big\} \to 0.$$

Proof. A single edge cannot form a core; two edges form a core if and only if they map identically. Hence $D \subseteq F$, and on $F \setminus D$ the core has at least three edges. Since $d = o(M)$, eventually $d \le \varepsilon M$, so every core of size $e \le d$ falls within the range of Lemma D.1; use the bound that keeps the factor in d. The $e = 3$ term is $O_k(d^3 M^{-3-b_k})$. For $e \ge 4$, use the envelope without the ceiling $(C_k d/M)^e (e/M)^{(k-2)e/2}$, whose ratio of consecutive terms is at most $C_k(d/M)((e+1)/M)^{(k-2)/2}$, tending to zero uniformly on $e \le d = o(M)$, so $\sum_{e=4}^{d} B_e \le C_k d^4/M^{2k}$. Adding the two gives the first bound. From $\lambda \asymp_k d^2/M^k$ and $b_k \ge k-1$, the ratio of the first bound to λ is $O_k(dM^{k-b_k-3} + d^2/M^k) \to 0$.

A specified pair of edges maps identically with probability $1/K$, so $\mathbb{E}N = \lambda$. Two distinct edge pairs are simultaneously duplicated with probability K^{-2}: if they share an edge, three edges map identically; if they do not, the two pairs are independent. Hence $\mathbb{E}[N(N-1)] \le \lambda^2$, and from $\mathbf{1}_D \ge N - N(N-1)/2$ we get $\lambda - \lambda^2/2 \le \Pr(D) \le \lambda$. The density of Q with respect to the original distribution is exactly N/λ; by the triangle inequality,

$$\|\mathcal{L}(\mathcal{G} \mid D) - Q\|_{TV} \le \frac{1}{2}\mathbb{E}\Big|\frac{\mathbf{1}_D}{\Pr(D)} - \frac{\mathbf{1}_D}{\lambda}\Big| + \frac{1}{2\lambda}\mathbb{E}(N - \mathbf{1}_D) = \frac{\lambda - \Pr(D)}{\lambda} \le \frac{\lambda}{2}.$$

Since $D \subseteq F$, $\|\mathcal{L}(\mathcal{G} \mid F) - \mathcal{L}(\mathcal{G} \mid D)\|_{TV} = \Pr(F \setminus D)/\Pr(F) \le \Pr(F \setminus D)/\Pr(D)$. As $\lambda \to 0$, $\Pr(D) \ge \lambda/2$, and by the first bound this term is $O_k(dM^{k-b_k-3} + d^2/M^k)$. Adding the two bounds gives the second bound. The proof uses only the small-core counting, so it holds for all $2 \le d = o(M)$. □

D.2 failure structure at fixed subcritical load

Lemma D.3 (Low load)**.** *Let $d/M \to \alpha \in (0, \alpha_c)$. Then $\Pr(F \setminus P_2) = O(M^{-b_k})$, $\Pr(F) = \lambda\{1 + O(M^{-\delta_k})\}$, and $\|\mathcal{L}(\mathcal{G} \mid F) - Q\|_{TV} = O(M^{-\delta_k})$, where P_2 denotes the event that the actual core consists of exactly two duplicate edges and $\delta_k = \min\{k-2,\, b_k - (k-2)\} \ge 1$.*

Proof. We first show that for every fixed $\varepsilon > 0$ and $\beta > 0$, $\Pr(|\text{core}| \ge \varepsilon M) = o(M^{-\beta})$. This is a variant of a known result: Kim [24], Theorem 1.7, proves for the binomial hypergraph $H(n, p; k)$ that $\Pr(|\text{core}| \ge \delta n)$ is exponentially small at subcritical load, and [19] also states the same conclusion citing [11], whose model assumes a given bounded degree distribution. The mapping in this paper allows duplicate edges, and the probability of a duplicate edge at linear load is only polynomially small, so we prove the required bound directly by message passing of fixed depth, following the finite-round neighborhood analysis and concentration used for LDPC codes [33, Theorem 2], and handle the cell degree, which has no fixed upper bound in this model, separately. Let Z_T be the number of edges all of whose incoming messages are 1 at round T; then $|\text{core}| \le Z_T$ on every graph. Local convergence to a Poisson branching tree gives $\mathbb{E}Z_T/M \to \alpha q_T^k$, where $q_0 = 1$ and $q_{t+1} = 1 - \exp(-k\alpha q_t^{k-1})$. When $\alpha < \alpha_c$, this recursion has no positive fixed point, so one can choose T with $\alpha q_T^k < \varepsilon/4$. On the set H of graphs whose maximum degree does not exceed $\lceil \log M \rceil$, changing h edge coordinates changes Z_T by at most $L_M h$, where L_M is of polylogarithmic order. By the typical bounded differences inequality [9, Proposition 2], $\Pr(Z_T \ge \varepsilon M) \le \Pr(H^c) + \exp\{-cM/L_M^2\} = o(M^{-\beta})$.

On $F \setminus P_2$, the actual core has at least three edges. The part of size below εM is controlled by Lemma D.1, and the rest by the linear core bound above, giving $\Pr(F \setminus P_2) = O(M^{-b_k})$. The remaining steps are the same as the second paragraph of the proof of Lemma D.2: the Bonferroni inequality gives $\Pr(F) = \lambda + O(\lambda^2 + M^{-b_k})$; $\|\mathcal{L}(\mathcal{G} \mid D) - Q\|_{TV} \le \lambda/2$; and since $P_2 \subseteq D \subseteq F$, $\|\mathcal{L}(\mathcal{G} \mid F) - \mathcal{L}(\mathcal{G} \mid D)\|_{TV} \le \Pr(F \setminus P_2)/\Pr(D)$. From $\lambda \asymp_k M^{2-k}$, the two terms combine into $O(M^{-\delta_k})$. This is a total variation bound on the entire labeled mapping. □

D.3 Superpolynomial concentration of the overlap graph

Lemma D.4 (Superpolynomial graph concentration)**.** *Fix $k \ge 3$, and let d/M lie in a fixed compact interval of $(0, \infty)$. Write $m_{ij} = H_{ij}$, $\Omega = \sum_{i<j} m_{ij}$, $\Omega_2 = \sum_{i<j} m_{ij}^2$, $\ell_i = l_i = \sum_{j \ne i} H_{ij}$, and $\Omega_\ell = \sum_i \ell_i^2$. There is a deterministic set of graphs $\mathcal{G}_M$, whose complement has probability smaller than any fixed negative power of M, on which these three statistics deviate from their exact means by at most $O(\sqrt{M}(\log M)^4)$, and*

$$\Omega_2 = \binom{d}{2}\left\{\frac{k^2}{M} + \frac{k^2(k-1)^2}{M(M-1)}\right\} + O(\sqrt{M}(\log M)^4), \qquad \max_i l_i \le k\lceil \log M \rceil.$$

Proof. Write D_v for the degree of cell v, let $L = \lceil \log M \rceil$ and $\mathcal{D}_L = \{\max_v D_v \le L\}$. Since $D_v \sim \mathrm{Bin}(d, k/M)$, $\Pr(\mathcal{D}_L^c) \le M(eC/L)^L$, which is superpolynomially small.

Lipschitz bound. For two graphs in $\mathcal{D}_L$, successively replace the edges with differing labels by the edges of the other graph. The incidence set of every intermediate graph is contained in the union of the two endpoint graphs, so no vertex degree exceeds $2L$. Replacing one edge in such an intermediate graph affects only the $O(L)$ edge labels overlapping the old or new edge, and each $m_{ij} \le k$, so Ω and Ω_2 each change by $O(L)$. For the replaced edge $\ell_i = O(L)$, whose square changes by $O(L^2)$; the ℓ_j of the other $O(L)$ affected labels change by $O(1)$, each square changing by $O(L)$. Thus Ω_ℓ changes in total by $O(L^2)$. Hence all three are CL^2-Lipschitz in the Hamming distance over edge labels on $\mathcal{D}_L$.

Concentration. For functions that are Hamming Lipschitz on the good set, the typical bounded differences inequality [37] gives concentration; here we use the form of Combes [9, Proposition 2]: extend f from $\mathcal{D}_L$ as a Lipschitz extension, truncate it to a deterministic nonnegative polynomial range, and then apply McDiarmid's inequality over the independent edges. The bad set has superpolynomially small probability, and both f and its truncated extension are at most $O(M^3)$, so the change in the expectation is also superpolynomially small; hence

$$\Pr(|f - \mathbb{E}f| > C\sqrt{M}(\log M)^4) \le \Pr(\mathcal{D}_L^c) + 2e^{-c(\log M)^4}, \qquad f = \Omega, \Omega_2, \Omega_\ell.$$

Deterministic means. For the overlap H of independent uniform k-subsets, write $h_1 = \mathbb{E}H = k^2/M$ and $h_2 = \mathbb{E}H^2 = k^2/M + k^2(k-1)^2/[M(M-1)]$. Exactly,

$$\mathbb{E}\Omega = \tbinom{d}{2} h_1, \qquad \mathbb{E}\Omega_2 = \tbinom{d}{2} h_2, \qquad \mathbb{E}\Omega_\ell = d\{(d-1)h_2 + (d-1)(d-2)h_1^2\}.$$

The last identity holds because, given S_i, the m_{ij} are independent and their conditional first and second moments do not depend on S_i. Taking the good set as the intersection of $\mathcal{D}_L$ with the three deviation events gives the stated concentration bound. Since $l_i \le kL$, the lemma follows. □

E The Gaussian limit after failure and the Gaussian rule

This appendix proves Theorem 4.2 and Corollary 4.4. Write D_v for the degree of cell v and $l_i = \sum_{j \ne i} H_{ij}$. The structure of the proof is as follows: first prove a Gaussian limit for the quadratic form in the random signs on a deterministic good graph (Lemma E.1), uniformly over good graphs; then show that the good graphs still carry almost all of the probability after conditioning on failure; for the $d = o(M)$ and overload regimes use the Gaussian limit under deterministic coefficients (Lemma E.2), which is uniform in the coefficients and holds under both the random mapping and the planted distribution; finally branch on the limit of d/M and assemble the pieces by a subsequence argument.

E.1 CLT for the random signs on a good graph

Lemma E.1. *Fix $k \ge 3$. Let a deterministic sequence of graphs satisfy $\Omega_2/M \to \beta \in (0, \infty)$ and $\max_i l_i = O(\log M)$. Then $\Omega_2^{-1/2} \sum_{i<j} H_{ij}\varepsilon_i\varepsilon_j \Rightarrow N(0,1)$, and the convergence is uniform over graphs satisfying the same set of constant bounds.*

Proof. There are at most $\Omega_2 = O(M)$ nonzero summands, each bounded by k. Two summands are dependent only when they share a key, so the maximum degree of the dependency graph is at most $2\max_i l_i = O(\log M)$. The covariance of distinct summands is zero (Appendix C), so the variance of the sum is exactly Ω_2. For a fixed order $r \ge 3$, a joint cumulant is nonzero only when the group of summands is connected in the dependency graph [22]; the number of connected groups is $O_r(M(\log M)^{r-1})$, and the cumulant of each group is bounded. Dividing by $\Omega_2^{r/2} \asymp M^{r/2}$ gives $O_r(M^{1-r/2}(\log M)^{r-1}) \to 0$. The first cumulant is zero and the second

is one; the standard Gaussian is determined by its moments, and the method of moments gives convergence. These bounds depend only on the range of β and the constant in $\max_i l_i$, so they are uniform over graphs satisfying the same set of constant bounds; if the convergence of the CDF were not uniform, one could extract a deterministic good graph along which convergence fails, contradicting the argument above.

The difference between $\widehat{d} - d$ and $\frac{2}{k(1-k/M)}$ times this sum is

$$-\frac{k}{M(1-k/M)}\Big[\Big(\sum_i \varepsilon_i\Big)^2 - d\Big],$$

which depends only on the signs and has L^2 norm $O(d/M)$, that is $O(M^{-1/2})$ after dividing by $s(d) \asymp d/\sqrt{M}$, for every graph. On the good graph of Lemma D.4, Ω_2 equals its exact mean plus $O(\sqrt{M}(\log M)^4)$, so the ratio of $\frac{4\Omega_2}{k^2(1-k/M)^2}$ to $s(d)^2$ tends to one, and the normalization by $\sqrt{\Omega_2}$ in this lemma can be replaced by normalization by $s(d)$. Hence the standardized X of $\widehat{d}$ has the same uniform Gaussian limit on the good graph. By Eq. (C.2), the difference between $\widetilde{d}$ and $\widehat{d}$ is $\frac{k}{M}[(\sum_i \varepsilon_i)^2 - \widehat{d}]$. Given the graph, $(\sum_i \varepsilon_i)^2 - d$ has L^2 norm $\sqrt{2d(d-1)}$; $\widehat{d}$ is conditionally unbiased with conditional variance at most $(M-1)s(d)^2 = 2d(d-1)$. By the triangle inequality, the conditional L^2 norm of the difference is at most $\frac{2k}{M}\sqrt{2d(d-1)}$, which is at most $2k\sqrt{M-1}/M \to 0$ after dividing by $s(d)$, so the same holds for $\widetilde{d}$. □

E.2 failure conditioning at positive load ratio

For $d \geq 2$, a specified pair of duplicate edges already forms a core, so

$$\Pr(F) \geq \Pr(S_1 = S_2) = \binom{M}{k}^{-1}.$$

By Lemma D.4, $\Pr(\mathcal{G} \notin \mathcal{G}_M \mid F) \leq \Pr(\mathcal{G} \notin \mathcal{G}_M)\binom{M}{k} \to 0$. The error of the conditional CDF is at most the uniform error on the good graph plus this conditional exceptional probability, so at positive load ratio

$$\sup_x \big|\Pr(X \leq x \mid F) - \Phi(x)\big| \to 0.$$

The same argument applies to any event that depends only on the mapping and has probability at least some fixed negative power of M. What is divided by $\Pr(F)$ here is the superpolynomially small exceptional probability; the CLT error on the good graph is simply averaged after conditioning, so the proof does not need a CLT convergence rate strong enough to overcome a rare failure. A typical floor failure is caused by a pair of duplicate edges, which only sets one H_{ij} to k; the change in Ω_2 is $O(1)$, far smaller than the deviation allowed by Lemma D.4, so it lies inside $\mathcal{G}_M$.

The conditional variance also tends to $s(d)^2$: on the good graph the ratio tends to one by Eq. (4.2) and Lemma D.4; on any graph the ratio is at most $M-1$, and the contribution of bad graphs is controlled by a superpolynomial probability.

E.3 Gaussian limit under deterministic coefficients

The following Gaussian limit holds uniformly over any deterministic coefficient vector $(s_i) \in \{\pm 1\}^d$, so it applies to random-sign encoding by using it given iid coding coefficients and then averaging over the signs.

Lemma E.2 (Gaussian limit under deterministic coefficients)**.** *For deterministic coefficients $(s_i) \in \{\pm 1\}^d$, let $C_v = \sum_{i:v\in S_i} s_i$, compute $\widehat{d}$ by Eq. (2.1), and obtain X by Eq. (4.3). If $d/\sqrt{M} \to \infty$, then $X \Rightarrow N(0,1)$, with convergence uniform in the coefficients and without requiring d/M to have a finite upper bound. Under the planted distribution Q, we likewise have $X \Rightarrow N(0,1)$, uniformly in the coefficients and the label pair.*

Proof. Write $Y = \widehat{d} - d$, $W_{ij} = H_{ij} - k^2/M$, and $v_i = \mathbf{1}_{S_i} - (k/M)\mathbf{1}$. Exposing the edges in the order of their labels, the martingale difference is $\Delta_j = \frac{2}{k(1-k/M)} s_j v_j^\top \sum_{i<j} s_i v_i$, with $Y = \sum_j \Delta_j$. The sum of conditional variances is

$$\mathcal{V}_d = s(d)^2 + \frac{8}{k(1-k/M)(M-1)} \sum_{i<\ell} (d-\ell) s_i s_\ell W_{i\ell},$$

$$\operatorname{Var} \mathcal{V}_d = \frac{64}{(M-1)^3} \sum_{\ell=2}^{d} (\ell-1)(d-\ell)^2 = O\Big(\frac{d^4}{M^3}\Big).$$

Also $s(d)^4 \approx 4d^4/M^2$, so the variance of the normalized bracket $\mathcal{V}_d/s(d)^2$ is $O(1/M)$, converging to one in L^2. Given the first $j-1$ edges, the components of the vector $\sum_{i<j} s_i v_i$ sum to zero, so $v_j^\top \sum_{i<j} s_i v_i$ equals the sum of its components over the k cells of S_j, whose fourth conditional moment is at most $k^3 \cdot \frac{k}{M}$ times the sum of the fourth powers of all components. The component on cell v, $\sum_{i<j} s_i(\mathbf{1}\{v \in S_i\} - k/M)$, is a sum of $j-1$ independent, centered, bounded terms, each with variance at most k/M; by Rosenthal's inequality [34], its fourth moment is at most $C_k(j/M + j^2/M^2)$. Hence $\mathbb{E}\Delta_j^4 \le C_k(j/M + j^2/M^2)$; summing over $j \le d$ and dividing by $s(d)^4$, the fourth-order Lyapunov remainder satisfies

$$\frac{\sum_j \mathbb{E}\Delta_j^4}{s(d)^4} \le C_k\Big(\frac{M}{d^2} + \frac{1}{d}\Big) \to 0,$$

and by the martingale CLT [20], $X \Rightarrow N(0,1)$. Both bounds depend only on $|s_i| = 1$, so the convergence is uniform in the coefficients, and no boundedness of d/M is needed.

Under the planted distribution, the statistic is invariant under label permutation, so we may fix the duplicate edge pair as 1, 2, with $S_1 = S_2$. Write $Y_{\ge 3}$ for $\widehat{d}$ computed from the remaining $d-2$ edges alone, minus $d-2$; then

$$Y = Y_{\ge 3} + 2s_1 s_2 + \frac{2}{k(1-k/M)}(s_1 + s_2) \sum_{j \ge 3} s_j \Big(|S_1 \cap S_j| - \frac{k^2}{M}\Big).$$

The last term has variance $O(d/M)$. The deterministic term $2s_1 s_2$ and the last term, divided by $s(d)$, have mean squares $O(M/d^2)$ and $O(1/d)$ respectively, both tending to zero; $Y_{\ge 3}$ is handled by the unconditional result. Combining with the total variation bound of Lemma D.2 or D.3 returns to the failure-conditional distribution. The $d = o(M)$ and overload branches of the next subsection use exactly this conclusion. □

E.4 Proof of Theorem 4.2 (the Gaussian limit after failure) and Corollary 4.4

Assume $d/\sqrt{M} \to \infty$. For any subsequence, take a further subsequence along which d/M converges in $[0, \infty]$.

When the limit is a positive finite value, use the previous subsection.

When the limit is zero, Lemma D.2 gives that the TV distance between the failure-conditional distribution and the planted distribution Q tends to zero, valid for all $2 \le d = o(M)$; it is a purely graph-theoretic result, which still holds after adding independent signs. The conclusions of Lemma E.2 under the random mapping and the planted distribution are both uniform in deterministic signs: the fourth-order Lyapunov remainder of the edge-exposure martingale is $O_k(M/d^2 + 1/d)$, and the normalized mean square of the correction term from the planted duplicate edge pair is also $O_k(M/d^2 + 1/d)$, both tending to zero when $d \gg \sqrt{M}$. Applying these uniform conclusions given iid coding coefficients and averaging over the signs yields the conditional Gaussian limit.

When the limit is infinite, eventually $d > M$. Each edge deleted in the full peeling can be assigned a cell of degree one at the moment of deletion; that cell is subsequently empty and

cannot serve again as a witness for a deletion, so full peeling requires $d \le M$. In this branch F occurs with certainty, and we directly use the conclusion of Lemma E.2 under the random mapping.

All three branches give uniform convergence of the conditional CDF to Φ, and a subsequence contradiction gives Eq. (4.5) for the full sequence. Corollary 4.4 follows from the fact that the Gaussian quantile and the standardized chi-square quantile have the same limit, together with $\{U < d\} \subseteq \{\widehat{d} < t(d)\}$. □

E.5 Monotonicity of the acceptance lower bound $t(n)$ in Definition 4.3

Here $t(n) = n + q\,s(n)$ with $q = [\chi^2_{M-1,u} - (M-1)]/\sqrt{2(M-1)}$. When $q \ge 0$, t is clearly increasing. When $q < 0$, differentiating for $n \ge 2$, $\frac{d}{dn}s(n) \le \frac{\sqrt{2}(2n-1)}{2\sqrt{n(n-1)(M-1)}} \le \frac{3}{2\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{M-1}}$, so a sufficient condition is $|q|\sqrt{2/(M-1)} < \sqrt{8}/3$. The Chernoff bound for the chi-square lower tail [26, Lemma 1] gives $|q| \le \sqrt{2\log(1/u)} \le 3.035$ (for $u \ge 0.01$), and the condition holds when $M \ge 64$. □

F Proofs of the discrete limit distribution and the Skellam rule

This appendix proves Theorems 5.3, 5.4, and 5.6. Writing the coding coefficients as iid ε_i, by Eq. (5.2) the identity $\widetilde{d} - d = \frac{2}{k}\sum_{i<j} H_{ij}\varepsilon_i\varepsilon_j$ holds sample by sample. A final subsection discusses the extension that stores several copies of the sign count per cell.

F.1 Exact moments for a specified duplicate edge pair

Let $S_1 = S_2$, with the remaining $d-2$ edges independent and uniform. When the two coding coefficients are opposite, the two edges cancel completely, leaving the remaining $d-2$ elements; when they are equal, they combine into a single edge with coefficient ± 2. For $d \ge 2$,

$$\mathbb{E}[\widehat{d} \mid S_1 = S_2] = d, \qquad \mathrm{Var}(\widehat{d} \mid S_1 = S_2) = \frac{2d(d-1)}{M-1} + 4 - \frac{4}{M-1}.$$

The variance of the opposite component is $2(d-2)(d-3)/(M-1)$, and that of the equal component is $2(d-2)(d+5)/(M-1)$; averaging and then adding the 4 arising from the difference $d \mp 2$ between the two component means gives the result.

F.2 From the true failure to the planted case

Lemma D.2 gives $\|\mathcal{L}(\mathcal{G} \mid F) - Q\|_{TV} \to 0$ for all $2 \le d = o(M)$, where Q first picks a pair of edge labels uniformly and then conditions on this pair mapping identically. On $2 \le d \le C\sqrt{M}$, both λ and d^2/M^k are $O_k(M^{1-k})$; since $b_k = \lceil 3(k-2)/2 \rceil$, $k - b_k - 3 = -\lceil k/2 \rceil \le -2$, so $dM^{k-b_k-3} = O_k(d/M^2)$, and this distance is at most $O_k(d/M^2 + M^{1-k})$. Adding independent signs and applying Eq. (5.2) do not increase the TV distance. The statistic is invariant under label permutation, so we may fix the duplicate edge pair as 1, 2; their common edge and the remaining $d-2$ edges are independent and uniform. When $d = 2$, the null distribution holds exactly under the planted case; empty sums below are treated as zero.

F.3 c bounded: three classes of events that can be ignored

The probability that the common edge of the duplicate edge pair intersects any remaining edge is $O_k(d/M)$ by the union bound. The probability that two remaining edges share at least two cells is $O_k(M^{-2})$, summing over all pairs of remaining edges to $O_k(d^2/M^2)$. Consider the intersection

graph of the remaining edges: for any three specified remaining edges, the probability that one of them intersects both of the other two is

$$\Big(1 - \binom{M-k}{k} \Big/ \binom{M}{k}\Big)^2 = O_k(M^{-2}),$$

the equality following by conditioning on the common edge: the other two edges are independent, and the intersection probability does not depend on the position of the common edge. Summing over ordered triples, the probability that the intersection graph is not a matching is $O_k(d^3/M^2)$. Hence, except for the probability in Eq. (5.6), the duplicated base edge and the ordinary edges are completely separated, and every overlap among the remaining edges is a single-cell overlap within a matching.

F.4 Poisson approximation of the overlaps among the remaining edges

Take the intersection indicator for each pair of remaining edges; the labels of the two edges form a dependency graph, where indicator sets with disjoint labels are independent. Each indicator has mean $p = 1 - \binom{M-k}{k}/\binom{M}{k} = k^2/M + O_k(M^{-2})$, and each indicator has $O(d)$ neighbors; the joint probability of two indicators sharing an edge equals p^2 exactly by the computation of the previous subsection. The local dependence bound of Arratia, Goldstein, and Gordon [2, Theorem 1] (with the neighborhood containing the indicator itself, indicators outside the neighborhood depending only on the remaining independent edges, and the third term zero) gives: the TV distance is at most a universal constant times the sum of products of means within the neighborhood plus the sum of joint probabilities over distinct neighbors, both of which are $O_k((d^2 + d^3)/M^2)$ in this model. Hence the total number of intersecting edge pairs among the remaining edges has TV distance $O_k((d^2 + d^3)/M^2)$ from $\mathrm{Pois}(\binom{d-2}{2}p)$. Replacing the parameter by $\binom{d-2}{2}k^2/M = 2\lambda_d$, the TV distance between two Poisson distributions is at most the difference of their parameters (writing the Poisson with the larger parameter as the Poisson with the smaller parameter plus an independent Poisson increment, the two differ only when the increment is nonzero), an additional error of $O_k(d^2/M^2)$.

F.5 Independence of the overlap sign marks and the Skellam limit

On the matching event, each overlap uses a pair of iid signs not shared with any other overlap, so these sign products are mutually independent and symmetric, and independent of the sign product $\varepsilon_1\varepsilon_2$ of the duplicate edge pair. Given the mapping, $\widetilde{d}$ depends only on the number of overlaps among the remaining edges and on these independent ± 1 marks. Formally, first keep the true number of overlaps, and regenerate that many independent marks for each mapping: on the matching event the conditional distribution of the old and new marked sums is the same; on the remaining mappings couple arbitrarily, the distribution difference being at most the probability of the exceptional event. Then apply the same marking kernel to the Poisson approximation of the count, which does not increase the TV distance. By the thinning property of the Poisson distribution [25], $\mathrm{Pois}(2\lambda_d)$ independent ± 1 marks split by sign into two independent $\mathrm{Pois}(\lambda_d)$ counts, and the difference of the positive and negative counts is a Skellam-distributed [35] Z_d; adding $2\varepsilon_1\varepsilon_2$ of the duplicate edge pair gives J_{λ_d} in Eq. (5.4), that is, $\widetilde{d}$ is close to the null distribution $d + J_{\lambda_d}$ Eq. (5.5). This step uses only the Poisson approximation of the unconditional count; the matching event is used only for the coupling error.

Adding the errors of the four steps gives $O_k(d/M + d^2/M^2 + d^3/M^2 + d/M^2 + M^{1-k})$. Since $d \le M$ and $k \ge 3$, every term except d/M and d^3/M^2 is absorbed by these two, yielding Eq. (5.6). When $d/\sqrt{M} \to c$, $\lambda_d \to k^2c^2/4$; the Poisson parameter is continuous, and J_{λ_d} converges in total variation to $J_{k^2c^2/4}$, giving Theorem 5.3(ii); since the ± 2 term and the Skellam term are independent, $\mathrm{Var}\, J_\lambda = 4 + 8\lambda/k^2$, and the limiting variance is $4 + 2c^2$. For (iii), as $\lambda \to 0$, $\Pr(Z \ne 0) \le 2\lambda \to 0$ and $J_\lambda \Rightarrow 2\varepsilon_1\varepsilon_2$; as $\lambda \to \infty$, $Z/\sqrt{2\lambda} \Rightarrow \mathcal{N}(0,1)$, while

$\sqrt{8\lambda}/k = \sqrt{2}\,c$, and the ± 2 term divided by $\sqrt{2}\,c$ tends to zero, so $J_{k^2c^2/4}/(\sqrt{2}\,c) \Rightarrow \mathcal{N}(0,1)$. $\widehat{d}$ has the same weak limit, because the coefficient of the AMS term in Eq. (C.2) is k/M, while $(\sum_i \varepsilon_i)^2 = O_p(d) = O_p(\sqrt{M})$. □

F.6 Sparse limit for fixed small d

Proposition F.1 (Limit when $d = o(\sqrt{M})$)**.** *Fix $k \geq 3$, and let $2 \leq d = o(\sqrt{M})$. Given failure, both $\widehat{d} - d$ and $\widetilde{d} - d$ converge weakly to the equally weighted ± 2. The conditional miss rate of the Gaussian rule tends to zero for bounded $d = 2, 3$, and to $1/2$ for fixed $d \geq 4$ or $d \to \infty$.*

Proof. Theorem 5.3 gives the equally weighted ± 2 limit of $\widetilde{d}$ when $d = o(\sqrt{M})$; by Eq. (C.2), the correction of $\widehat{d}$ is $O_p(d^2/M) = o_p(1)$, so it has the same weak limit. On the event that the edges other than the duplicate edge pair are pairwise disjoint and also disjoint from the duplicate edge pair, $\widetilde{d}$ equals $d + 2\varepsilon_1\varepsilon_2$ exactly; the conditional probability of this event tends to one. The Gaussian rule has $t(d) = d - O(s(d))$ with $s(d) \to 0$: the negative component rejects the true candidate, and the positive component accepts it. For $d = 2, 3$, the observation under the negative component is below $t(2)$, returning $+\infty$ and not counted as a miss; for $d \geq 4$, the negative component still accepts the candidate $d - 2$, causing a miss. When $d = 2$, failure is exactly equivalent to a duplicate edge, and $\Pr(\widehat{d} = 0 \mid F) = \Pr(\widehat{d} = 4 \mid F) = 1/2$ holds for every $M > k$; this also shows that no upper bound obtained by multiplying $\widehat{d}$ by a fixed constant can reduce the post-failure risk below $1/2$. □

F.7 Proof of the uniform CDF approximation

Proof. The branch $d/\sqrt{M} \to \infty$: Appendix E gives the conditional Gaussian limit of $\widetilde{d}$ after standardizing by $s(d)$. The null distribution is that of $d + J_{\lambda_d}$. As $\lambda_d \to \infty$, the Z_d in the Skellam term can be written as a sum of iid increments that grows with λ_d, and the CLT gives $(2/k)Z_d/\sqrt{8\lambda_d/k^2} \Rightarrow \mathcal{N}(0,1)$; moreover $8\lambda_d/k^2 = 2(d-2)(d-3)/M$ has ratio to $s(d)^2$ tending to one, and the ± 2 term divided by $s(d)$ tends to zero, so $J_{\lambda_d}/s(d) \Rightarrow \mathcal{N}(0,1)$. Both, after standardization, tend to the continuous Φ; when the limit is continuous, weak convergence implies uniform convergence of the CDF (Pólya's theorem), so the Kolmogorov distance tends to zero.

Uniformity: suppose there exist $\delta > 0$ and a sequence (M, d) along which the distance is at least δ. Take a subsequence with $d/\sqrt{M} \to c \in [0, \infty]$. For $c < \infty$, this subsequence eventually falls in a range $d \leq C\sqrt{M}$, where Theorem 5.3 gives TV distance tending to zero; for $c = \infty$, the previous paragraph gives the same. Both contradict δ. □

F.8 Proof of the conditional miss rate of the Skellam rule

Proof. Write G_n for the CDF of the null distribution at candidate n and Y for the observation. The rule rejects n if and only if $G_n(Y) \leq u$. The following is the standard argument for the conservativeness of test inversion with a discrete p-value [31]. If $Y \sim G_n$, then $\Pr(G_n(Y) \leq u) \leq u$: when no lattice point satisfies $G_n(y) \leq u$, the rejection event is empty; otherwise let y^* be the largest lattice point with $G_n(y) \leq u$, so the rejection event is $\{Y \leq y^*\}$ and has probability $G_n(y^*) \leq u$. The rejection region $\{Y \leq y^*\}$ is a lower set, so the difference between the rejection probability under the true distribution and under G_n is at most their Kolmogorov distance. Taking $n = d$, by Theorem 5.4 the rejection probability of the true candidate is at most $u + o(1)$ uniformly over all $d \geq 2$. The event $\{U < d\}$ implies that the true candidate is rejected, yielding Eq. (5.9). This proof requires neither that the acceptance set be an interval nor that the rejection probability of a candidate be monotone in n. □

Rejecting by $\leq u$ using the left limit $G_n(Y-) = \Pr(Y' < Y)$ (with $Y' \sim G_n$) would yield a different rule: the equally weighted ± 2 has left limit zero at -2, so the minimal value point

carrying probability $1/2$ would be rejected, and the conditional miss rate would tend to $1/2$ when $d \geq 4$ and $d = o(\sqrt{M})$. At that point G_n equals $1/2 > u$, so it is accepted.

When $d = 3$, the observation need not lie on a value point of G_3, but $\widetilde{d} \geq 1$ always. When the three coefficients have the same sign, $\sum_v C_v^2 \geq 3k$. Otherwise, label the edge whose sign differs from the other two as edge 3, so the sum of squares is $3k + 2(H_{12} - H_{13} - H_{23})$; among the k cells of edge 3, the number shared with edge 1 or edge 2 is at least $H_{13} + H_{23} - H_{12}$, so $H_{13} + H_{23} - H_{12} \leq k$, and the sum of squares is at least k. Hence $G_3(\widetilde{d}) \geq G_3(1) = 1/2 > u$, and the true candidate is always accepted.

F.9 Combining the two rules by the first-round observation

The combined rule takes the upper bound of the Gaussian rule when $\widetilde{d}$ is not below the switch point, and the upper bound of the Skellam rule otherwise. Assume that the ratio of the switch point to $\sqrt{M}$ tends to infinity as $M \to \infty$; we show that the combined rule satisfies Eq. (5.9).

Proof. Suppose there exist $\delta > 0$ and a sequence (M, d) along which the combined rule has $\Pr(U < d \mid F) \geq u + \delta$. Take a subsequence with $d/\sqrt{M} \to c \in [0, \infty]$.

For $c < \infty$, by Theorem 5.4 the distribution of $\widetilde{d} - d$ after failure and that of J_{λ_d} have distance tending to zero, and λ_d is bounded, so $\widetilde{d} = d + O_p(1) = O_p(\sqrt{M})$, and the probability of lying above the switch point tends to zero. The combined rule and the Skellam rule differ only on this event, so by Theorem 5.6 the miss probability is at most $u + o(1)$.

For $c = \infty$, a miss means that the rule selected by the observation rejected the true candidate d. The Skellam rule rejects d if and only if $(\widetilde{d} - d)/s(d)$ is below the lower u-quantile of the standardized G_d; by the proof of Theorem 5.4, $J_{\lambda_d}/s(d) \Rightarrow \mathcal{N}(0, 1)$, and this quantile tends to $\Phi^{-1}(u)$. The Gaussian rule rejects d if and only if $(\widehat{d} - d)/s(d) < (t(d) - d)/s(d)$; the right side also tends to $\Phi^{-1}(u)$, and $(\widehat{d} - \widetilde{d})/s(d) \to 0$ (Appendix C.5). By Theorem 4.2, the conditional distribution of the standardized statistic tends to the continuous $\mathcal{N}(0, 1)$, so the events on which the two rules reject d differ only on a boundary region of probability tending to zero. The rejection event of the combined rule is contained in the union of the two, and its probability is at most the rejection probability of the Skellam rule plus $o(1)$, namely $u + o(1)$.

Both cases contradict $\delta > 0$. □

F.10 Extension to multiple sign counts

If each cell stores L copies of the count, with the copies using independent sign hashes and sharing the mapping, then given $\mathcal{G}$ the $\widehat{d}$ of the copies are mutually independent, each with mean d; averaging independent copies of the estimate is exactly how AMS reduces variance [1]. From $\mathbb{E}[\widehat{d}^2 \mid \mathcal{G}] \leq 3d^2$ and $e^{-x} \leq 1 - x + x^2/2$,

$$\mathbb{E}\Big[\exp\Big(-\frac{\widehat{d}}{5d}\Big) \;\Big|\; \mathcal{G}\Big] \leq \frac{43}{50},$$

and applying the exponential Markov inequality (Chernoff bound) to $U = \frac{2}{L}\sum_\ell \widehat{d}^{(\ell)}$ gives $\Pr(U < d \mid \mathcal{G}) \leq (\frac{43}{50} e^{1/10})^L$. This bound holds for every deterministic mapping, hence for every rare failure; when $L = 272$, the right side is below 10^{-6}. When only an extra scalar AMS accumulator is stored, the L count sums are independent given d, with joint distribution a product of binomial distributions, allowing an exact discrete candidate test, and the finite-sample guarantee $\Pr(U < d \mid F) \leq u$ holds directly.

Repeating counts reduces the absolute error but does not change the ratio of the conditional scale to the unconditional scale. Fix $d \geq 4$ and L, and let $M \to \infty$: if Gaussian inversion is still applied to the average of L copies using the unconditional standard deviation $s(n)/\sqrt{L}$, the actual conditional miss rate tends to $2^{-L}\sum_{j<L/2}\binom{L}{j}$, namely $1/2, 1/4, 5/16, 93/256$ for $L = 1, 2, 4, 8$. Hence increasing the number of copies cannot replace the correct post-failure

distribution, which is why the main text uses a single count together with the discrete limit distribution.

G Exact evaluation and search cutoffs

This appendix describes the computation of the Skellam rule (Definition 5.5): the exact evaluation of $G_n(\widetilde{d})$, the statement and proof of two search cutoffs, and the numerical implementation of the exact CDF.

The guarantee of the Skellam rule does not require the accepted candidates to form an interval, and the implementation takes the largest accepted candidate as specified by the definition. The candidates have no theoretical upper bound, so an explicit cutoff is needed: the cutoff of Proposition G.2 is about 1.035 times the observation $\widetilde{d}$ at $M = 16384$ and $u = 0.01$. The search checks candidates downward from the cutoff and returns the first accepted one, matching an unbounded scan. The number of candidates requiring an exact CDF and the length of the Poisson window are both proportional to c, so the cost of a single sizing is roughly proportional to λ_d. The Poisson table depends only on (M, k) and the candidate n, and can be reused across syncs by the same receiver; measured results appear in Appendix H.9.

G.1 Computing $G_n(\widetilde{d})$ with integer thresholds

Expanding Eq. (5.5) according to the two cases $\varepsilon_1\varepsilon_2 = \pm 1$,

$$G_n(\widetilde{d}) = \frac{1}{2}\Pr\Big(Z_n \le \frac{k}{2}(\widetilde{d} - n - 2)\Big) + \frac{1}{2}\Pr\Big(Z_n \le \frac{k}{2}(\widetilde{d} - n + 2)\Big).$$

Let $R = \sum_v C_v^2$ (an integer), so $\widetilde{d} = R/k$. A random variable following G_n is at most the observation if and only if $Z_n \le \frac{k}{2}(R/k - n \mp 2)$, that is, $Z_n \le \lfloor (R - kn \mp 2k)/2 \rfloor$. Hence $G_n(\widetilde{d})$ equals

$$\frac{1}{2}\Pr\Big(Z_n \le \Big\lfloor \frac{R - kn - 2k}{2} \Big\rfloor\Big) + \frac{1}{2}\Pr\Big(Z_n \le \Big\lfloor \frac{R - kn + 2k}{2} \Big\rfloor\Big),$$

where both floors are computed over integers, so floating-point arithmetic does not change the decision at lattice boundaries.

G.2 Search cutoff from the Chernoff bound

Proposition G.1 (Chernoff cutoff)**.** *If $e^{-9M/80} < u$, then every candidate satisfying*

$$n \ge \max\Big\{4\Big(\widetilde{d} + 2\Big),\ \frac{M}{2k},\ 4\Big\} \tag{G.1}$$

is rejected. If $e^{-9M/80} \ge u$, the rule returns $+\infty$ directly.

At $M = 64$, $e^{-9M/80} \approx 7.5 \times 10^{-4}$, which holds for both $u = 0.01$ and 0.05; the branch returning $+\infty$ disappears as $M \to \infty$ for any fixed u, and does not affect Theorem 5.6.

Proof. The lower tail of the null distribution is governed by the component with shift -2, so it suffices to show that this component satisfies $\Pr(n - 2 + \frac{2}{k}Z_n \le R/k) \le e^{-9M/80}$ under Eq. (G.1). From $n \ge 4(R/k + 2)$, $R/k - n + 2 \le -\frac{3}{4}n$, so we need $Z_n \le -\frac{3k}{8}n$. The sum of the two Poisson parameters of Z_n satisfies $2\lambda_n \le k^2n^2/(2M)$. By the moment generating function of the Skellam distribution $\mathbb{E}e^{-tZ_n} = \exp\{2\lambda_n(\cosh t - 1)\}$ and the exponential Markov inequality (Chernoff bound), for $t > 0$,

$$\Pr\Big(Z_n \le -\frac{3kn}{8}\Big) \le \exp\Big\{-\frac{3kn}{8}t + 2\lambda_n(\cosh t - 1)\Big\}.$$

Take $t = M/(2kn)$; since $n \geq M/(2k)$, $t \leq 1$. Using $\cosh t - 1 \leq \frac{3}{5}t^2$ (for $0 \leq t \leq 1$, by comparing the power series term by term with t^2 and using $\cosh 1 - 1 < 3/5$), the exponent is at most $-\frac{3M}{16} + \frac{k^2n^2}{2M} \cdot \frac{3}{5} \cdot \frac{M^2}{4k^2n^2} = -\frac{3M}{16} + \frac{3M}{40} = -\frac{9M}{80}$. The condition $n \geq 4$ ensures that the expression for λ_n is valid. □

G.3 Smaller cutoff from the Bernstein bound

The coefficient 4 in Proposition G.1 makes the scan range about $3U$ candidates. The Bernstein bound for the Skellam distribution gives a smaller cutoff.

Proposition G.2 (Bernstein cutoff)**.** *If $M > 4\log(1/u)$, then every candidate satisfying*

$$n \geq \frac{\widetilde{d} + 2 + \frac{4}{3k}\log(1/u)}{1 - 2\sqrt{\log(1/u)/M}} \tag{G.2}$$

is rejected.

At $M = 16384$ and $u = 0.01$, the coefficient of the observation $\widetilde{d}$ in Eq. (G.2) is about 1.035, reducing the scanned candidates from about $3U$ to one or two hundred; the number of candidates requiring an exact CDF is unchanged. The search checks downward one by one, starting from the smaller of the two cutoffs minus one.

Proof. For $0 < t < 3$, $\log \mathbb{E}e^{-tZ_n} = 2\lambda_n(\cosh t - 1) = 2\lambda_n \sum_{j\geq 1} t^{2j}/(2j)! \leq \lambda_n t^2 \sum_{j\geq 0}(t/3)^j = \lambda_n t^2/(1 - t/3)$, because $2/(2j)! \leq 3^{-(2j-2)}$. This shows that $-Z_n$ is a sub-gamma variable with variance factor $2\lambda_n$ and scale $1/3$; the standard Bernstein argument [6] gives

$$\Pr(Z_n \leq -x) \leq \exp\Big[-\frac{x^2}{4\lambda_n + 2x/3}\Big], \qquad x > 0.$$

A sufficient condition for the right side to be at most u is $x \geq 2\sqrt{\lambda_n \log(1/u)} + \frac{2}{3}\log(1/u)$. Taking $x = \frac{k}{2}(n - R/k - 2)$ and using $\lambda_n \leq k^2n^2/(4M)$, the sufficient condition becomes $\frac{k}{2}(n - R/k - 2) \geq kn\sqrt{\log(1/u)/M} + \frac{2}{3}\log(1/u)$, which rearranges to Eq. (G.2); $M > 4\log(1/u)$ ensures the denominator is positive. When $\lambda_n = 0$, the lower tail is zero and the conclusion holds directly. □

Numerically, the cutoff is computed at risk $u(1 - 10^{-9})$, below the $u - 10^{-12}$ used by the acceptance test in the implementation, and one is added after rounding up. Hence rounding errors cannot place the cutoff above a candidate that should be accepted.

G.4 Per-candidate Chernoff prescreening

For each candidate, if $j = \lfloor (R - kn + 2k)/2 \rfloor < 0$, first compute the Chernoff bound of the Skellam lower tail, namely the exponential Markov bound of the moment generating function above at the optimal parameter $t = \operatorname{asinh} r$, where $r = -j/(2\lambda_n)$:

$$\Pr(Z_n \leq j) \leq \exp\Big\{-\Big[(-j)\operatorname{asinh} r - 2\lambda_n\big(\sqrt{1 + r^2} - 1\big)\Big]\Big\}.$$

Of the two components, the one with shift -2 has the larger CDF; if its Chernoff bound is already at most $u(1 - 10^{-9})$, the average is at most u, and the candidate is rejected directly. Prescreening rejects only candidates already proven to be rejected, and turning prescreening off and computing exact CDFs for all candidates gives identical results one by one.

G.5 Windowed Poisson table for the exact CDF

$\Pr(Z_n \le j) = \sum_m \Pr(P_2 = m) \Pr(P_1 \le j + m)$, with P_1, P_2 independent $\mathrm{Pois}(\lambda_n)$. The implementation builds one Poisson table per candidate, containing only the window $[\lambda_n - w, \lambda_n + w]$ with $w = 14\sqrt{\lambda_n} + 40$: starting from the mode, the pmf is recurred outward on both sides by $p(i+1) = p(i)\lambda_n/(i+1)$, normalized within the window, and accumulated to obtain the CDF. By the Chernoff bound for the Poisson tail, the Poisson mass outside the window on each side is below 10^{-40}, smaller than double-precision rounding; normalization also avoids the relative error of applying $\log\Gamma$ term by term at large λ_n. The initial implementation built the table from zero up to $\lambda_n + 14\sqrt{\lambda_n}$; at $M_1 = 16384$ and load 3, λ_n is about 2×10^5 to 4×10^5, a single sizing taking about 1.3 s and exhausting memory in the statistical experiments; after windowing a single sizing takes about 36 ms, with identical results one by one.

The Poisson table depends only on (M, k) and n, and can be shared across syncs and risk levels. The implementation maintains a cache keyed by (M, k); when full, it evicts the least recently used tables until half the capacity is free; the cache capacity is measured in entries, each entry being two float64.

The cutoffs, prescreening, and windowed table above do not change the output of the rule. We compared against a reference implementation with 70 decimal digits case by case (4536 general cases and 286 bounded-c cases, covering lattice points, parity boundaries, and $+\infty$), and compared the windowed implementation with cutoffs against the full-table implementation on 9779 cases, with identical results one by one.

H Experimental Setup and Supplementary Results

This appendix first states the validation goal and how each experiment is run, then gives the setup and supplementary results item by item in the order of Table H.1.

H.1 Validation Goals and Experiment Overview

The experiments of Section 6 verify the following conclusions. The lower-tail shape of the post-failure statistic (Theorems 5.3 and 4.2) and the conditional miss rate and extra capacity of the two rules at finite size (Theorem 5.6 and Corollary 4.4) are measured by the statistical experiments (Exp-STAT). For some configurations of the statistical experiments, sampling draws from the planted distribution in place of the failure-conditioned distribution, and the planted sampling control (Exp-PLANT) tests the finite-size error of this substitution (Lemma D.2). Random-sign encoding does not change decoding (Proposition 3.1), verified by the decoding-equivalence test (Exp-DEC) on the full IBLT; the post-failure estimate is conditionally unbiased given the mapping for every sign composition (Propositions 4.1 and 3.2), verified by the conditional-unbiasedness test (Exp-UNB). The conditional miss rate of the Skellam rule, beyond the statistical experiments, is tested on independent samples by the worst-configuration test (Exp-WORST) on the configurations closest to the risk upper bound. The reliability and communication cost of the two-round protocol (Theorem 5.7) are measured by the protocol experiments (Exp-PROTO), and the time and memory of sizing and decoding are given by the computation-cost measurement (Exp-COST).

The configurations, seeds, sampling sources, stopping rules, and budgets of the statistical experiments, the worst-configuration test, the planted sampling control, and the protocol experiments were frozen before launch based on a single feasibility pilot; the selection rule for the worst-configuration test was fixed before the statistical experiments began, and the specific configurations and sample sizes were derived from that rule after the statistical experiments finished (Appendix H.7). The conditional-unbiasedness test and the decoding-equivalence test come from separate independent runs, and the computation-cost measurement was carried out separately from the other runs (Appendix H.9).

Table H.1: Experimental setup. Tags are referenced by the main text and appendix; the last column lists the main-text and appendix subsections where the results appear.

Experiment	Goal	Parameter range	Samples and stopping rule	Location
Exp-STAT statistical experiments	Lower-tail shape (Theorems 5.3, 4.2); conditional miss rate and extra capacity (Theorem 5.6)	$k = 3, 4, 5$, $M \in \{256, 1024, 4096, 16384\}$, $c \in \{0.5, \ldots, 64\}$, load $\in \{0.95, \ldots, 3\}$ and $d = 2, 3, 6$, load at most 3; $M = 64$ boundary under the same rule	189 + 42 configurations, about 20,000 failures each; fixed number of trials	6.1, 6.3, 6.4; H.2, H.6, H.7
Exp-PLANT planted sampling control	Finite-size error of the planted distribution (Lemma D.2)	Critical-load group: $k = 3, 4, 5$, $M = 64$–1024, load 0.85–0.95, direct failure probability at least 10^{-3}; low-load group: for $k = 3$, $M = 1024$–16384, load 0.04–0.61; for $k = 4, 5$, $M = 256, 1024$, load 0.04–0.71	Critical-load group: 21 configurations, 65,000 failures each for direct and planted; low-load group: 12 configurations, direct trial count fixed by λ, about 65,000 failures for $k = 3$ and about 20,000 to 65,000 for $k = 4, 5$, 65,000 planted each	6.1; H.3
Exp-DEC decoding-equivalence test	Decoding equivalence (Proposition 3.1)	Grid: $k \in \{3, 4, 5, 6, 10\}$, $M \in \{64, 1024, 16384\}$, load 0.5–3, five sign compositions; near threshold: $M \in \{1024, 16384, 65536\}$, load 0.90–1.05; rejected candidates: $M = 16384$, load 0.5–1, nine sign compositions	Grid: 375 configurations, 500,000,000 paired trials in total; near threshold: 13,000,000; rejected candidates: 6,750,000	6.2; H.4
Exp-UNB conditional-unbiasedness test	Conditional unbiasedness (Propositions 4.1, 3.2)	$k = 3$–6, $M = 64$–262144, critical window and its left side with load 0.4–0.95; Figures 6.1, 6.2: $M = 16384$, failure probability about 0.1	203 configurations; Figure 6.1: 4,000,000 mappings, Figure 6.2: 4 configurations	6.2; H.5
Exp-WORST worst-configuration test	Risk guarantee on the worst configurations (Theorem 5.6)	12 configurations (Table H.8); another 10 configurations for Figure 6.7 under the same design	56,000–330,000 failures per configuration; stop when the block-ordered cumulative failure count reaches the target	6.4, 6.5; H.7
Exp-PROTO protocol experiments	Protocol reliability and communication cost (Theorem 5.7)	$k = 3, 4, 5$, $M_1 = 256$–16384, load $\in \{0.95, 1, 1.05, 1.5, 3\}$, one-sided and balanced, $u \in \{0.01, 0.05\}$	120 configurations, 13,940,000 trials in total per level of u	6.5; H.8
Exp-COST computation-cost measurement	Time and memory of sizing and decoding	$M = 16384$, $k = 3, 4, 5$, c from 4 to 314; protocol load 1.0–3	10 repetitions per item; candidate-check counts run separately, 30 trials per configuration	6.6; H.9

H.2 Sampling Method

The statistical experiments choose the sampling source by the cost estimated in the pilot: configurations whose 20,000 direct failures are estimated to take at most 20 CPU core-hours use direct sampling, with the number of trials set to $\lceil 2 \times 10^4 / \widehat{\Pr}(F) \rceil$; the remaining configurations use 20,000 planted samples. Figure H.1 shows the position and sampling source of each configuration on the two axes of c and load. The six points at load 3 ($d = \mathrm{round}(3\alpha_c M)$, where rounding makes the load slightly above 3) were run separately with independent seeds after the statistical experiments finished, with the same sampling method and target number of failures as the other configurations.

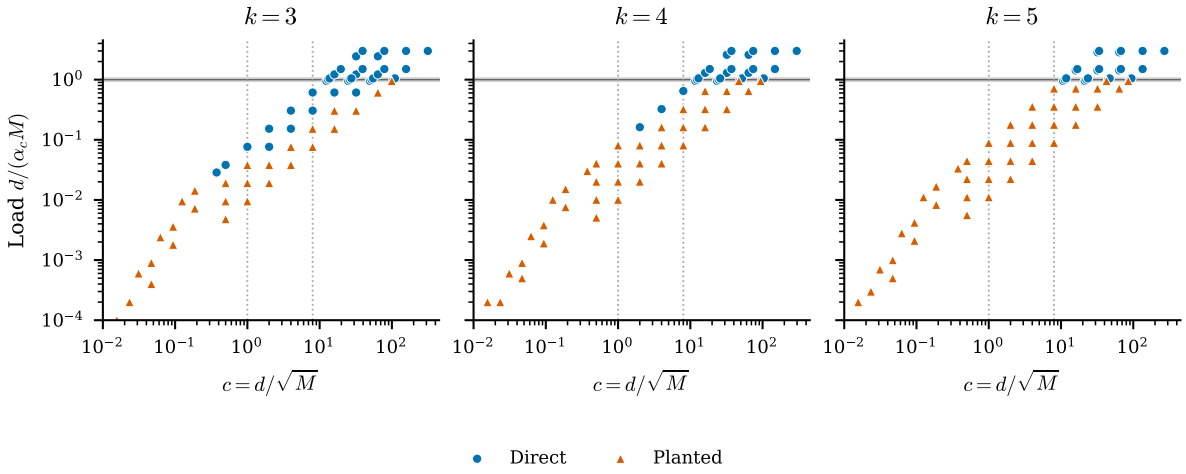


Figure H.1: The 189 configurations of the statistical experiments. Point shape distinguishes the sampling source; vertical lines are the grouping boundaries of c, and the horizontal shading marks load 0.95–1.05.

Planted sampling uniformly selects a pair of difference keys, makes them share a uniformly random k-cell set, generates the mappings of the remaining keys independently and uniformly, and draws the signs independently. This construction produces a peeling failure every time. Direct sampling needs on average $1/\Pr(F)$ mappings to obtain one failure, while planted sampling directly generates a mapping drawn from Q; its asymptotic relation to the true failure-conditioned distribution is given by Lemma D.2, and the finite-size comparison is in Appendix H.3. Under iid random signs the statistic is label-invariant, so these experiments equivalently fix the last two labels as the duplicate edge pair.

The runs used by the conditional-unbiasedness test implement direct sampling and splitting on an NVIDIA H100, combined with CPU sampling. The GPU direct sampler processes one graph per thread block and executes peeling in synchronized rounds; each round first identifies degree-one cells, then claims and deletes the corresponding edges through atomic operations, ensuring each edge is deleted once. Random numbers use Philox, organized by seed and trial index, so the random stream is independent of the GPU block partition. The CPU implementation re-peels the same exported batch of mappings independently and compares the final residual core graph by graph; the initial direct sampler found no difference on 14,100 graphs, and the failure-probability statistical controls on 15 configurations all passed the preset criterion. Synchronized peeling and sequential CPU peeling have the same final core, and path statistics that depend on the peeling order are handled by the CPU implementation.

H.3 Finite-Size Error of the Planted Distribution

The planted sampling control (Exp-PLANT) compares the failure samples generated at finite size by the planted distribution and the failure-conditioned distribution (Lemma D.2), comparing three quantities: the fraction of pure duplicate edge pairs among direct failures, the full-range KS distance of the post-failure X on the two samples, and the miss/u of the Skellam rule on the two samples. The KS distance is compared with the asymptotic 95% critical value, which is 0.0075 with 65,000 failures on each side and 0.011 with 20,000 on the direct side; since X takes discrete values, this comparison is conservative. The control is split into two groups. The 12 configurations of the low-load group are taken from the planted configurations of the statistical experiments and directly test the planted samples used by the main text; the 21 configurations of the critical-load group have load 0.85 to 0.95, where direct failures are common, and are used to observe how the planted approximation departs with M. Figure H.2 summarizes the two groups.

Most of the 105 planted configurations of the statistical experiments are far from the critical load, so the expected number of duplicate edge pairs $\lambda = \binom{d}{2}/\binom{M}{k}$ is small, and by Lemmas D.2 and D.3 the failure probability is of the same order as λ. The low-load group selects from these the 12 configurations that direct sampling can afford, with the direct-sampling trial count fixed by λ before sampling (Table H.2). The four configurations with $k = 3$ cover $c = 1$ to 64 and load 0.04 to 0.61, with about 65,000 failures on each side: at $u = 0.05$ the equivalence tests with bound ± 0.1 and one-sided level 0.05/4 all pass, and at $u = 0.01$ the difference intervals all contain zero; the KS distances are 0.0048 to 0.0079, three below the critical value, and $(3, 4096, 1024)$ is 0.0079 ($p = 0.034$), giving no detectable difference after correction for 12 tests. The eight configurations with $k = 4, 5$ cover $c = 0.5$ to 16 and load 0.04 to 0.71, where failures are rarer, with about 20,000 direct failures each, of which $(k, M, d) = (4, 256, 16)$ and $(5, 256, 128)$ have about 65,000, and 65,000 planted samples. At $u = 0.05$ the point estimates of the differences are all within ± 0.04 and the 95% intervals all contain zero; the equivalence tests with bound ± 0.1 and one-sided level 0.05/8 pass in 5 cases, and the interval endpoints of the other 3 ($(4, 1024, 512)$, $(5, 256, 32)$, $(5, 256, 64)$) are within ± 0.13. At $u = 0.01$ the difference intervals of 7 configurations contain zero, and $(4, 1024, 512)$ is -0.16 (interval $[-0.30, -0.02]$), meaning the planted sample has higher miss. The KS distances are 0.0048 to 0.0071, all below their respective critical values. The fraction of pure duplicate edge pairs among direct failures over the 12 configurations is 0.988 to 1.000, consistent with the mechanism of Lemma D.2.

Table H.2: Direct and planted sampling for the low-load group (Skellam rule, $u = 0.05$; miss/u is direct / planted, difference is the 95% interval of direct minus planted; KS is the full-range KS distance of X, with 95% critical value 0.0075 at 65,000 direct failures and 0.011 at 20,000).

k	M	d	c	load	direct failure count	pure duplicate edge pairs fraction	miss/u	difference	KS
3	1024	32	1	0.04	65061	1.000	0.94 / 0.94	$[-0.05, +0.04]$	0.0059
3	4096	512	8	0.15	64977	1.000	0.95 / 0.96	$[-0.06, +0.03]$	0.0048
3	4096	1024	16	0.31	65136	0.999	0.92 / 0.94	$[-0.07, +0.03]$	0.0079
3	16384	8192	64	0.61	65530	0.999	0.98 / 0.99	$[-0.06, +0.04]$	0.0051
4	256	8	0.5	0.04	20071	0.998	0.54 / 0.52	$[-0.03, +0.07]$	0.0071
4	256	16	1	0.08	65839	0.995	0.68 / 0.69	$[-0.05, +0.03]$	0.0057
4	1024	128	4	0.16	20067	0.997	0.95 / 0.95	$[-0.06, +0.07]$	0.0049
4	1024	256	8	0.32	20265	0.994	0.98 / 0.97	$[-0.06, +0.08]$	0.0048
4	1024	512	16	0.65	20038	0.988	0.94 / 0.97	$[-0.10, +0.04]$	0.0062
5	256	32	2	0.18	20137	0.999	1.00 / 0.98	$[-0.05, +0.09]$	0.0068
5	256	64	4	0.36	19885	0.997	0.91 / 0.95	$[-0.11, +0.03]$	0.0048
5	256	128	8	0.71	65470	0.995	0.99 / 1.01	$[-0.07, +0.02]$	0.0063

Direct sampling is infeasible for the remaining planted configurations: among the 77 planted configurations with $k = 4, 5$, 76 have λ below 10^{-6} with the smallest about 10^{-19}, and the feasibility pilot observed no failure in 68 of 75 configurations; collecting 65,000 failures needs at least 6.5×10^{10} trials, and about half of the configurations need more than 10^{15}, beyond the feasible range of direct sampling on CPU and GPU. On these configurations the distance between the planted distribution and the failure-conditioned distribution is controlled by Lemma D.2, whose error term tends to zero when $d = o(M)$; the region where direct sampling is hardest is precisely the region where this approximation is most accurate.

The critical-load group observes the applicable boundary of the planted approximation. The statistical experiments have five planted configurations at load 0.95, all with $M \geq 4096$; this group takes smaller M and compares the two samples in the region where failure is mostly caused by larger stopping sets. The sample size is determined by the equivalence test on the miss/u difference at $u = 0.05$: bound ± 0.1, one-sided level $0.05/21$, power 80% when the difference is zero, and 65,000 failures on each side. Table H.3 gives the results for these 21 configurations.

Table H.3: Direct and planted sampling for the critical-load group (Skellam rule; miss/u is direct / planted; KS is the full-range KS distance of X, 95% critical value 0.0075).

k	M	load	pure duplicate edge pairs	$u = 0.01$		$u = 0.05$		KS
			fraction	miss/u	difference	miss/u	difference	
3	64	0.86	0.09	0.44 / 0.48	-0.04 ± 0.07	0.87 / 0.88	-0.01 ± 0.04	0.0026
3	64	0.90	0.05	0.49 / 0.52	-0.03 ± 0.08	0.82 / 0.87	-0.05 ± 0.04	0.0085
3	64	0.95	0.02	0.44 / 0.51	-0.07 ± 0.07	0.87 / 0.94	-0.07 ± 0.05	0.0088
4	64	0.85	0.01	0.56 / 0.49	$+0.08 \pm 0.08$	0.89 / 0.92	-0.03 ± 0.05	0.0105
4	64	0.89	0.01	0.54 / 0.65	-0.11 ± 0.08	0.89 / 0.96	-0.07 ± 0.05	0.0097
4	64	0.95	0.00	0.39 / 0.55	-0.16 ± 0.07	0.81 / 0.93	-0.13 ± 0.04	0.0131
5	64	0.85	0.00	0.58 / 0.69	-0.11 ± 0.09	0.96 / 1.07	-0.11 ± 0.05	0.0130
5	64	0.89	0.00	0.57 / 0.69	-0.12 ± 0.09	1.00 / 1.13	-0.13 ± 0.05	0.0142
5	64	0.96	0.00	0.55 / 0.72	-0.16 ± 0.09	0.89 / 1.04	-0.15 ± 0.05	0.0118
3	256	0.85	0.63	0.75 / 0.62	$+0.13 \pm 0.09$	0.89 / 0.88	$+0.01 \pm 0.04$	0.0063
3	256	0.90	0.13	0.75 / 0.66	$+0.09 \pm 0.09$	0.95 / 0.88	$+0.07 \pm 0.05$	0.0080
3	256	0.95	0.02	0.77 / 0.64	$+0.14 \pm 0.09$	0.92 / 0.91	$+0.01 \pm 0.05$	0.0060
4	256	0.90	0.01	0.71 / 0.70	$+0.01 \pm 0.09$	0.93 / 0.98	-0.05 ± 0.05	0.0045
4	256	0.95	0.00	0.72 / 0.73	-0.01 ± 0.09	0.93 / 0.97	-0.05 ± 0.05	0.0052
5	256	0.90	0.00	0.71 / 0.75	-0.04 ± 0.09	0.92 / 0.94	-0.02 ± 0.05	0.0052
5	256	0.95	0.00	0.73 / 0.80	-0.07 ± 0.09	0.95 / 0.94	$+0.00 \pm 0.05$	0.0027
3	1024	0.85	0.97	0.79 / 0.78	$+0.01 \pm 0.10$	0.92 / 0.91	$+0.02 \pm 0.05$	0.0033
3	1024	0.90	0.96	0.83 / 0.86	-0.03 ± 0.10	0.93 / 0.95	-0.02 ± 0.05	0.0037
3	1024	0.95	0.08	0.84 / 0.84	$+0.01 \pm 0.10$	0.96 / 0.96	-0.00 ± 0.05	0.0091
4	1024	0.95	0.00	0.89 / 0.91	-0.02 ± 0.10	0.97 / 0.94	$+0.03 \pm 0.05$	0.0054
5	1024	0.95	0.00	0.85 / 0.79	$+0.06 \pm 0.10$	0.96 / 0.95	$+0.01 \pm 0.05$	0.0039

On the five tested $M = 1024$ configurations, the equivalence tests with bound ± 0.1 all pass at $u = 0.05$, and the difference intervals at both levels of u contain zero, including configurations where the fraction of pure duplicate edge pairs among direct failures is 0.97 and configurations where it is close to zero. Among the nine $M = 64$ configurations, seven fail to confirm the difference within the bound, on which the planted sample has higher miss, with the difference interval reaching -0.20; the two configurations with $k = 3$, $M = 256$ agree at $u = 0.05$, and at $u = 0.01$ the direct sample has miss/u higher by about 0.13 (interval $[+0.04, +0.22]$).

The KS distance decreases as M grows. For the five $M = 1024$ configurations it is 0.0033 to 0.0091, four below the critical value; the configuration $(k, M, d) = (3, 1024, 796)$ that exceeds it differs in the middle of the distribution, with a maximum difference of only 0.0039 in the lower tail between the 0.2% and 15% quantiles of the pooled sample. For the seven $M = 256$ configurations it is 0.0027 to 0.0080, six below the critical value; for the nine $M = 64$ configurations it is

0.0026 to 0.0142, eight above the critical value. The configuration among $M = 1024$ with pure-duplicate-edge-pair fraction 0.97 and those close to zero have similarly small KS distance: when the failure mechanism differs the distribution of X can still be close, and the pure-duplicate-edge-pair fraction does not suffice to predict the distribution error of X, which also depends on the overlap fluctuation on the $s(d)$ scale.

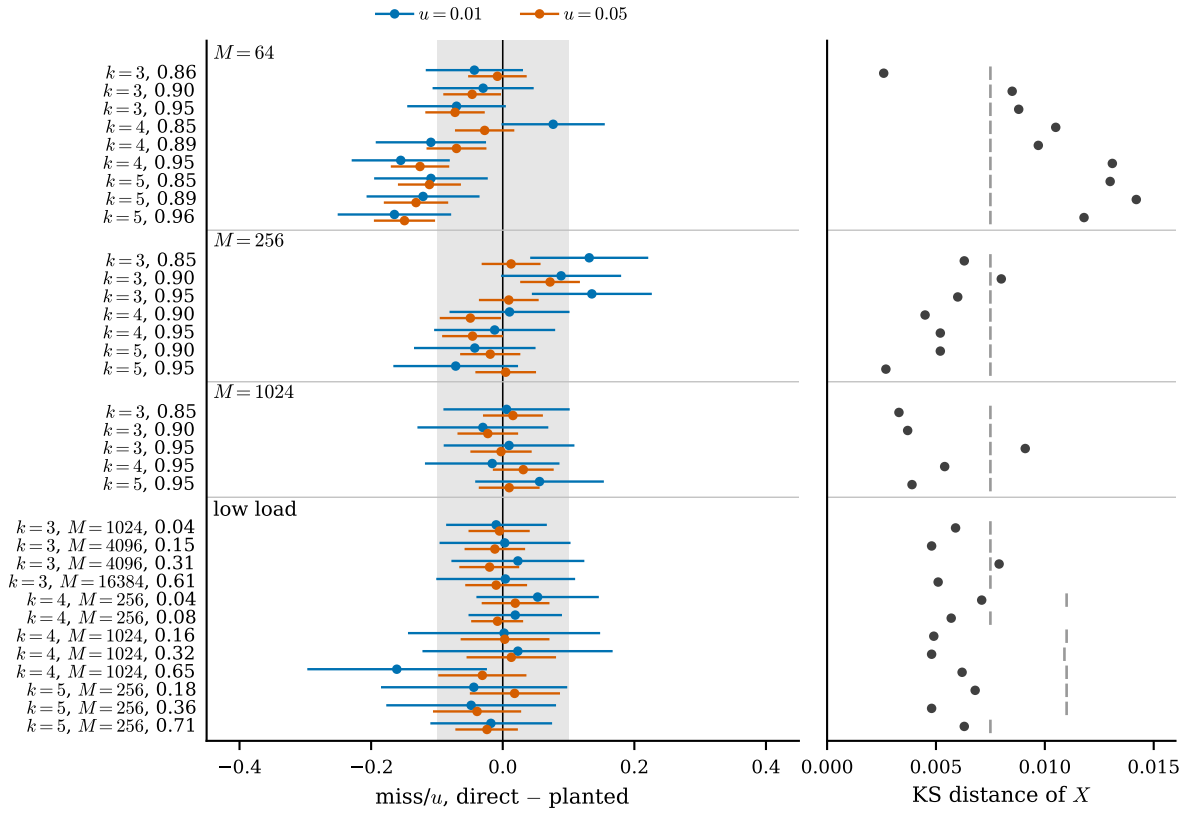


Figure H.2: Comparison of direct and planted sampling. The upper three panels are the critical-load group (grouped by M), and the bottom is the low-load group; the vertical axis labels k and load. Left: the miss/u difference of the Skellam rule (direct minus planted) with its 95% interval, shaded ± 0.1; right: the full-range KS distance of X, with gray ticks marking the 95% critical value at that sample size.

H.4 Random-Sign Encoding Does Not Change Decoding

The decoding-equivalence test (Exp-DEC) uses the real IBLT encoder and decoder of the two-round protocol directly, builds the two parties' tables and subtracts them, and compares the first-round decoding of standard encoding and random-sign encoding. The test consists of three groups of paired trials, which use the same construction and verification.

Each trial regenerates non-repeating keys and the seeds of the cell hash, sign hash, and checksum hash; the two encodings share the same batch of keys and hashes. We first verify cell by cell that the three fields after cancellation of the common elements equal the result of encoding the difference directly, then run decoding. The independent reference uses only the unsigned incidence graph of the true difference and computes the residual 2-core by the degree-one rule. Whenever the actual decoder accepts a key, it checks the true residual degree, key, and direction and updates the reference state; the checks cover the trajectories of both eventual success and failure. After termination, we verify that the recovered key set is exactly the difference outside

the 2-core with all directions correct, reconstruct the three fields from the true residual and compare cell by cell, and check that all fields are zero if and only if the residual core is empty.

The first group is a grid covering the whole load range: $k \in \{3, 4, 5, 6, 10\}$, $M \in \{64, 1024, 16384\}$, load $d/(\alpha_c M) \in \{0.5, 0.95, 1, 1.05, 3\}$, and positive-direction ratio d_+/d with target values $0, 1/4, 1/2, 3/4, 1$; d and d_+ are rounded to integers from the target values. Each k uses its corresponding peeling threshold, giving 375 configurations in total. The number of common elements is $\min\{1000, M\}$. At $M = 64, 1024, 16384$ each configuration runs 3,000,000, 900,000, and 100,000 trials respectively, 500,000,000 in total, and 100,000,000 for each k.

Table H.4 gives the results by k. Over all configurations, the numbers of paired success/failure disagreements, non-pure acceptances, incorrect recoveries, and residual-core disagreements are all zero. The two decoders together checked 211,516,840,988 acceptance operations, all satisfying the true degree-one and direction conditions.

The second group densifies the load near the threshold, and Table 6.1 of the main text comes from this group. The configurations are $k \in \{3, 4, 5, 6, 10\}$, $M \in \{1024, 16384, 65536\}$, load taking 10 values between 0.90 and 1.05, and $d_-/d \in \{0, 1/2\}$. Each configuration runs 100,000, 20,000, and 10,000 trials at $M = 1024, 16384, 65536$ respectively, 13,000,000 in total, all checks passing. At $M = 16384$ the failure probability at load 0.99 is 0.020 to 0.060 and at 1.01 is 0.993 to 0.998; at $M = 65536$ load 0.99 and 1.01 give almost no failure and almost all failure respectively. On failure the decoder still recovers the difference outside the 2-core, with an average recovery fraction at $M = 65536$, load 1.02 of 0.18 ($k = 10$) to 0.50 ($k = 3$).

The third group of paired trials counts the rejected candidates separately for each encoding, namely the cells whose count is ± 1 but that fail the checksum or the cell-ownership check. The configurations are $M = 16384$, $k \in \{3, 4, 5, 6, 10\}$, load $0.5, 0.95, 1$, and d_-/d taking nine values from 0 to 1 in steps of 1/8, with 50,000 trials per configuration, 6,750,000 in total, all checks passing. Figure H.3 gives the results for $k = 3$ to 6. Standard encoding has no rejected candidates under one-sided differences because the count equals the degree; the closer the difference is to balanced, the more opposite-sign cancellation occurs and the more rejected candidates there are. The number of rejected candidates under random-sign encoding differs by at most 0.3% across the nine sign compositions, and equals the value of standard encoding in the balanced case, consistent with Proposition 3.2: the coding coefficient is an iid random sign independent of (d_+, d_-). The results for $k = 10$ are the same, with both encodings giving about 1.27 rejected candidates per difference key at load 1. Across the three groups the two decoders together rejected 2.8×10^{11} non-pure candidates with count ± 1 without a single false acceptance; the cell-ownership check rejected zero candidates, and all wrong candidates were rejected at the checksum step.

Table H.4: First-round paired decoding results.

k	paired trials	successes	failures
3	100,000,000	36,532,515	63,467,485
4	100,000,000	35,900,798	64,099,202
5	100,000,000	33,530,250	66,469,750
6	100,000,000	32,553,378	67,446,622
10	100,000,000	30,233,256	69,766,744

H.5 The Post-Failure Estimate Is Conditionally Unbiased over the Sign Composition

Among the 203 configurations of the conditional-unbiasedness test (Exp-UNB), 196 have at least 1000 failure samples. On these configurations the $|\mathbb{E}[X \mid F]|$ under random-sign encoding is

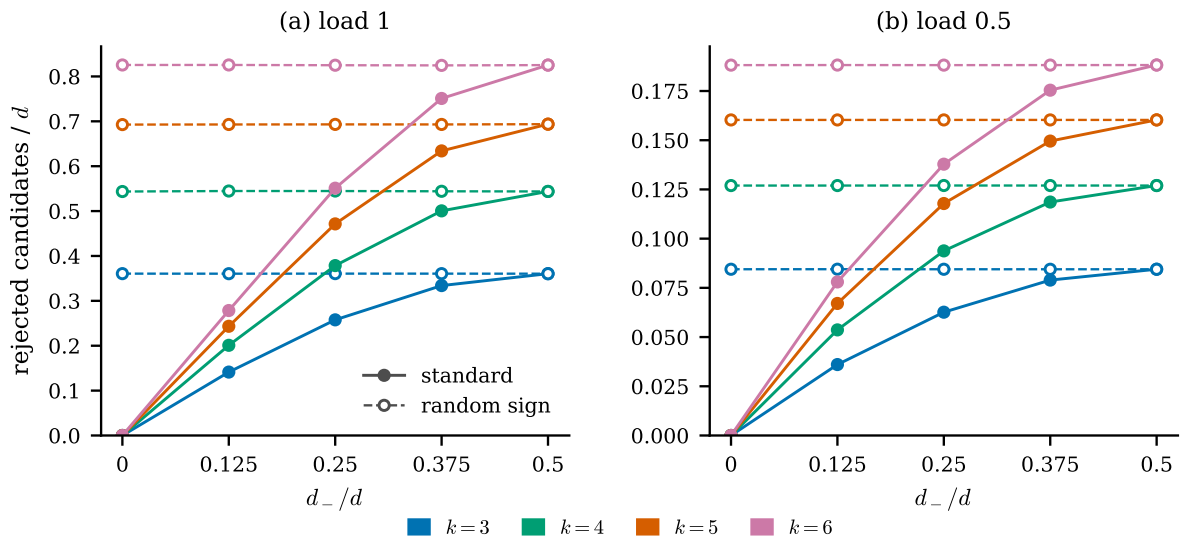


Figure H.3: The number of rejected candidates per first-round decoding divided by d ($M = 16384$, the two encodings share the same batch of mappings on the same configuration). Solid lines with filled markers are standard encoding, dashed lines with open markers are random-sign encoding.

at most 0.032, with 93% within two standard errors, consistent with the exact unbiasedness of Proposition 4.1; the conditional variance to the left of the critical window is 0.972–1.049. Among the 84 direct-sampling configurations of the statistical experiments, $|\mathbb{E}[X \mid F]|$ under random-sign encoding is at most 0.019, with 98% within two standard errors. In the statistical experiments at load 1.0 and failure probability about 0.67, the conditional mean under standard encoding with one-sided differences is $+0.20$, $+0.02$, and -0.08 for $k = 3, 4, 5$ (standard error 0.007).

Table H.5 lists the values behind Figure 6.2 of the main text. For each (k, M, d) it records the five compositions under both encodings, with the ten datasets sharing the same batch of mappings and both encodings using all failure records; the failure probability of the four configurations is 0.099–0.102. The 20 values under random-sign encoding all have absolute value at most 0.005, all within two standard errors of zero. The data of Figure 6.1 are generated by the sampling code of the interactive experiment page Peeling Lens, which computes X for three encodings on each random mapping, with the random signs from an independent generator. This configuration has $c \approx 104$. Under standard encoding $\mathbb{E}[X \mid F] = +0.599 \pm 0.001$ for one-sided differences and -0.002 for balanced differences; under random-sign encoding it is $+0.001 \pm 0.001$, with conditional variance 1.007, and the failure samples are close in distribution to the full sample, consistent with the large-c form described by Theorem 4.2.

H.6 Distance between the Post-Failure Lower Tail and the Limit Distribution

This section uses the failure records of the statistical experiments (Exp-STAT). Figure 6.3 computes $\widetilde{d} = \sum_v C_v^2/k$ from the raw integer sums of squares, keeps all empirical value points, and standardizes them on the same scale as G_d of the main text. The KS distance is computed on the left and right sides of all value points of the two distributions. By Eq. (C.2) of Appendix C.5, the difference between $\widehat{d}$ and $\widetilde{d}$ is about $0.02s(d)$ at $M = 16384$. This perturbation is small relative to the standard deviation but shifts the discrete atoms, so the distribution check uses the $\widetilde{d}$ corresponding to Theorem 5.3.

Table H.5: Values behind Figure 6.2: the mean of the post-failure X ($M = 16384$, failure probability about 0.1). Standard errors: 0.0018 for $k = 3$, 0.0031–0.0032 for $k = 4, 5, 6$.

			d_-/d				
k	d	encoding	0 (one-sided)	0.1	0.25	0.4	0.5 (balanced)
3	13306	standard	+0.638	+0.405	+0.158	+0.028	+0.003
		random sign	+0.002	−0.001	0.000	−0.004	+0.002
4	12567	standard	+0.084	+0.052	+0.016	−0.002	−0.003
		random sign	−0.001	−0.002	+0.005	+0.001	−0.001
5	11418	standard	−0.218	−0.137	−0.054	−0.009	+0.002
		random sign	+0.003	−0.005	0.000	−0.002	−0.004
6	10363	standard	−0.387	−0.244	−0.096	−0.021	−0.004
		random sign	+0.002	+0.003	+0.005	+0.001	+0.001

Table H.6 computes the full KS distance from the raw integer records of the statistical experiments. The discrete-limit column compares $\widetilde{d}$ with G_d, taking $c = 1, 2, 4$ and $k = 3, 4, 5$, nine configurations for each M, with direct and planted samples for $M \le 1024$ and planted samples for $M \ge 4096$; the Gaussian control in the same column uses $(\widetilde{d} - d)/s(d)$. The Gaussian column on the right compares $X = (\widehat{d} - d)/s(d)$ with $N(0, 1)$, taking the 75 direct-sampling configurations with $c \ge 8$, including the critical window and overload. Each configuration has about 20,000 failures, and at a fixed sample size the per-configuration 95% error bound of DKW is about 0.010. The discrete approximation has visible finite-size error on small tables, decreasing overall as M grows; the quantile error of the Gaussian also decreases as M grows.

Table H.6: Full KS distance after failure. The discrete limit uses $\widetilde{d}$ and the Gaussian uses $\widehat{d}$. The quantile error is in units of $s(d)$, taken as the maximum absolute value over each configuration.

	discrete limit ($c = 1, 2, 4$)			Gaussian ($c \ge 8$, direct sampling)		
M	KS	KS of Gaussian ($c = 1$)	configs	KS	1% quantile error	configs
256	0.019–0.034	0.237–0.252	9	0.010–0.021	0.18	23
1024	0.008–0.023	0.224–0.244	9	0.005–0.015	0.10	23
4096	0.004–0.016	0.215–0.230	9	0.005–0.014	0.06	17
16384	0.003–0.012	0.214–0.230	9	0.003–0.010	0.05	12

The conditional variance reflects the same structure. Among the 84 direct-sampling configurations of the statistical experiments, the conditional variance under random-sign encoding is 0.98–1.06 when $c \ge 8$, and 0.985–1.020 within the critical window. When c is small the conditional variance is clearly above one, about 1.12 at $c = 4$, about 1.5 at $c = 2$, and above 10 when $c < 1$, consistent with the $1 + 2/c^2$ obtained by converting the variance $4 + 2c^2$ of the limit distribution to the X scale. The excess comes from the ± 2 shift of the duplicate edge pair, which is non-negligible relative to $s(d)$.

H.7 Conditional Miss Rate and Extra Capacity at Finite Size

Among the 189 configurations of the statistical experiments, 84 use direct sampling and 105 use planted sampling. The maximum miss/u of the Skellam rule on the direct-sampling configurations is 1.04 (the same at both levels of u), occurring within the critical window; the maximum of 1.03 on the planted configurations is computed under the planted distribution. Each configuration has about 20,000 failures, corresponding to about 200 misses at $u = 0.01$, and the standard

error of miss/u is about 0.07, so a value like 1.04 slightly above 1 cannot be distinguished from 1. On the low-load configurations with $c < 8$, the median miss/u of the Skellam rule at $u = 0.01$ is at most 0.76. Every sample accepts at least one candidate. When no candidate is accepted, $U = +\infty$ and this is not counted as a miss. The Gaussian rule has miss/u 0.83–1.19 over all configurations when $c \geq 8$; at $c = 1$, $u = 0.01$ it is about 11 for $k = 3$ and about 8.5 for $k = 4, 5$. When c is small and the observation falls near $d - 2$, the observations for $d = 2, 3$ are below the acceptance threshold of every candidate, so the Gaussian rule returns $+\infty$ and no miss occurs. On the direct-sampling configurations with $c \geq 8$ in the statistical experiments, the median miss/u of the Skellam rule at the two levels of u is 0.79–0.86 and 0.95–0.97.

At $u = 0.01$ and $c \geq 8$, the median extra capacity of the Skellam rule is 25%, 11.3%, 5.4%, and 2.6% at $M = 256$, 1024, 4096, 16384, with the 90% quantile at most 41%, 18%, 8.5%, and 4.1%; the medians of the Gaussian rule are 23.5%, 10.8%, 5.3%, and 2.6%, differing from the Skellam rule by at most 1.5 percentage points. At $c = 1, 2, 4$ the median of $U - d$ is 5, 8, 15 under the Skellam rule and 3, 6, 14 under the Gaussian rule. For fixed small d, the 90% quantile of the upper bound at $d = 2$ exceeds d by 4–5. Among the 42 configurations with $M = 64$, the maximum miss/u of the Skellam rule is 1.08 at $u = 0.05$ ($k = 4$, $d = 6$, 95% Wilson interval 1.02–1.15) and 1.04 at $u = 0.01$.

Table H.7 lists the miss/u of the Gaussian rule at fixed c on $M = 16384$, corresponding to the triangles with $c \leq 16$ in Figure 6.4 of the main text. As c decreases the excess grows rapidly, being about 8–11 at $c = 1$ and above 33 at $c = 0.5$ for $u = 0.01$; for $c \leq 4$ the values at $M = 4096$ and 16384 differ by at most 8%, and the excess does not vanish as the table size grows.

Table H.7: The miss/u of the Gaussian rule at fixed c ($M = 16384$).

	$u = 0.01$			$u = 0.05$		
c	$k = 3$	$k = 4$	$k = 5$	$k = 3$	$k = 4$	$k = 5$
0.5	36.4	33.2	33.2	9.37	8.81	9.39
1	11.3	8.74	8.27	4.09	4.24	4.60
2	3.06	2.48	2.60	2.00	1.70	1.94
4	1.51	1.41	1.50	1.21	1.19	1.20
8	1.12	1.03	1.08	1.07	1.08	1.06
16	0.98	0.93	0.99	0.98	1.01	1.00

The samples of the worst-configuration test (Exp-WORST) are independent of the statistical experiments, and the selection rule and sample sizes are fixed before sampling. The configurations are selected as follows: first take the five configurations whose miss/u exceeds 1 in exploration, then fill the remaining slots in descending order of the larger of the one-sided 95% exact upper bounds divided by u over the two levels of u in the statistical experiments. Four of the top-ranked configurations have only planted samples, for which direct sampling cannot obtain failures at feasible cost, so they are passed over by the rule to later configurations; their miss/u under Q is at most 1.02. For each configuration and each level of u, the previously estimated 95% Wilson upper bound (at most 1) is used as the planning value, and the number of failures is set so that the single-item passing probability is at least $1 - 0.2/24$, taking the larger of the two levels, so that the probability that all 24 judgments pass is at least 80% at the planning value. Each judgment passes when the one-sided simultaneous confidence upper bound is at most $1.1u$; the bound uses the Clopper–Pearson interval with a Bonferroni correction over the 24 judgments. All 24 judgments pass, with the largest upper bound $1.024u$; 22 of the 24 two-sided 95% intervals lie entirely below 1, and the other 2 cover 1. Only one test configuration uses $M = 256$. Sampling runs in blocks, and the analysis uses all blocks up to the first time the block-ordered cumulative failure count reaches the target; the stopping depends only on the failure count. At the two test-configuration points with $k = 3$, $M = 1024$, the miss/u of the

Gaussian rule is 11.1 ($u = 0.01$) and 4.1 ($u = 0.05$) at $c = 1$, and 2.9 and 1.8 at $c = 2$. Table H.8 lists the load, failure count, miss/u, and one-sided simultaneous confidence upper bound of each test configuration; the segments in Figure 6.5 of the main text are two-sided 95% intervals.

Table H.8: The miss/u of the Skellam rule on the Exp-WORST configurations and the one-sided simultaneous confidence upper bound (in parentheses, per-item level 0.05/24).

configuration	load	failures	$u = 0.01$	$u = 0.05$
$k = 3$, $M = 1024$, $d = 32$	0.04	55,509	0.53 (0.62)	0.97 (1.02)
$k = 3$, $M = 1024$, $d = 64$	0.08	289,001	0.90 (0.95)	0.91 (0.94)
$k = 3$, $M = 1024$, $d = 192$	0.23	289,394	0.79 (0.83)	0.95 (0.97)
$k = 4$, $M = 256$, $d = 128$	0.65	105,689	0.74 (0.82)	0.93 (0.97)
$k = 4$, $M = 1024$, $d = 751$	0.95	298,439	0.85 (0.90)	0.95 (0.98)
$k = 3$, $M = 4096$, $d = 3352$	1.00	333,476	0.93 (0.98)	0.97 (1.00)
$k = 5$, $M = 4096$, $d = 3018$	1.05	332,350	0.90 (0.95)	0.99 (1.01)
$k = 5$, $M = 1024$, $d = 755$	1.05	332,353	0.83 (0.87)	0.96 (0.98)
$k = 4$, $M = 16384$, $d = 16384$	1.29	293,250	0.95 (1.01)	0.98 (1.01)
$k = 3$, $M = 4096$, $d = 5029$	1.50	332,350	0.91 (0.96)	0.98 (1.00)
$k = 4$, $M = 16384$, $d = 18980$	1.50	290,808	0.96 (1.02)	0.98 (1.00)
$k = 4$, $M = 16384$, $d = 37959$	3.00	289,952	0.95 (1.01)	0.97 (1.00)

Among the 12 configurations in the upper row of Figure 6.7, two ($k = 3$, $M = 4096$ and $k = 4$, $M = 16384$) are test configurations; the other 10 follow the same design (sample size and stopping rule fixed before sampling), each with about 290,000 direct-sampling failures. Under random-sign encoding the common elements cancel in the difference table, the sign composition does not affect the post-failure distribution, and the $|b| \leq 1$ used by the protocol under balanced differences does not change the upper bound either, so the protocol experiments, the statistical experiments, and these direct samples estimate the same conditional miss rate and are pooled by configuration in the figure. On these 10 configurations the one-sided upper bound of the Skellam rule is at most $1.025u$ at $u = 0.01$ and $1.003u$ at $u = 0.05$.

H.8 Reliability and Communication Cost of the Two-Round Protocol

Each trial generates the common elements and difference elements separately, with all differences on the A side when one-sided and half on each side when balanced. The common elements cancel in the difference table but still participate in table building and cell-by-cell subtraction as usual. Trials that succeed in the first round need no upper bound; only first-round-failure trials enter the comparison, and the two levels of u are computed separately on the same batch of trials. The 64 cells in the second-round table length are an additional lower bound set on top of the table length of Section 3.2; the second round decodes independently on the new table, consistent with the analysis of Theorem 5.7. The second-round recovery results all match the true difference element by element. The two rounds use independent cell hashes, and share the sign hash and checksum hash. Bytes are counted at 20 bytes per cell (4 bytes for the count line and 8 bytes each for keySum and checksum), with random-sign encoding adding an 8-byte global counter. The upper bounds share the same batch of keys and hashes, and for each first-round-failure trial we record U_b, the second-round size, and the second-round outcome for each bound, with the paired comparison aligned by trial. The number of trials per configuration is shared between the two levels of u, set to $\lceil 2 \times 10^4 / \widehat{\Pr}(F_1) \rceil$ with a cap of 3,000,000. At load 0.95 and $M_1 \geq 4096$, the eight configurations with $k = 4, 5$ have no first-round failure in the pilot or the 100,000 trials, and enter only the failure-rate and byte statistics. The interval of the success-rate difference is computed by the Newcombe hybrid score method for paired proportion differences.

Table H.9 summarizes the protocol results of Section 6.5 of the main text by M_1 and u. The final two-round failure rate is at most 0.3% at $M_1 = 256$; the conditional miss rate on the balanced configurations is consistent with the statistical experiments of Section 6.4. The second-round failure after $U_b \geq d$ is 0.06–0.08% at $M_1 = 256$. The oracle's second-round table is smaller, and in per-trial pairing the success rate of the Skellam rule is at most 0.35 percentage points higher than the oracle on some $M_1 = 256$ configurations.

Table H.9: Two-round protocol summary ($k = 3, 4, 5$). miss/u is the median / maximum over the balanced configurations (at least 1000 first-round failures); the final failure is the maximum over configurations; the second-round size and total bytes are medians relative to the oracle. The oracle does not depend on u and is listed only in the upper half.

M_1, u	upper bound	miss/u	final failure	second-round size	total bytes
256, 0.01	Skellam	0.73 / 0.87	0.24%	+25.6%	+14.7%
	Gaussian	1.00 / 1.12	0.25%	+23.9%	+13.7%
	oracle	–	0.51%	0	0
1024, 0.01	Skellam	0.88 / 1.00	0.06%	+11.4%	+6.5%
	Gaussian	1.04 / 1.17	0.08%	+11.1%	+6.3%
	oracle	–	0.09%	0	0
4096, 0.01	Skellam	0.92 / 1.01	0.02%	+5.4%	+3.1%
	Gaussian	1.01 / 1.09	0.03%	+5.3%	+3.0%
	oracle	–	0.03%	0	0
16384, 0.01	Skellam	0.96 / 1.07	0.01%	+2.6%	+1.5%
	Gaussian	1.02 / 1.13	0.01%	+2.6%	+1.5%
	oracle	–	0.02%	0	0
256, 0.05	Skellam	0.93 / 1.01	0.29%	+16.9%	+9.7%
	Gaussian	1.01 / 1.07	0.27%	+16.3%	+9.4%
1024, 0.05	Skellam	0.95 / 1.05	0.13%	+7.8%	+4.5%
	Gaussian	1.02 / 1.09	0.11%	+7.7%	+4.4%
4096, 0.05	Skellam	0.97 / 1.04	0.02%	+3.8%	+2.2%
	Gaussian	1.01 / 1.07	0.02%	+3.8%	+2.1%
16384, 0.05	Skellam	1.00 / 1.05	0.01%	+1.9%	+1.1%
	Gaussian	1.01 / 1.06	0.02%	+1.9%	+1.1%

The baseline uses the calibration rule from [38]: for $k = 3$ that work bins by first-round size and takes the worst 1% quantile $q_{0.01}$ of the post-failure $\widehat{d}/d$ over several loads and sign compositions (0.8045, 0.8997, 0.9482 for $M_1 = 256, 1024, 4096$). Here we keep only its upper-bound part $U = \lceil \widehat{d}/q_{0.01} \rceil$, attached to the same second-round table length as the two rules of this paper; the capacity coefficient, the deduction of elements already recovered in the first round, and the two-round joint peeling of that protocol are outside the comparison. Table H.10 compares this rule under the two encodings, with and without accounting for $|b|$.

Under the same calibration factor, the conditional miss rate of the standard-encoding one-sided configurations is between $0.05u$ and $1.03u$, and that of random-sign encoding is between $0.77u$ and $1.04u$; at fixed load 0.95–1.0, the one-sided and balanced of standard encoding are 0.05–$0.40u$ and 0.86–$0.94u$, and those of random-sign encoding are 0.80–$0.95u$ and 0.85–$1.01u$. After accounting for $|b|$, the one-sided miss of both encodings is zero. In per-trial pairing, at fixed calibration factor and with or without $|b|$, the success-rate difference of random-sign encoding relative to standard encoding is within $\pm 1 \times 10^{-3}$, with 2 of the 30 items having intervals that exclude zero (both at $M_1 = 256$, load 1.0), and the median total-byte difference is about 0.01%.

Table H.10: Calibration rule versus the Skellam rule ($k = 3$, $u = 0.01$). conditional miss rate /u is the median / maximum over balanced and the range over one-sided; the final failure is the maximum; total bytes are the median relative to the oracle.

M_1	rule	balanced	one-sided	final failure	total bytes
256	standard encoding, calibration factor	0.92 / 1.02	0.09–0.90	0.27%	+13.9%
	standard encoding, calibration factor, with $\lvert b\rvert$	0.92 / 1.02	0	0.26%	+13.9%
	random sign, calibration factor	0.87 / 0.97	0.85–0.93	0.23%	+13.9%
	random sign, Skellam rule	0.72 / 0.80	0	0.24%	+14.6%
1024	standard encoding, calibration factor	0.88 / 0.94	0.05–1.03	0.07%	+6.4%
	standard encoding, calibration factor, with $\lvert b\rvert$	0.88 / 0.94	0	0.07%	+6.4%
	random sign, calibration factor	0.98 / 1.03	0.80–1.04	0.08%	+6.4%
	random sign, Skellam rule	0.89 / 0.92	0	0.06%	+6.5%
4096	standard encoding, calibration factor	0.84 / 0.86	0.36–0.89	0.02%	+3.1%
	standard encoding, calibration factor, with $\lvert b\rvert$	0.84 / 0.86	0	0.02%	+3.1%
	random sign, calibration factor	0.90 / 0.94	0.77–0.95	0.02%	+3.1%
	random sign, Skellam rule	0.97 / 1.01	0	0.01%	+3.1%

H.9 Computation Cost

The host of the computation-cost measurement (Exp-COST) is a dual-socket Intel Xeon Silver 4410Y (nominal 2.0 GHz, 24 physical cores and 48 logical CPUs), Linux, with a single-threaded Go implementation. Before timing, the host's task queue is paused, and after all the current batch's tasks on it finish, the program is pinned to one core; the frequency and system load of that core are recorded every 10 seconds, and the frequency stays at 2.001 GHz during the run.

The Poisson tables are kept in a main-memory cache managed by the program. One cache entry corresponds to one value in the Poisson table of a candidate n and stores two double-precision numbers, the probability and the cumulative probability of that value, 16 bytes in total; the capacity is 16,000,000 entries (about 256 MB). When the cache exceeds its limit, the program evicts the least recently used tables in recency order until the occupancy plus the size of the table to insert is at most half the capacity, then inserts the new table. We time sizing under three cache conditions: cold means each sizing starts from an empty cache; session means the same receiver performs several consecutive synchronizations (possibly one or two rounds of the protocol) at a fixed (M, k), with the cache kept between synchronizations and each processing a new observation; warm means recomputing the same observation once immediately.

Sizing timing uses the failure records of the statistical experiments at $M = 16384$, taking the first 200 per configuration, with the order of the four modes cold, session, warm, and Gaussian rule rotated over repetitions and each mode repeated 10 times. Protocol timing uses $M_1 = 16384$ and 100,000 common elements, with the same set of 40 trials per configuration repeated 10 times and the branch order rotated over repetitions; the first-round decoding time is counted per trial. Figure H.4 and Table 6.2 come from two runs on the same host and settings, and the protocol part differs by less than 1% from the first run.

Appendix G predicts that the sizing cost is roughly proportional to λ_d, that is, growing about as c^2. Figure H.4 fixes M, with load proportional to c. cold grows roughly as the 1.9-th power of the load, that is, about as $c^{1.9}$, from 0.06–0.12 ms at load about 0.08 ($c = 8$) to 69–84 ms at load 3; session is about one fifth of cold, and warm about one sixth to one seventh. At load 3 the Poisson tables requested cumulatively by session exceed the cache limit, the program evicts some old tables, and session rises to 41–55 ms. The sizing of the Gaussian rule is 0.06–0.29 μs. In the full protocol, the total time of the Gaussian rule and the oracle differs by less than 1%. The Poisson-table memory has a session peak of 49–86 MB at load 1.05, 128–174 MB at load 1.5, and reaches the 256 MB limit at load 3.

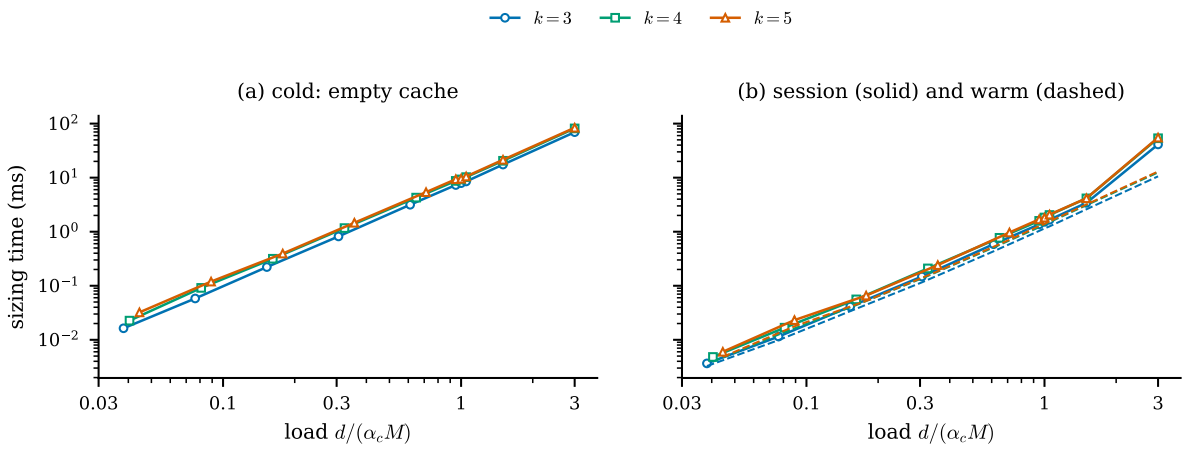


Figure H.4: Single sizing time of the Skellam rule ($M = 16384$, single thread, cache limit 16,000,000 entries). (a) cold; (b) session (solid) and warm (dashed).

The decoder treats cells with count ± 1 as candidate pure cells and confirms them with the checksum. Under standard encoding with one-sided differences all keys have the same sign, so a cell with count ± 1 necessarily contains only one key and every candidate passes. When the difference has both positive and negative elements, for example three keys with signs $+1, +1, -1$ landing in the same cell, the count is also $+1$, and such candidates are excluded by the checksum. The numbers of pure-cell checks, checksum rejections, and sign hashes per recovered key come from a single separate run (30 trials per configuration, 3 repetitions): random-sign encoding checks on average 1.77 candidates ($k = 3$) and 2.53 candidates ($k = 4$); standard encoding checks 1.00 under one-sided differences and matches random-sign encoding under balanced differences. The first-round decoding time ratio (random sign / standard) is 0.99–1.06 under balanced differences and 1.26–2.42 under one-sided differences at load 1.0–1.5. At load 3 the one-sided ratio is 3.1, because standard encoding stops almost immediately under one-sided differences; the absolute times of both are only 0.04–0.16 ms, less than 1% of the total protocol time.

The exact Skellam CDF can also be replaced by the Gaussian approximation at large parameters: over the 33 configurations in the exploration phase whose Poisson mean exceeds 200, the samples whose U changes are at most 1.8%, and the miss count is unchanged. All results in this paper use the exact rule.